

11. List five specifications for a second-order underdamped system.
12. For Question 11 how many specifications completely determine the response?
13. What pole locations characterize (1) the underdamped system, (2) the overdamped system, and (3) the critically damped system?
14. Name two conditions under which the response generated by a pole can be neglected.
15. How can you justify pole-zero cancellation?
16. Does the solution of the state equation yield the output response of the system? Explain.
17. What is the relationship between  $(s\mathbf{I} - \mathbf{A})$ , which appeared during the Laplace transformation solution of the state equations, and the state-transition matrix, which appeared during the classical solution of the state equation?
18. Name a major advantage of using time-domain techniques for the solution of the response.
19. Name a major advantage of using frequency-domain techniques for the solution of the response.
20. What three pieces of information must be given in order to solve for the output response of a system using state-space techniques?
21. How can the poles of a system be found from the state equations?

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## Problems

1. Derive the output responses for all parts of Figure 4.7. [Section: 4.4]
2. Find the output response,  $c(t)$ , for each of the systems shown in Figure P4.1. Also find the time constant, rise time, and settling time for each case. [Sections: 4.2, 4.3]

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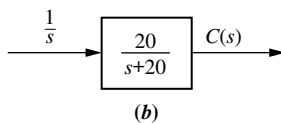
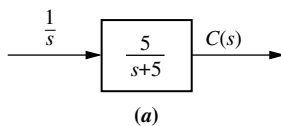


FIGURE P4.1

3. Plot the step responses for Problem 2 using MATLAB.
4. Find the capacitor voltage in the network shown in Figure P4.2 if the switch closes at  $t = 0$ . Assume zero

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initial conditions. Also find the time constant, rise time, and settling time for the capacitor voltage. [Sections: 4.2, 4.3]

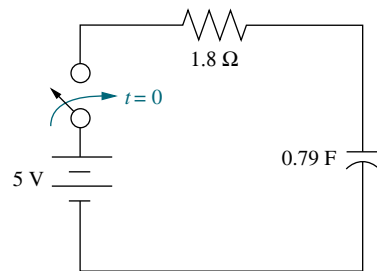


FIGURE P4.2

5. Plot the step response for Problem 4 using MATLAB. From your plots, find the time constant, rise time, and settling time.
6. For the system shown in Figure P4.3, (a) find an equation that relates settling time of the velocity of

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the mass to  $M$ ; (b) find an equation that relates rise time of the velocity of the mass to  $M$ . [Sections: 4.2, 4.3]

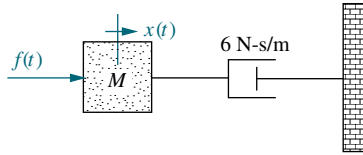


FIGURE P4.3

7. Plot the step response for Problem 6 using MATLAB. From your plots, find the time constant, rise time, and settling time. Use  $M = 1$  and  $M = 2$ .
8. For each of the transfer functions shown below, find the locations of the poles and zeros, plot them on the  $s$ -plane, and then write an expression for the general form of the step response without solving for the inverse Laplace transform. State the nature of each response (overdamped, underdamped, and so on). [Sections: 4.3, 4.4]

a.  $T(s) = \frac{2}{s+2}$

b.  $T(s) = \frac{5}{(s+3)(s+6)}$

c.  $T(s) = \frac{10(s+7)}{(s+10)(s+20)}$

d.  $T(s) = \frac{20}{s^2 + 6s + 144}$

e.  $T(s) = \frac{s+2}{s^2 + 9}$

f.  $T(s) = \frac{(s+5)}{(s+10)^2}$

9. Use MATLAB to find the poles of [Section: 4.2]

$$T(s) = \frac{s^2 + 2s + 2}{s^4 + 6s^3 + 4s^2 + 7s + 2}$$

10. Find the transfer function and poles of the system represented in state space here: [Section: 4.10]

$$\dot{\mathbf{x}} = \begin{bmatrix} 8 & -4 & 1 \\ -3 & 2 & 0 \\ 5 & 7 & -9 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -4 \\ -3 \\ 4 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 2 & 8 & -3 \end{bmatrix} \mathbf{x}; \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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11. Repeat Problem 10 using MATLAB. [Section: 4.10]

12. Write the general form of the capacitor voltage for the electrical network shown in Figure P4.4. [Section: 4.4]

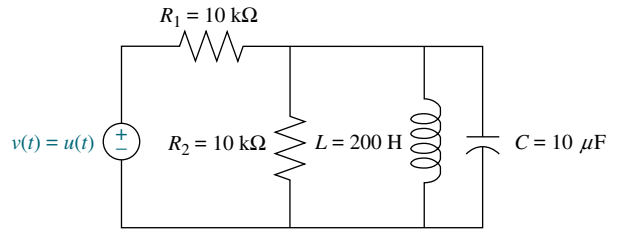


FIGURE P4.4

13. Use MATLAB to plot the capacitor voltage in Problem 12. [Section: 4.4]

14. Solve for  $x(t)$  in the system shown in Figure P4.5 if  $f(t)$  is a unit step. [Section: 4.4]

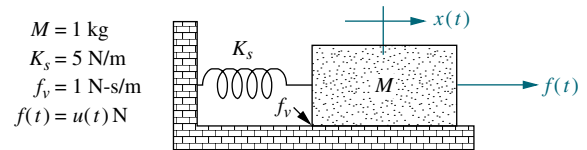


FIGURE P4.5

15. The system shown in Figure P4.6 has a unit step input. Find the output response as a function of time. Assume the system is underdamped. Notice that the result will be Eq. (4.28). [Section: 4.6]

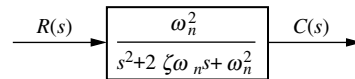


FIGURE P4.6

16. Derive the relationship for damping ratio as a function of percent overshoot, Eq. (4.39). [Section: 4.6]
17. Calculate the exact response of each system of Problem 8 using Laplace transform techniques, and compare the results to those obtained in that problem. [Sections: 4.3, 4.4]
18. Find the damping ratio and natural frequency for each second-order system of Problem 8 and show that the value of the damping ratio conforms to the type of response (underdamped, overdamped, and so on) predicted in that problem. [Section: 4.5]
19. A system has a damping ratio of 0.5, a natural frequency of 100 rad/s, and a dc gain of 1. Find the response of the system to a unit step input. [Section: 4.6]

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20. For each of the second-order systems that follow, find  $\zeta$ ,  $\omega_n$ ,  $T_s$ ,  $T_p$ ,  $T_r$ , and %OS. [Section: 4.6].

a.  $T(s) = \frac{16}{s^2 + 3s + 16}$

b.  $T(s) = \frac{0.04}{s^2 + 0.02s + 0.04}$

c.  $T(s) = \frac{1.05 \times 10^7}{s^2 + 1.6 \times 10^3 s + 1.05 \times 10^7}$

21. Repeat Problem 20 using MATLAB. Have the computer program estimate the given specifications and plot the step responses. Estimate the rise time from the plots. [Section: 4.6].

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22. Use MATLAB's LTI Viewer and obtain settling time, peak time, risetime, and percent overshoot for each of the systems in Problem 20. [Section: 4.6].

Gui Tool  
GUIT

23. For each pair of second-order system specifications that follow, find the location of the second-order pair of poles. [Section: 4.6].

a. %OS = 12%;  $T_s = 0.6$  second

b. %OS = 10%;  $T_p = 5$  seconds

c.  $T_s = 7$  seconds;  $T_p = 3$  seconds

24. Find the transfer function of a second-order system that yields a 12.3% overshoot and a settling time of 1 second. [Section: 4.6]

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25. For the system shown in Figure P4.7, do the following: [Section: 4.6]

a. Find the transfer function  $G(s) = X(s)/F(s)$ .

b. Find  $\zeta$ ,  $\omega_n$ , %OS,  $T_s$ ,  $T_p$ , and  $T_r$ .

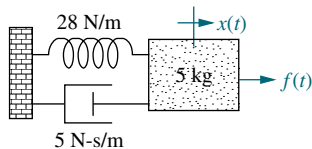


FIGURE P4.7

26. For the system shown in Figure P4.8, a step torque is applied at  $\theta_1(t)$ . Find

a. The transfer function,  $G(s) = \theta_2(s)/T(s)$ .

b. The percent overshoot, settling time, and peak time for  $\theta_2(t)$ . [Section: 4.6]

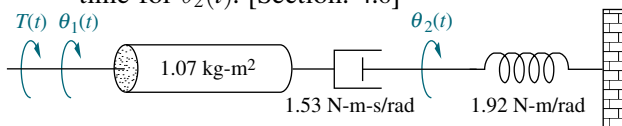


FIGURE P4.8

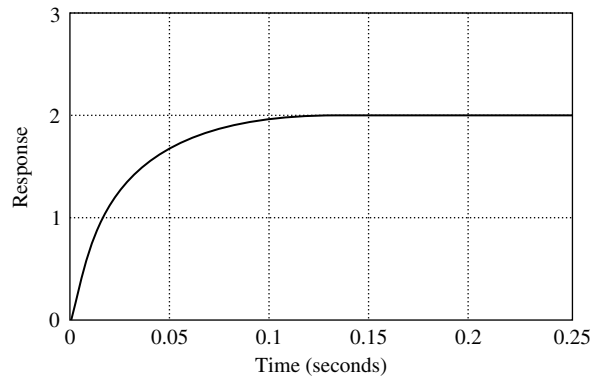
27. Derive the unit step response for each transfer function in Example 4.8. [Section: 4.7].

28. Find the percent overshoot, settling time, rise time, and peak time for

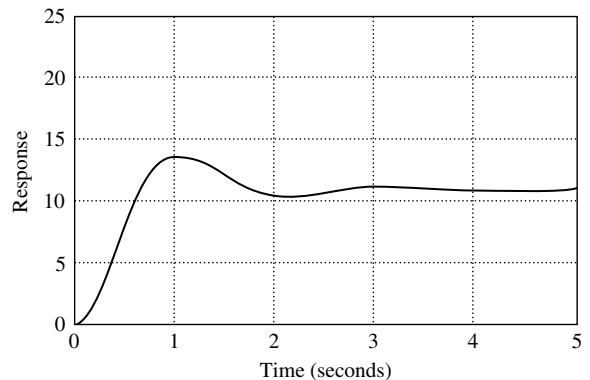
$$T(s) = \frac{14.145}{(s^2 + 0.842s + 2.829)(s + 5)} \quad [\text{Section: 4.7}]$$

29. For each of the unit step responses shown in Figure P4.9, find the transfer function of the system. [Sections: 4.3, 4.6].

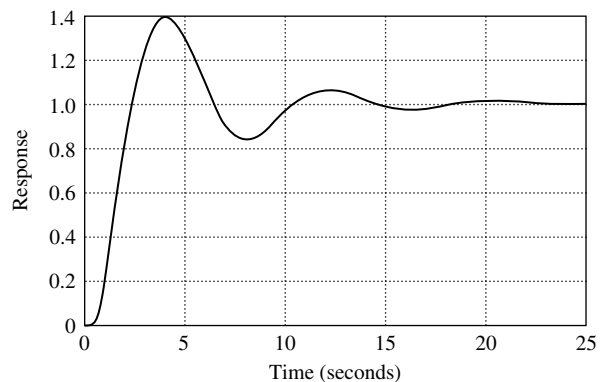
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(a)



(b)



(c)

FIGURE P4.9

30. For the following response functions, determine if pole-zero cancellation can be approximated. If it can, find percent overshoot, settling time, rise time, and peak time. [Section: 4.8].

a.  $C(s) = \frac{(s+3)}{s(s+2)(s^2+3s+10)}$

b.  $C(s) = \frac{(s+2.5)}{s(s+2)(s^2+4s+20)}$

c.  $C(s) = \frac{(s+2.1)}{s(s+2)(s^2+s+5)}$

d.  $C(s) = \frac{(s+2.01)}{s(s+2)(s^2+5s+20)}$

31. Using MATLAB, plot the time response of Problem 30a and from the plot determine percent overshoot, settling time, rise time, and peak time. [Section: 4.8]

MATLAB  
ML

32. Find peak time, settling time, and percent overshoot for only those responses below that can be approximated as second-order responses. [Section: 4.8].

a.  $c(t) = 0.003500 - 0.001524e^{-4t}$   
 $-0.001976e^{-3t}\cos(22.16t)$   
 $-0.0005427e^{-3t}\sin(22.16t)$

b.  $c(t) = 0.05100 - 0.007353e^{-8t}$   
 $-0.007647e^{-6t}\cos(8t)$   
 $-0.01309e^{-6t}\sin(8t)$

c.  $c(t) = 0.009804 - 0.0001857e^{-5.1t}$   
 $-0.009990e^{-2t}\cos(9.796t)$   
 $-0.001942e^{-2t}\sin(9.796t)$

d.  $c(t) = 0.007000 - 0.001667e^{-10t}$   
 $-0.008667e^{-2t}\cos(9.951t)$   
 $-0.0008040e^{-2t}\sin(9.951t)$

33. For each of the following transfer functions with zeros, find the component parts of the unit step response: (1) the derivative of the response without a zero and (2) the response without a zero, scaled to the negative of the zero value. Also, find and plot the total response. Describe any nonminimum-phase behavior. [Section: 4.8].

a.  $G(s) = \frac{s+2}{s^2+3s+36}$

b.  $G(s) = \frac{s-2}{s^2+3s+36}$

34. Use MATLAB's Simulink to obtain the step response of a system,

Simulink  
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$$G(s) = \frac{1}{s^2 + 3s + 10}$$

under the following conditions:  
[Section: 4.9]

- a. The system is linear and driven by an amplifier whose gain is 10.  
 b. An amplifier whose gain is 10 drives the system. The amplifier saturates at  $\pm 0.25$  volts. Describe the effect of the saturation on the system's output.  
 c. An amplifier whose gain is 10 drives the system. The amplifier saturates at  $\pm 0.25$  volts. The system drives a 1:1 gear train that has backlash. The dead-band width of the backlash is 0.02 rad. Describe the effect of saturation and backlash on the system's output.

35. A system is represented by the state and output equations that follow. Without solving the state equation, find the poles of the system. [Section: 4.10]

State Space  
SS

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & -1 \\ -3 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u(t)$$

$$y = [3 \quad 2] \mathbf{x}$$

36. A system is represented by the state and output equations that follow. Without solving the state equation, find [Section: 4.10]

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- a. the characteristic equation;  
 b. the poles of the system

State Space  
SS

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 6 & 5 \\ 1 & 4 & 2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u(t)$$

$$y = [1 \quad 2 \quad 0] \mathbf{x}$$

37. Given the following state-space representation of a system, find  $Y(s)$ : [Section: 4.10]

State Space  
SS

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 2 \\ -3 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \sin 3t$$

$$y = [1 \quad 2] \mathbf{x}; \quad \mathbf{x}(0) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

38. Given the following system represented in state space, solve for  $Y(s)$  using the

State Space  
SS

Laplace transform method for solution of the state equation: [Section: 4.10]

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ -2 & -4 & 1 \\ 0 & 0 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} e^{-t}$$

$$y = [0 \ 0 \ 1] \mathbf{x}; \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

39. Solve the following state equation and output equation for  $y(t)$ , where  $u(t)$  is the unit step. Use the Laplace transform method. [Section: 4.10]

State Space  
**SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 0 \\ -1 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

$$y = [0 \ 1] \mathbf{x}; \mathbf{x}(0) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

40. Solve for  $y(t)$  for the following system represented in state space, where  $u(t)$  is the unit step. Use the Laplace transform approach to solve the state equation. [Section: 4.10]

State Space  
**SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u(t)$$

$$y = [0 \ 1 \ 1] \mathbf{x}; \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

41. Use MATLAB to plot the step response of Problem 40. [Section: 4.10]

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42. Repeat Problem 40 using MATLAB's Symbolic Math Toolbox and Eq. (4.96). In addition, run your program with an initial condition,

Symbolic Math  
**SM**

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}. \text{ [Section : 4.10]}$$

43. Using classical (not Laplace) methods only, solve for the state-transition matrix, the state vector, and the output of the

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system represented here: [Section: 4.11]

State Space  
**SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -1 & -5 \end{bmatrix} \mathbf{x}; y = [1 \ 2] \mathbf{x};$$

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

44. Using classical (not Laplace) methods only, solve for the state-transition matrix, the state vector, and the output of the system represented here, where  $u(t)$  is the unit step: [Section: 4.11].

State Space  
**SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y = [3 \ 4] \mathbf{x}; \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

45. Solve for  $y(t)$  for the following system represented in state space, where  $u(t)$  is the unit step. Use the classical approach to solve the state equation. [Section: 4.11]

State Space  
**SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -6 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t)$$

$$y = [1 \ 0 \ 0] \mathbf{x}; \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

46. Repeat Problem 45 using MATLAB's Symbolic Math Toolbox and Eq. (4.109). In addition, run your program with an initial condition,

Symbolic Math  
**SM**

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}. \text{ [Section : 4.11]}$$

47. Using methods described in Appendix H.1 located at [www.wiley.com/college/nise](http://www.wiley.com/college/nise) simulate the following system and plot the step response. Verify the expected values of percent overshoot, peak time, and settling time.

State Space  
**SS**

$$T(s) = \frac{1}{s^2 + 0.8s + 1}$$

48. Using methods described in Appendix H.1 located at [www.wiley.com/college/nise](http://www.wiley.com/college/nise), simulate the following system and plot the output,  $y(t)$ , for a step input:

State Space  
SS

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ -10 & -7 & 1 \\ 0 & 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [1 \ 1 \ 0] \mathbf{x}; \quad \mathbf{x}(0) = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

49. A human responds to a visual cue with a physical response, as shown in Figure P4.10. The transfer function that relates the output physical response,  $P(s)$ , to the input visual command,  $V(s)$ , is

$$G(s) = \frac{P(s)}{V(s)} = \frac{(s + 0.5)}{(s + 2)(s + 5)}$$

(Stefani, 1973). Do the following:

- Evaluate the output response for a unit step input using the Laplace transform.
  - Represent the transfer function in state space.
  - Use MATLAB to simulate the system and obtain a plot of the step response.
50. Industrial robots are used for myriad applications. Figure P4.11 shows a robot used to move 55-pound bags of salt pellets; a vacuum head lifts the bags before positioning. The robot can move as many as 12 bags per minute (Schneider, 1992). Assume a

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FIGURE P4.11 Vacuum robot lifts two bags of salt.

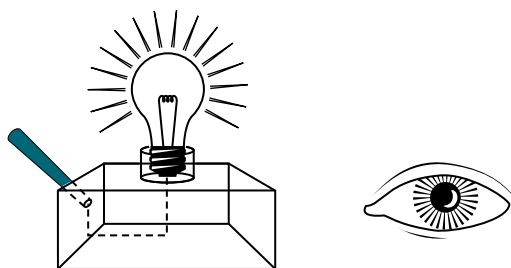
model for the open-loop swivel controller and plant of

$$G_e(s) = \frac{\omega_o(s)}{V_i(s)} = \frac{K}{(s + 10)(s^2 + 4s + 10)}$$

where  $\omega_o(s)$  is the Laplace transform of the robot's output swivel velocity and  $V_i(s)$  is the voltage applied to the controller.

- Evaluate percent overshoot, settling time, peak time, and rise time of the response of the open-loop swivel velocity to a step-voltage input. Justify all second-order assumptions.
- Represent the open-loop system in state space.

State Space  
SS



Step 1: Light source on

Step 2: Recognize light source

Step 3: Respond to light source

FIGURE P4.10 Steps in determining the transfer function relating output physical response to the input visual command

- c. Use MATLAB or any other computer program to simulate the system and compare your results to a.

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51. Anesthesia induces muscle relaxation (paralysis) and unconsciousness in the patient. Muscle relaxation can be monitored using electromyogram signals from nerves in the hand; unconsciousness can be monitored using the cardiovascular system's mean arterial pressure. The anesthetic drug is a mixture of isoflurane and atracurium. An approximate model relating muscle relaxation to the percent isoflurane in the mixture is

$$\frac{P(s)}{U(s)} = \frac{7.63 \times 10^{-2}}{s^2 + 1.15s + 0.28}$$

where  $P(s)$  is muscle relaxation measured as a fraction of total paralysis (normalized to unity) and  $U(s)$  is the percent mixture of isoflurane (Linkens, 1992). [Section: 4.6]

- Find the damping ratio and the natural frequency of the paralysis transient response.
  - Find the maximum possible percent paralysis if a 2% mixture of isoflurane is used.
  - Plot the step response of paralysis if a 1% mixture of isoflurane is used.
  - What percent isoflurane would have to be used for 100% paralysis?
52. To treat acute asthma, the drug theophylline is infused intravenously. The rate of change of the drug concentration in the blood is equal to the difference between the infused concentration and the eliminated concentration. The infused concentration is  $i(t)/V_d$ , where  $i(t)$  is the rate of flow of the drug by weight and  $V_d$  is the apparent volume and depends on the patient. The eliminated concentration is given by  $k_{10}c(t)$ , where  $c(t)$  is the current concentration of the drug in the blood and  $k_{10}$  is the elimination rate constant. The theophylline concentration in the blood is critical—if it is too low, the drug is ineffective; if too high, the drug is toxic (Jannett, 1992). You will help the doctor with your calculations.
- Derive an equation relating the desired blood concentration,  $C_D$ , to the required infusion rate by weight of the drug,  $I_R$ .
  - Derive an equation that will tell how long the drug must be administered to reach the desired

blood concentration. Use both rise time and settling time.

- Find the infusion rate of theophylline if  $V_D = 600$  ml,  $k_{10} = 0.07 \text{ h}^{-1}$ , and the required blood level of the drug is 12 mcg/ml (“mcg” means micrograms). See (Jannett, 1992); for a description of parameter values.
  - Find the rise and settling times for the constants in c.
53. Upper motor neuron disorder patients can benefit and regain useful function through the use of functional neuroprostheses. The design requires a good understanding of muscle dynamics. In an experiment to determine muscle responses, the identified transfer function was (Zhou, 1995)

$$M(s) = \frac{2.5e^{-0.008s}(1 + 0.172s)(1 + 0.008s)}{(1 + 0.07s)^2(1 + 0.05s)^2}$$

Find the unit step response of this transfer function.

54. When electrodes are attached to the mastoid bones (right behind the ears) and current pulses are applied, a person will sway forward and backward. It has been found that the transfer function from the current to the subject's angle (in degrees) with respect to the vertical is given by (Nashner, 1974)

$$\frac{\theta(s)}{I(s)} = \frac{5.8(0.3s + 1)e^{-0.1s}}{(s + 1)(s^2/1.2^2 + 0.6s/1.2 + 1)}$$

- Determine whether a dominant pole approximation can be applied to this transfer function.
  - Find the body sway caused by a 250  $\mu\text{A}$  pulse of 150 msec duration.
55. A MOEMS (optical MEMS) is a MEMS (Micro Electromechanical Systems) with an optical fiber channel that takes light generated from a laser diode. It also has a photodetector that measures light intensity variations and outputs voltage variations proportional to small mechanical device deflections. Additionally, a voltage input is capable of deflecting the device. The apparatus can be used as an optical switch or as a variable optical attenuator, and it does not exceed 2000  $\mu\text{m}$  in any dimension. Figure P4.12 shows input-output signal pairs used to identify the parameters of the system. Assume a second-order transfer function and find the system's transfer function (Borovic, 2005).



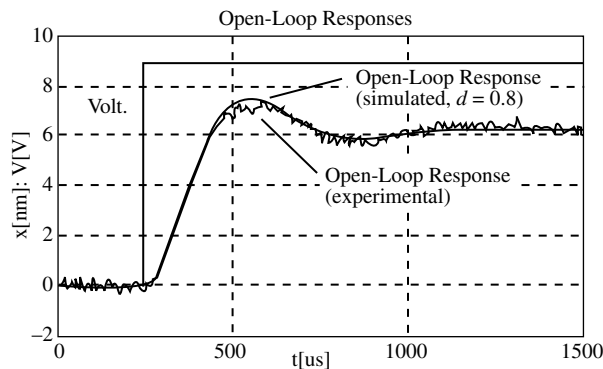


FIGURE P4.12

56. The response of the deflection of a fluid-filled catheter to changes in pressure can be modeled using a second-order model. Knowledge of the parameters of the model is important because in cardiovascular applications the undamped natural frequency should be close to five times the heart rate. However, due to sterility and other considerations, measurement of the parameters is difficult. A method to obtain transfer functions using measurements of the amplitudes of two consecutive peaks of the response and their timing has been developed (*Glantz, 1979*). Assume that Figure P4.13 is obtained from catheter measurements. Using the information shown and assuming a second-order model excited by a unit step input, find the corresponding transfer function.

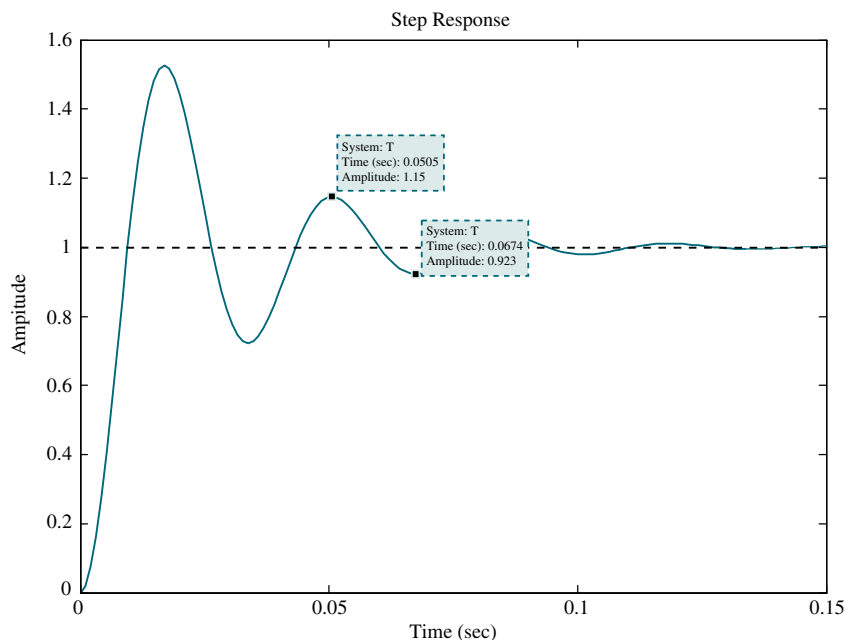


FIGURE P4.13

57. Several factors affect the workings of the kidneys. For example, Figure P4.14 shows how a step change in arterial flow pressure affects renal blood flow in rats. In the “hot tail” part of the experiment, peripheral thermal receptor stimulation is achieved by inserting the rat’s tail in heated water. Variations between different test subjects are indicated by the vertical lines. It has been argued that the “control” and “hot tail” responses are identical except for their steady-state values (*DiBona, 2005*).

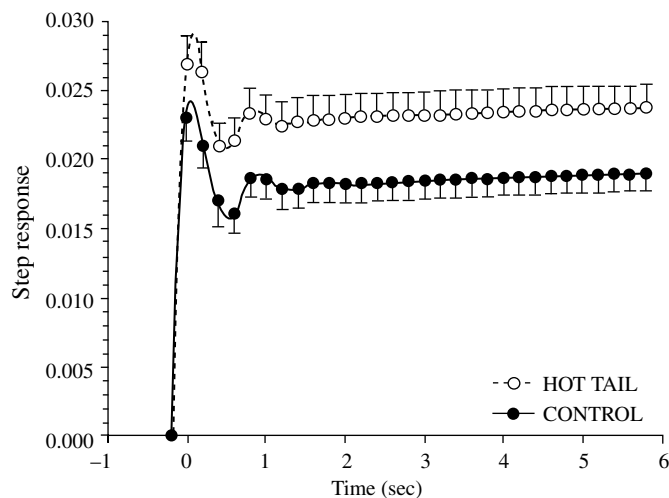


FIGURE P4.14



- a. Using Figure P4.14, obtain the normalized ( $c_{\text{final}} = 1$ ) transfer functions for both responses.
- b. Use MATLAB to prove or disprove the assertion about the “control” and “hot tail” responses.

MATLAB  
ML

58. The transfer function of a nano-positioning device capable of translating biological samples within a few  $\mu\text{m}$  uses a piezoelectric actuator and a linear variable differential transformer (LDVT) as a displacement sensor. The transfer function from input to displacement has been found to be (Salapaka, 2002)

$$G(s) = \frac{9.7 \times 10^4 (s^2 - 14400s + 106.6 \times 10^6)}{(s^2 + 3800s + 23.86 \times 10^6)(s^2 + 240s + 2324.8 \times 10^3)}$$

Use a dominant-pole argument to find an equivalent transfer function with the same numerator but only three poles.

MATLAB  
ML

Use MATLAB to find the actual size and approximate system unit step responses, plotting them on the same graph. Explain the differences between both responses given that both pairs of poles are so far apart.

59. At some point in their lives most people will suffer from at least one onset of low back pain. This disorder can trigger excruciating pain and temporary disability, but its causes are hard to diagnose. It is well known that low back pain alters motor trunk patterns; thus it is of interest to study the causes for these alterations and their extent. Due to the different possible causes of this type of pain, a “control” group of people is hard to obtain for laboratory studies. However, pain can be stimulated in healthy people and muscle movement ranges can be compared. Controlled back pain can be induced by injecting saline solution directly into related muscles or ligaments. The transfer function from infusion rate to pain response was obtained experimentally by injecting a 5% saline solution at six different infusion rates over a period of 12 minutes. Subjects verbally rated their pain every 15 seconds on a scale from 0 to 10, with 0 indicating no pain and 10 unbearable pain. Several trials were averaged and the data was fitted to the following transfer function:

$$G(s) = \frac{9.72 \times 10^{-8}(s + 0.0001)}{(s + 0.009)^2(s^2 + 0.018s + 0.0001)}$$

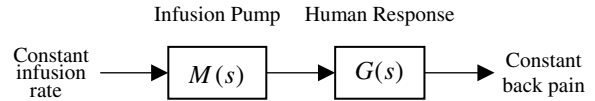


FIGURE P4.15

For experimentation it is desired to build an automatic dispensing system to make the pain level constant as shown in Figure P4.15. It follows that ideally the injection system transfer function has to be

$$M(s) = \frac{1}{G(s)}$$

to obtain an overall transfer function  $M(s)G(s) \approx 1$ . However, for implementation purposes  $M(s)$  must have at least one more pole than zeros (Zedka, 1999). Find a suitable transfer function,  $M(s)$  by inverting  $G(s)$  and adding poles that are far from the imaginary axis.

60. An artificial heart works in closed loop by varying its pumping rate according to changes in signals from the recipient's nervous system. For feedback compensation design it is important to know the heart's open-loop transfer function. To identify this transfer function, an artificial heart is implanted in a calf while the main parts of the original heart are left in place. Then the atrial pumping rate in the original heart is measured while step input changes are effected on the artificial heart. It has been found that, in general, the obtained response closely resembles that of a second-order system. In one such experiment it was found that the step response has a %OS = 30% and a time of first peak  $T_p = 127$  sec (Nakamura, 2002). Find the corresponding transfer function.
61. An observed transfer function from voltage potential to force in skeletal muscles is given by (Ionescu, 2005)

$$T(s) = \frac{450}{(s + 5)(s + 20)}$$

- a. Obtain the system's impulse response.
- b. Integrate the impulse response to find the step response.
- c. Verify the result in Part b by obtaining the step response using Laplace transform techniques.
62. In typical conventional aircraft, longitudinal flight model linearization results in transfer functions with

two pairs of complex conjugate poles. Consequently, the natural response for these airplanes has two modes in their natural response. The “short period” mode is relatively well-damped and has a high-frequency oscillation. The “plugoid mode” is lightly damped and its oscillation frequency is relatively low. For example, in a specific aircraft the transfer function from wing elevator deflection to nose angle (angle of attack) is (McRuer, 1973)

$$\frac{\theta(s)}{\delta_e(s)} = \frac{26.12(s + 0.0098)(s + 1.371)}{(s^2 + 8.99 \times 10^{-3}s + 3.97 \times 10^{-3})(s^2 + 4.21s + 18.23)}$$

- Find which of the poles correspond to the short period mode and which to the phugoid mode.
- Perform a “phugoid approximation” (dominant-pole approximation), retaining the two poles and the zero closest to the  $\omega$ -axis.
- Use MATLAB to compare the step responses of the original transfer function and the approximation.

MATLAB  
ML

- A crosslapper is a machine that takes as an input a light fiber fabric and produces a heavier fabric by laying the original fabric in layers rotated by 90 degrees. A feedback system is required in order to maintain consistent product width and thickness by controlling its carriage velocity. The transfer function from servomotor torque,  $T_m(s)$ , to carriage velocity,  $Y(s)$ , was developed for such a machine (Kuo, 2008). Assume that the transfer function is:

$$G(s) = \frac{Y(s)}{T_m(s)} = \frac{33s^4 + 202s^3 + 10061s^2 + 24332s + 170704}{s^7 + 8s^6 + 464s^5 + 2411s^4 + 52899s^3 + 167829s^2 + 913599s + 1076555}$$

- Use MATLAB to find the partial fraction residues and poles of  $G(s)$ .
- Find an approximation to  $G(s)$  by neglecting the second-order terms found in **a**.
- Use MATLAB to plot on one graph the step response of the transfer function

given above and the approximation found in **b**. Explain the differences between the two plots.

- Although the use of fractional calculus in control systems is not new, in the last decade there is increased interest in its use for several reasons. The most relevant are that fractional calculus differential equations may model certain systems with higher accuracy than integer differential equations, and that fractional calculus compensators might exhibit advantageous properties for control system design. An example of a transfer function obtained through fractional calculus is:

MATLAB  
ML

$$G(s) = \frac{1}{s^{2.5} + 4s^{1.7} + 3s^{0.5} + 5}$$

This function can be approximated with an integer rational transfer function (integer powers of  $s$ ) using Oustaloup's method (Xue, 2005). We ask you now to do a little research and consult the aforementioned reference to find and run an M-file that will calculate the integer rational transfer function approximation to  $G(s)$  and plot its step response.

- Mathematical modeling and control of pH processes are quite challenging since the processes are highly nonlinear, due to the logarithmic relationship between the concentration of hydrogen ions  $[H^+]$  and pH level. The transfer function from input pH to output pH is  $G_a(s) = \frac{Y_a(s)}{X_a(s)} = \frac{14.49e^{-4s}}{1478.26s + 1}$ .  $G_a(s)$  is a model for the anaerobic process in a wastewater treatment system in which methane bacteria need the pH to be maintained in its optimal range from 6.8 to 7.2 (Jiayu, 2009). Similarly, (Elarafi, 2008) used empirical techniques to model a pH neutralization plant as a second-order system with a pure delay, yielding the following transfer function relating output pH to input pH:

$$G_p(s) = \frac{Y_p(s)}{X_p(s)} = \frac{1.716 \times 10^{-5} e^{-30s}}{s^2 + 6.989 \times 10^{-3}s + 1.185 \times 10^{-6}}$$

- Find analytical expressions for the unit-step responses  $y_a(t)$  and  $y_p(t)$  for the two

processes,  $G_a(s)$  and  $G_p(s)$ .

(Hint: Use the time shift theorem in Table 2.2).

- b. Use Simulink to plot  $y_a(t)$  and  $y_p(t)$  on a single graph. Simulink  
SL

66. Using wind tunnel tests, insect flight dynamics can be studied in a very similar fashion to that of man-made aircraft. Linearized longitudinal flight equations for a bumblebee have been found in the unforced case to be

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -8.792 \times 10^{-3} & 0.56 \times 10^{-3} & -1.0 \times 10^{-3} & -13.79 \times 10^{-3} \\ -0.347 \times 10^{-3} & -11.7 \times 10^{-3} & -0.347 \times 10^{-3} & 0 \\ 0.261 & -20.8 \times 10^{-3} & -96.6 \times 10^{-3} & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix}$$

where  $u$  = forward velocity;  $w$  = vertical velocity,  $q$  = angular pitch rate at center of mass, and  $\theta$  = pitch angle between the flight direction and the horizontal (Sun, 2005).

- a. Use MATLAB to obtain the system's eigenvalues. MATLAB  
ML
- b. Write the general form of the state-transition matrix. How many constants would have to be found?

67. A dc-dc converter is a device that takes as an input an unregulated dc voltage and provides a regulated dc voltage as its output. The output voltage may be lower (buck converter), higher (boost converter), or the same as the input voltage. Switching dc-dc converters have a semiconductor active switch (BJT or FET) that is closed periodically with a duty cycle  $d$  in a pulse width modulated (PWM) manner. For a boost converter, averaging techniques can be used to arrive at the following state equations (Van Dijk, 1995): State Space  
SS

$$L \frac{di_L}{dt} = -(1-d)u_c + E_s$$

$$C \frac{du_c}{dt} = (1-d)i_L - \frac{u_c}{R}$$

where  $L$  and  $C$  are respectively the values of internal inductance and capacitance;  $i_L$  is the current through the internal inductor;  $R$  is the resistive load connected to the converter;  $E_s$  is the dc input voltage; and the capacitor voltage,  $u_c$ , is the converter's output.

- a. Write the converter's equations in the form

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$$

$$\mathbf{y} = \mathbf{Cx}$$

assuming  $d$  is a constant.

- b. Using the  $\mathbf{A}$ ,  $\mathbf{B}$ , and  $\mathbf{C}$  matrices of Part a, obtain the converter's transfer function  $\frac{U_c(s)}{E_s(s)}$ .

68. An IPMC (ionic polymer-metal composite) is a Nafion sheet plated with gold on both sides. An IPMC bends when an electric field is applied across its thickness. IPMCs have been used as robotic actuators in several applications and as active catheters in biomedical applications. With the aim of improving actuator settling times, a state-space model has been developed for a 20 mm  $\times$  10 mm  $\times$  0.2 mm polymer sample (Mallavarapu, 2001): State Space  
SS

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -8.34 & -2.26 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = [12.54 \quad 2.26] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

where  $u$  is the applied input voltage and  $y$  is the deflection at one of the material's tips when the sample is tested in a cantilever arrangement.

- a. Find the state-transition matrix for the system.
- b. From Eq. (4.109) in the text, it follows that if a system has zero initial conditions the system output for any input can be directly calculated from the state-space representation and the state-transition matrix using

$$y(t) = \mathbf{Cx}(t) = \int \mathbf{C}\Phi(t-\tau) \mathbf{B}u(\tau) d\tau$$

Use this equation to find the zero initial condition unit step response of the IPMC material sample.

- c. Use MATLAB to verify that your step response calculation in Part b is correct. MATLAB  
ML

## DESIGN PROBLEMS

69. Find an equation that relates 2% settling time to the value of  $f_v$  for the translational mechanical system shown in Figure P4.16. Neglect the mass of all components. [Section: 4.6] WileyPLUS  
WPCS  
Control Solutions

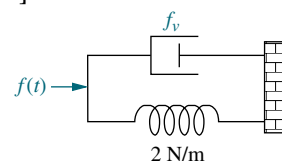


FIGURE P4.16

70. Consider the translational mechanical system shown in Figure P4.17. A 1-pound force,  $f(t)$ , is applied at  $t = 0$ . If  $f_v = 1$ , find  $K$  and  $M$  such that the response is characterized by a 4-second settling time and a 1-second peak time. Also, what is the resulting percent overshoot? [Section: 4.6]

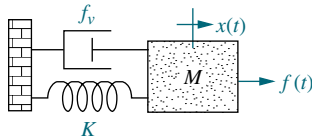


FIGURE P4.17

71. Given the translational mechanical system of Figure P4.17, where  $K = 1$  and  $f(t)$  is a unit step, find the values of  $M$  and  $f_v$  to yield a response with 17% overshoot and a settling time of 10 seconds. [Section: 4.6]
72. Find  $J$  and  $K$  in the rotational system shown in Figure P4.18 to yield a 30% overshoot and a settling time of 3 seconds for a step input in torque. [Section: 4.6]

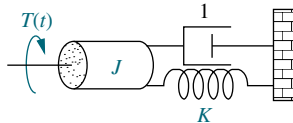


FIGURE P4.18

73. Given the system shown in Figure P4.19, find the damping,  $D$ , to yield a 30% overshoot in output angular displacement for a step input in torque. [Section: 4.6]

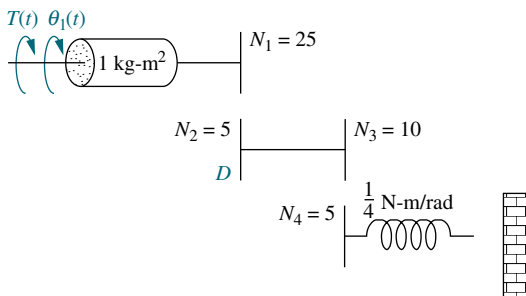


FIGURE P4.19

74. For the system shown in Figure P4.20, find  $N_1/N_2$  so that the settling time for a step torque input is 16 seconds. [Section: 4.6]

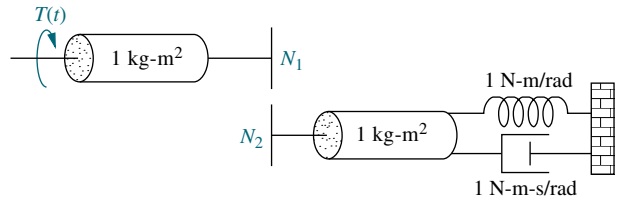


FIGURE P4.20

75. Find  $M$  and  $K$ , shown in the system of Figure P4.21, to yield  $x(t)$  with 10% overshoot and 15 seconds settling time for a step input in motor torque,  $T_m(t)$ . [Section: 4.6]

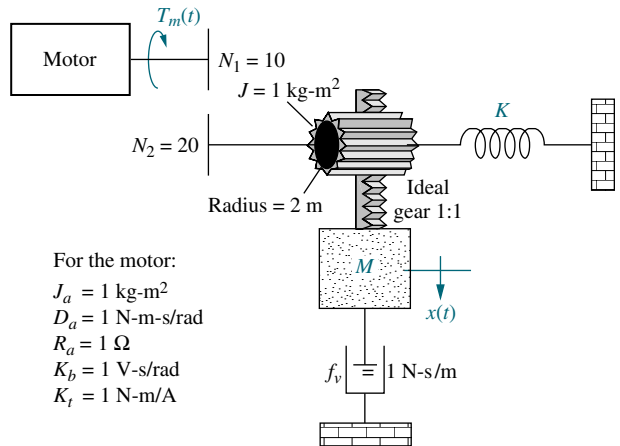


FIGURE P4.21

76. If  $v_i(t)$  is a step voltage in the network shown in Figure P4.22, find the value of the resistor such that a 20% overshoot in voltage will be seen across the capacitor if  $C = 10^{-6}$  F and  $L = 1$  H. [Section: 4.6]

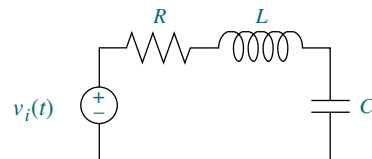


FIGURE P4.22

77. If  $v_i(t)$  is a step voltage in the network shown in Figure P4.22, find the values of  $R$  and  $C$  to yield a 20% overshoot and a 1 ms settling time for  $v_c(t)$  if  $L = 1$  H. [Section: 4.6]
78. Given the circuit of Figure P4.22, where  $C = 10\mu\text{F}$ , find  $R$  and  $L$  to yield 15% overshoot with a settling time of 7 ms for the capacitor voltage. The input,  $v(t)$ , is a unit step. [Section: 4.6]

79. For the circuit shown in Figure P4.23, find the values of  $R_2$  and  $C$  to yield 8% overshoot with a settling time of 1 ms for the voltage across the capacitor, with  $v_i(t)$  as a step input. [Section: 4.6]

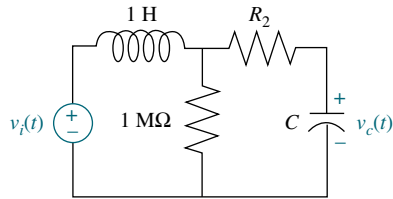


FIGURE P4.23

80. Hydraulic pumps are used as inputs to hydraulic circuits to supply pressure, just as voltage sources supply potential to electric circuits. Applications for hydraulic circuits can be found in the robotics and aircraft industries, where hydraulic actuators are used to move component parts. Figure P4.24 shows the internal parts of the pump. A barrel containing equally spaced pistons rotates about the  $i$ -axis. A swashplate, set at an angle, causes the slippers at the ends of the pistons to move the pistons in and out. When the pistons are moving across the intake port, they are extending, and when they are moving across the discharge port, they are retracting and pushing fluid from the port. The large and small actuators at the top and bottom, respectively, control the angle of the swashplate,  $\alpha$ . The swashplate angle affects the piston stroke length. Thus, by controlling the swashplate angle, the pump discharge flow rate can be regulated. Assume the state equation for the

State Space  
SS

hydraulic pump is

$$\dot{\mathbf{x}} = \begin{bmatrix} (3.45 - 14000K_c) & -0.255 \times 10^{-9} \\ 0.499 \times 10^{11} & -3.68 \end{bmatrix} \mathbf{x} + \begin{bmatrix} -3.45 + 14000K_c \\ -0.499 \times 10^{11} \end{bmatrix} \alpha_0,$$

$$\text{where } \mathbf{x} = \begin{bmatrix} \alpha \\ P_d \end{bmatrix}$$

and  $P_d$  is the pump discharge pressure (*Manring, 1996*). Find the value of controller flow gain,  $K_c$ , so that the damping ratio of the system's poles is 0.9.

### PROGRESSIVE ANALYSIS AND DESIGN PROBLEMS

81. **High-speed rail pantograph.** Problem 67c in Chapter 2 asked you to find  $G(s) = (Y_h(s) - Y_{cat}(s))/F_{up}(s)$  (*O'Connor, 1997*).

- Use the dominant poles from this transfer function and estimate percent overshoot, damping ratio, natural frequency, settling time, peak time, and rise time.
- Determine if the second-order approximation is valid.
- Obtain the step response of  $G(s)$  and compare the results to Part a.

MATLAB  
ML

82. **Control of HIV/AIDS.** In Chapter 3, Problem 31, we developed a linearized

State Space  
SS

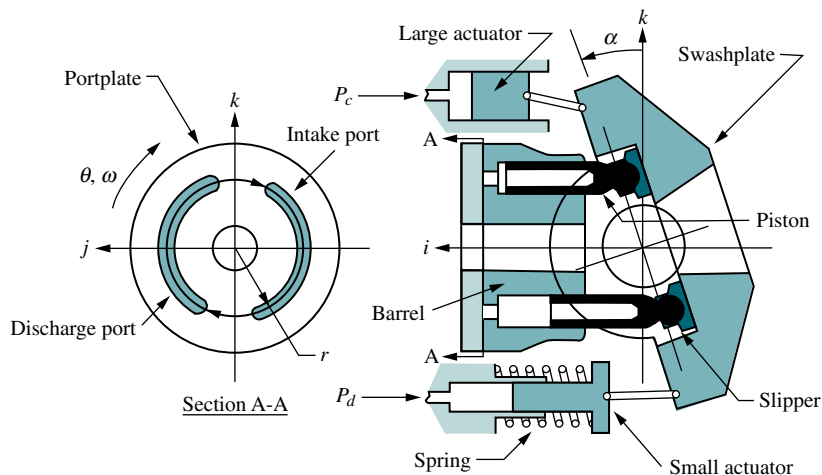


FIGURE P4.24 Pump diagram (Reprinted with permission of ASME.)

state-space model of HIV infection. The model assumed that two different drugs were used to combat the spread of the HIV virus. Since this book focuses on single-input, single-output systems, only one of the two drugs will be considered. We will assume that only RTIs are used as an input. Thus, in the equations of Chapter 3, Problem 31,  $u_2 = 0$  (Craig, 2004).

- a. Show that when using only RTIs in the linearized system of Problem 31 and substituting the typical parameter values given in the table of Problem 31c, the resulting state-space representation for the system is given by

$$\begin{bmatrix} \dot{T} \\ \dot{T}^* \\ \dot{v} \end{bmatrix} = \begin{bmatrix} -0.04167 & 0 & -0.0058 \\ 0.0217 & -0.24 & 0.0058 \\ 0 & 100 & -2.4 \end{bmatrix} \times \begin{bmatrix} T \\ T^* \\ v \end{bmatrix} + \begin{bmatrix} 5.2 \\ -5.2 \\ 0 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix}$$

- b. Obtain the transfer function from RTI efficiency to virus count; namely find  $\frac{Y(s)}{U_1(s)}$ .
- c. Assuming RTIs are 100% effective, what will be the steady-state change of virus count in a given infected patient? Express your answer in virus copies per ml of plasma. Approximately how much time will the medicine take to reach its maximum possible effectiveness?

83. **Hybrid vehicle.** Assume that the car motive dynamics for a hybrid electric vehicle (HEV) can be described by the transfer function

$$\frac{\Delta V(s)}{\Delta F_e(s)} = \frac{1}{1908s + 10}$$

where  $\Delta V$  is the change of velocity in m/sec and  $\Delta F_e$  is the change in excess motive force in  $N$  necessary to propel the vehicle.

- a. Find an analytical expression for  $\Delta v(t)$  for a step change in excess motive force  $\Delta F_e = 2650 N$ .
- b. Simulate the system using MATLAB. Plot the expression found in Part a together with your simulated plot.

MATLAB  
ML

## Cyber Exploration Laboratory

### Experiment 4.1

**Objective** To evaluate the effect of pole and zero location upon the time response of first- and second-order systems.

**Minimum Required Software Packages** MATLAB, Simulink, and the Control System Toolbox

### Prelab

- Given the transfer function  $G(s) = \frac{a}{s+a}$ , evaluate settling time and rise time for the following values of  $a$ : 1, 2, 3, 4. Also, plot the poles.
- Given the transfer function  $G(s) = \frac{b}{s^2 + as + b}$ :
  - Evaluate percent overshoot, settling time, peak time, and rise time for the following values:  $a = 4$ ,  $b = 25$ . Also, plot the poles.
  - Calculate the values of  $a$  and  $b$  so that the imaginary part of the poles remains the same but the real part is increased two times over that of Prelab 2a, and repeat Prelab 2a.

# F O U R

## Time Response

### SOLUTIONS TO CASE STUDIES CHALLENGES

#### Antenna Control: Open-Loop Response

The forward transfer function for angular velocity is,

$$G(s) = \frac{\omega_0(s)}{V_p(s)} = \frac{24}{(s+150)(s+1.32)}$$

a.  $\omega_0(t) = A + Be^{-150t} + Ce^{-1.32t}$

b.  $G(s) = \frac{24}{s^2 + 151.32s + 198}$ . Therefore,  $2\zeta\omega_n = 151.32$ ,  $\omega_n = 14.07$ , and  $\zeta = 5.38$ .

c.  $\omega_0(s) = \frac{24}{s(s^2 + 151.32s + 198)} =$

$$\frac{24}{s(s+150)(s+1.32)} = 0.12121 \frac{1}{s} + 0.0010761 \frac{1}{s+150} - 0.12229 \frac{1}{s+1.32}$$

Therefore,  $\omega_0(t) = 0.12121 + .0010761 e^{-150t} - 0.12229e^{-1.32t}$ .

d. Using  $G(s)$ ,

$$\ddot{\omega}_0 + 151.32 \dot{\omega}_0 + 198\omega_0 = 24v_p(t)$$

Defining,

$$x_1 = \omega_0$$

$$x_2 = \dot{\omega}_0$$

Thus, the state equations are,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -198x_1 - 151.32x_2 + 24v_p(t)$$

$$y = x_1$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -198 & -151.32 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 24 \end{bmatrix} v_p(t); y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}$$



e.

**Program:**

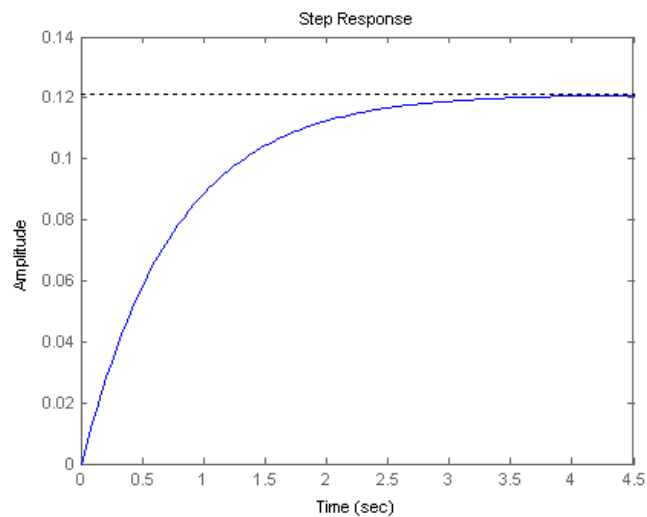
```
'Case Study 1 Challenge (e)'
num=24;
den=poly([-150 -1.32]);
G=tf(num,den)
step(G)
```

**Computer response:**

ans =

Case Study 1 Challenge (e)

Transfer function:

$$\frac{24}{s^2 + 151.3 s + 198}$$


### Ship at Sea: Open-Loop Response

a. Assuming a second-order approximation:  $\omega_n^2 = 2.25$ ,  $2\zeta\omega_n = 0.5$ . Therefore  $\zeta = 0.167$ ,  $\omega_n = 1.5$ .

$$T_s = \frac{4}{\zeta\omega_n} = 16; T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 2.12;$$

$$\%OS = e^{-\zeta\pi / \sqrt{1-\zeta^2}} \times 100 = 58.8\%; \omega_n T_r = 1.169 \text{ therefore, } T_r = 0.77.$$

$$\begin{aligned} \text{b. } \theta(s) &= \frac{2.25}{s(s^2 + 0.5s + 2.25)} = \frac{1}{s} - \frac{s + 0.5}{s^2 + 0.5s + 2.25} \\ &= \frac{1}{s} - \frac{(s + 0.25) + \frac{0.25}{\sqrt{2.1875}}}{(s + 0.25)^2 + 2.1875} \end{aligned}$$

$$= \frac{1}{s} - \frac{(s + 0.25) + 0.16903 \cdot 1.479}{(s + 0.25)^2 + 2.1875}$$

Taking the inverse Laplace transform,

$$\theta(t) = 1 - e^{-0.25t} (\cos 1.479t + 0.16903 \sin 1.479t)$$

c.

**Program:**

```
'Case Study 2 Challenge (C)'
'(a)'
numg=2.25;
deng=[1 0.5 2.25];
G=tf(numg,deng)
omegan=sqrt(deng(3))
zeta=deng(2)/(2*omegan)
Ts=4/(zeta*omegan)
Tp=pi/(omegan*sqrt(1-zeta^2))
pos=exp(-zeta*pi/sqrt(1-zeta^2))*100
t=0:.1:2;
[y,t]=step(G,t);
Tlow=interp1(y,t,.1);
Thi=interp1(y,t,.9);
Tr=Thi-Tlow
'(b)'
numc=2.25*[1 2];
denc=conv(poly([0 -3.57]),[1 2 2.25]);
[K,p,k]=residue(numc,denc)
'(c)'
[y,t]=step(G);
plot(t,y)
title('Roll Angle Response')
xlabel('Time(seconds)')
ylabel('Roll Angle(radians)')
```

**Computer response:**

ans =

Case Study 2 Challenge (C)

ans =

(a)

Transfer function:

```
      2.25
-----
s^2 + 0.5 s + 2.25
```

omegan =

1.5000

zeta =

0.1667

Ts =

16

#### 4-4 Chapter 4: Time Response

Tr =

2.1241

pos =

58.8001

Tr =

0.7801

ans =

(b)

K =

0.1260  
-0.3431 + 0.1058i  
-0.3431 - 0.1058i  
0.5602

p =

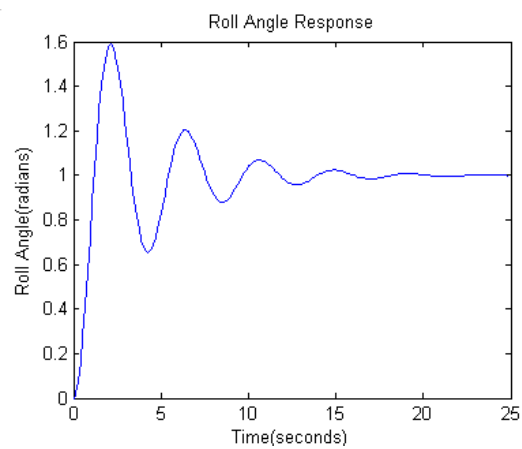
-3.5700  
-1.0000 + 1.1180i  
-1.0000 - 1.1180i  
0

k =

[]

ans =

(c)



## ANSWERS TO REVIEW QUESTIONS

1. Time constant
2. The time for the step response to reach 63% of its final value
3. The input pole
4. The system poles
5. The radian frequency of a sinusoidal response
6. The time constant of an exponential response
7. Natural frequency is the frequency of the system with all damping removed; the damped frequency of oscillation is the frequency of oscillation with damping in the system.
8. Their damped frequency of oscillation will be the same.
9. They will all exist under the same exponential decay envelop.
10. They will all have the same percent overshoot and the same shape although differently scaled in time.
11.  $\zeta$ ,  $\omega_n$ ,  $T_p$ , %OS,  $T_s$
12. Only two since a second-order system is completely defined by two component parameters
13. (1) Complex, (2) Real, (3) Multiple real
14. Pole's real part is large compared to the dominant poles, (2) Pole is near a zero
15. If the residue at that pole is much smaller than the residues at other poles
16. No; one must then use the output equation
17. The Laplace transform of the state transition matrix is  $(s\mathbf{I} - \mathbf{A})^{-1}$
18. Computer simulation
19. Pole-zero concepts give one an intuitive feel for the problem.
20. State equations, output equations, and initial value for the state-vector
21.  $\text{Det}(s\mathbf{I} - \mathbf{A}) = 0$

## SOLUTIONS TO PROBLEMS

1.

a. Overdamped Case:

$$C(s) = \frac{9}{s(s^2 + 9s + 9)}$$

Expanding into partial fractions,

$$C(s) = \frac{9}{s(s + 7.854)(s + 1.146)} = \frac{1}{s} + \frac{0.171}{(s + 7.854)} - \frac{1.171}{(s + 1.146)}$$

Taking the inverse Laplace transform,

$$c(t) = 1 + 0.171 e^{-7.854t} - 1.171 e^{-1.146t}$$

**b. Underdamped Case:**

$$C(s) = \frac{9}{s(s^2 + 3s + 9)} = \frac{K_1}{s} + \frac{K_2s + K_3}{(s^2 + 3s + 9)}$$

$$K_1 = \left. \frac{9}{(s^2 + 3s + 9)} \right|_{s \rightarrow 0} = 1$$

$K_2$  and  $K_3$  can be found by clearing fractions with  $K_1$  replaced by its value. Thus,

$$9 = (s^2 + 3s + 9) + (K_2s + K_3)s$$

or

$$9 = s^2 + 3s + 9 + K_2s^2 + K_3s$$

Hence  $K_2 = -1$  and  $K_3 = -3$ . Thus,

$$C(s) = \frac{9}{s} - \frac{(s + \frac{3}{2})}{(s + \frac{3}{2})^2 + \frac{27}{4}}$$

$$C(s) = \frac{1}{s} - \frac{(s + \frac{3}{2})}{(s + \frac{3}{2})^2 + \frac{27}{4}} - \frac{\frac{3}{2}}{(s + \frac{3}{2})^2 + \frac{27}{4}}$$

$$C(s) = \frac{1}{s} - \frac{(s + \frac{3}{2})}{(s + \frac{3}{2})^2 + \frac{27}{4}} - \frac{\frac{3}{\sqrt{27}} \sqrt{\frac{27}{4}}}{(s + \frac{3}{2})^2 + \frac{27}{4}}$$

$$c(t) = 1 - e^{-\frac{3}{2}t} \cos \sqrt{\frac{27}{4}} t - \frac{3}{\sqrt{27}} e^{-\frac{3}{2}t} \sin \sqrt{\frac{27}{4}} t$$

$$c(t) = 1 - \frac{2}{\sqrt{3}} e^{-3t/2} \cos(\sqrt{\frac{27}{4}} t - \phi)$$

$$= 1 - 1.155 e^{-1.5t} \cos(2.598t - \phi)$$

where

$$\phi = \arctan\left(\frac{3}{\sqrt{27}}\right) = 30^\circ$$

c. Oscillatory Case:

$$C(s) = \frac{9}{s(s^2 + 9)}$$

$$C(s) = \frac{9}{s(s^2 + 9)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{(s^2 + 9)}$$

The evaluation of the constants in the numerator are found the same way as they were for the underdamped case. The results are  $K_2 = -1$  and  $K_3 = 0$ . Hence,

$$C(s) = \frac{1}{s} - \frac{s}{(s^2 + 9)}$$

Therefore,

$$c(t) = 1 - \cos 3t$$

d. Critically Damped

$$C(s) = \frac{9}{s(s^2 + 6s + 9)}$$

$$C(s) = \frac{9}{s(s^2 + 6s + 9)} = \frac{K_1}{s} + \frac{K_2}{(s + 3)^2} + \frac{K_3}{(s + 3)}$$

The constants are then evaluated as

$$K_1 = \left. \frac{9}{(s^2 + 6s + 9)} \right|_{s \rightarrow 0} = 1; \quad K_2 = \left. \frac{9}{s} \right|_{s \rightarrow -3} = -3; \quad K_3 = \left. \frac{d}{ds} \left( \frac{9}{s} \right) \right|_{s \rightarrow -3} = -1$$

Now, the transform of the response is

$$C(s) = \frac{9}{s(s^2 + 6s + 9)} = \frac{1}{s} - \frac{3}{(s + 3)^2} - \frac{1}{(s + 3)}$$

$$c(t) = 1 - 3t e^{-3t} - e^{-3t}$$

2.

a.  $C(s) = \frac{5}{s(s+5)} = \frac{1}{s} - \frac{1}{s+5}$ . Therefore,  $c(t) = 1 - e^{-5t}$ .

Also,  $T = \frac{1}{5}$ ,  $T_r = \frac{2.2}{a} = \frac{2.2}{5} = 0.44$ ,  $T_s = \frac{4}{a} = \frac{4}{5} = 0.8$ .

b.  $C(s) = \frac{20}{s(s+20)} = \frac{1}{s} - \frac{1}{s+20}$ . Therefore,  $c(t) = 1 - e^{-20t}$ . Also,  $T = \frac{1}{20}$ ,

$T_r = \frac{2.2}{a} = \frac{2.2}{20} = 0.11$ ,  $T_s = \frac{4}{a} = \frac{4}{20} = 0.2$ .

3.

**Program:**

```

'(a)'
num=5;
den=[1 5];
Ga=tf(num,den)
subplot(1,2,1)
step(Ga)
title('(a)')
'(b)'
num=20;
den=[1 20];
Gb=tf(num,den)
subplot(1,2,2)
step(Gb)
title('(b)')

```

**Computer response:**

ans =

(a)

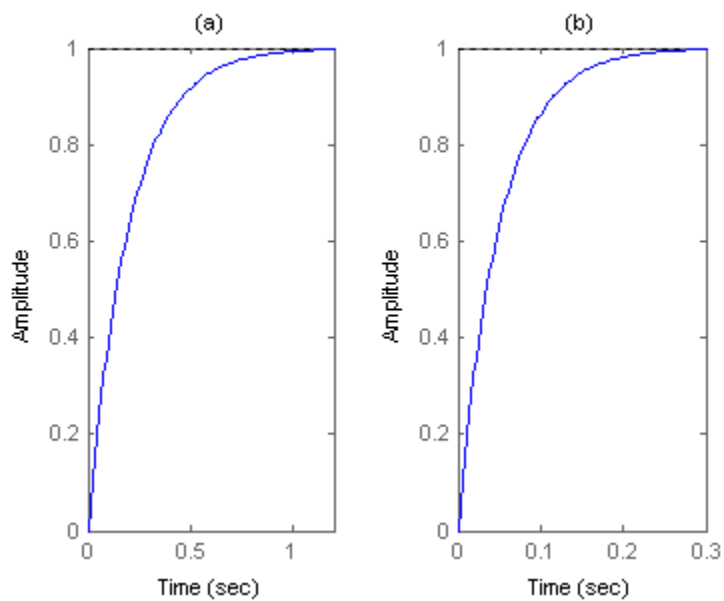
Transfer function:

$$\frac{5}{s + 5}$$

ans =

(b)

Transfer function:

$$\frac{20}{s + 20}$$




4.

Using voltage division,  $\frac{V_C(s)}{V_i(s)} = \frac{1/RC}{s + \frac{1}{RC}} = \frac{0.703}{s + 0.703}$ . Since  $V_i(s) = \frac{5}{s}$

$$V_c(s) = \frac{5}{s} \left( \frac{0.703}{s + 0.703} \right) = \frac{5}{s} - \frac{5}{s + 0.703}.$$

Therefore  $v_c(t) = 5 - 5e^{-0.703t}$ . Also,

$$T = \frac{1}{0.703} = 1.422; \quad T_r = \frac{2.2}{0.703} = 3.129; \quad T_s = \frac{4}{0.703} = 5.69.$$

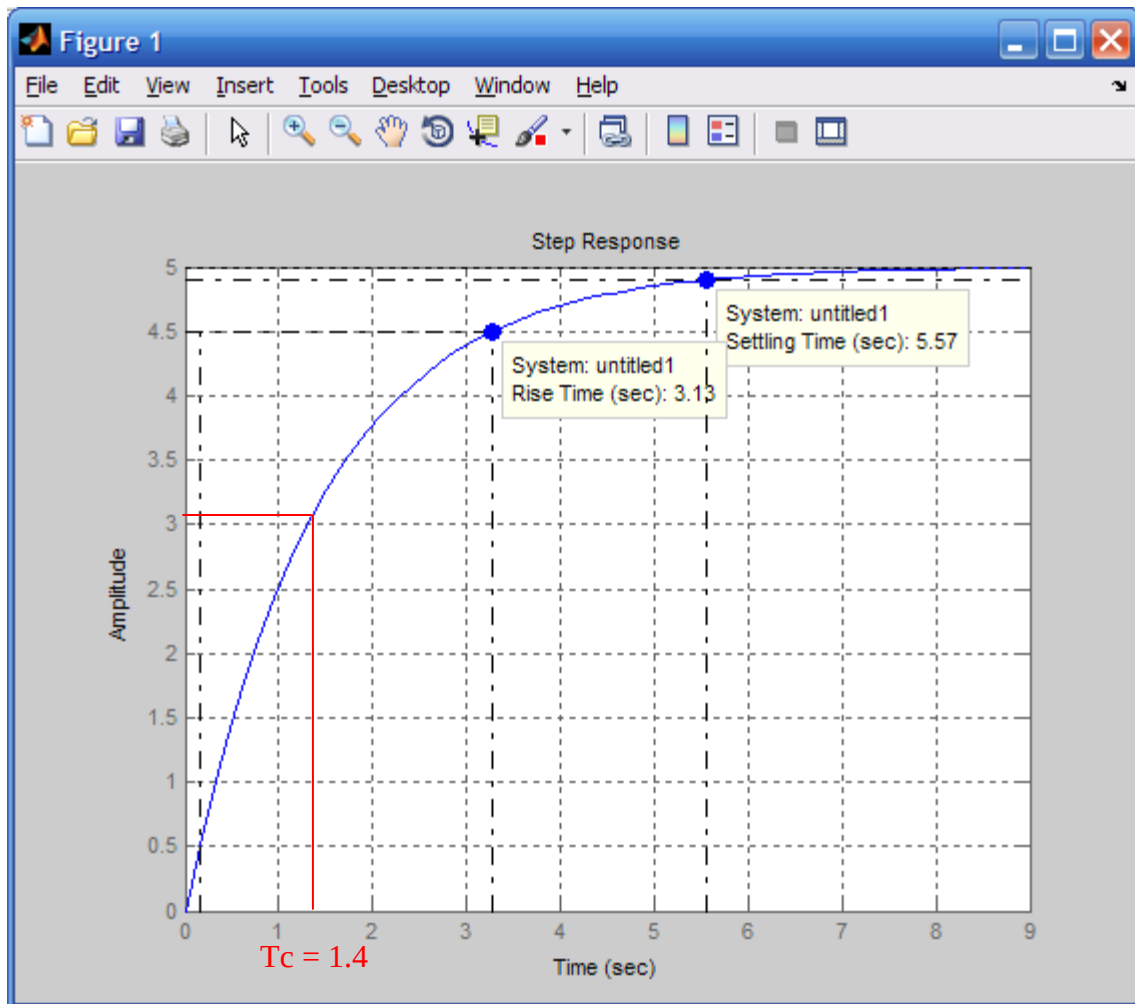
5.

**Program:**

```
clf
num=0.703;
den=[1 0.703];
G=tf(num,den)
step(5*G)
```

**Computer response:**

```
Transfer function:
      0.703
-----
s + 0.703
```



6.

Writing the equation of motion,

$$(Ms^2 + 6s)X(s) = F(s)$$

Thus, the transfer function is,

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + 6s}$$

Differentiating to yield the transfer function in terms of velocity,

$$\frac{sX(s)}{F(s)} = \frac{1}{Ms + 6} = \frac{1/M}{s + \frac{6}{M}}$$

Thus, the settling time,  $T_s$ , and the rise time,  $T_r$ , are given by

$$T_s = \frac{4}{6/M} = \frac{2}{3}M = 0.667M; \quad T_r = \frac{2.2}{6/M} = \frac{1.1}{3}M = 0.367M$$

7.

**Program:**

```

Clf
M=1
num=1/M;
den=[1 6/M];
G=tf(num,den)
step(G)
pause
M=2
num=1/M;
den=[1 6/M];
G=tf(num,den)
step(G)

```

**Computer response:**

M =

1

Transfer function:

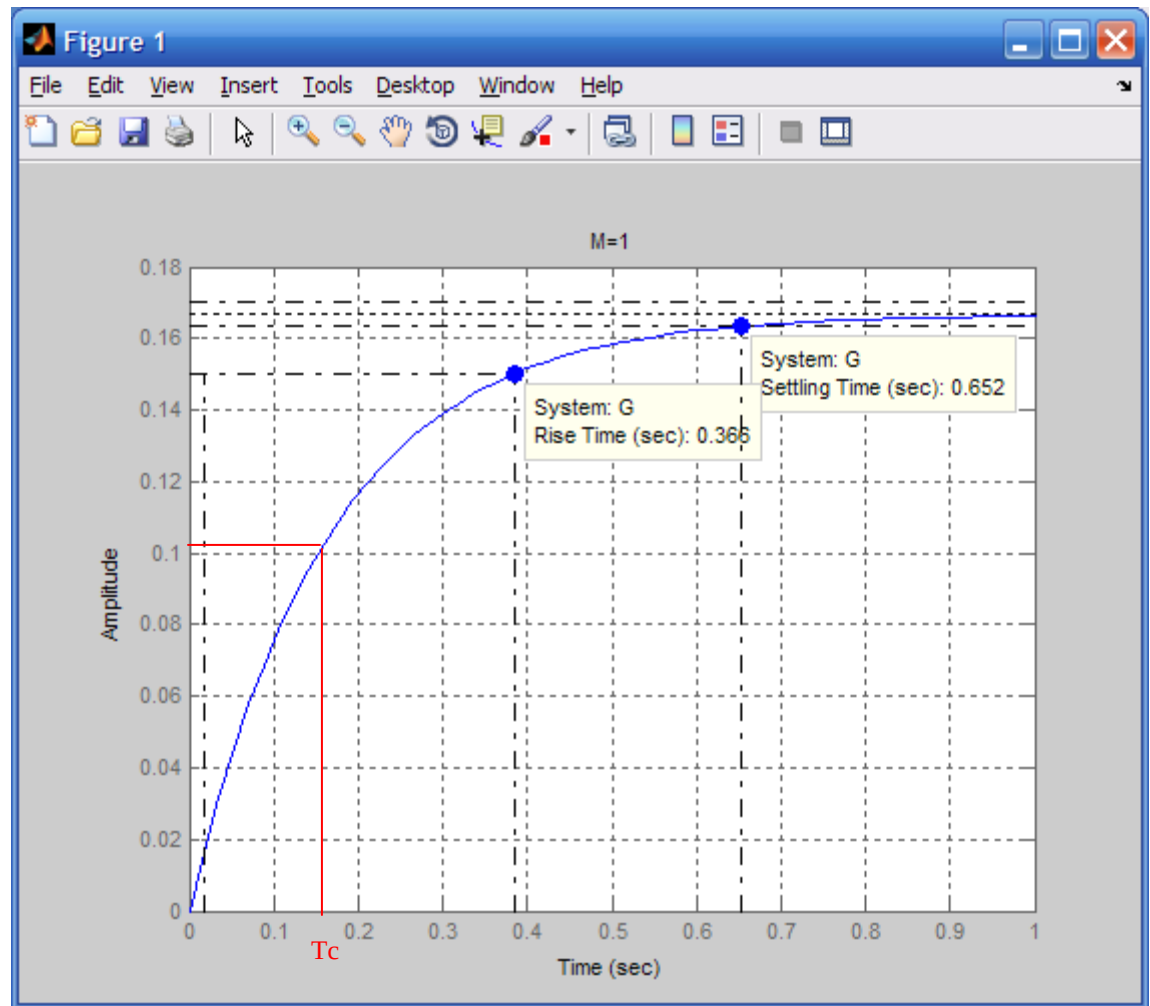
$$\frac{1}{s + 6}$$

M =

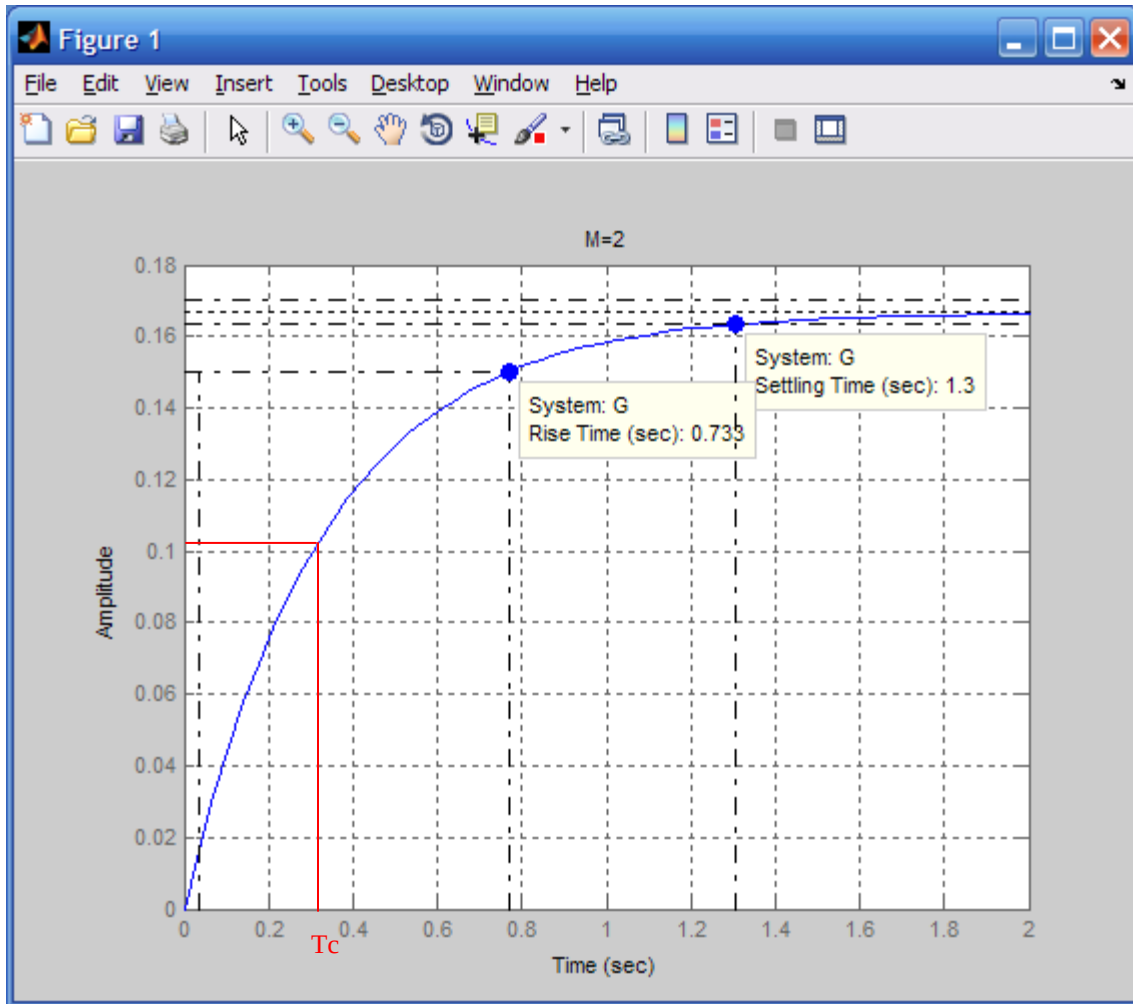
2

Transfer function:

$$\frac{0.5}{s + 3}$$



From plot, time constant = 0.16 s.



From plot, time constant = 0.33 s.

8.

- a. Pole: -2;  $c(t) = A + Be^{-2t}$ ; first-order response.
- b. Poles: -3, -6;  $c(t) = A + Be^{-3t} + Ce^{-6t}$ ; overdamped response.
- c. Poles: -10, -20; Zero: -7;  $c(t) = A + Be^{-10t} + Ce^{-20t}$ ; overdamped response.
- d. Poles:  $(-3+j3\sqrt{15})$ ,  $(-3-j3\sqrt{15})$ ;  $c(t) = A + Be^{-3t} \cos(3\sqrt{15}t + \phi)$ ; underdamped.
- e. Poles:  $j3$ ,  $-j3$ ; Zero: -2;  $c(t) = A + B \cos(3t + \phi)$ ; undamped.
- f. Poles: -10, -10; Zero: -5;  $c(t) = A + Be^{-10t} + Cte^{-10t}$ ; critically damped.

9.

**Program:**  
`p=roots([1 6 4 7 2])`

**Computer response:**  
`p =`

-5.4917  
 -0.0955 + 1.0671i  
 -0.0955 - 1.0671i  
 -0.3173

10.

$$G(s) = \mathbf{C} (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}$$

$$\mathbf{A} = \begin{bmatrix} 8 & -4 & 1 \\ -3 & 2 & 0 \\ 5 & 7 & -9 \end{bmatrix}; \mathbf{B} = \begin{bmatrix} -4 \\ -3 \\ 4 \end{bmatrix}; \mathbf{C} = [2 \quad 8 \quad -3]$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^3 - s^2 - 91s + 67} \begin{bmatrix} (s-2)(s+9) & -(4s+29) & (s-2) \\ -(3s+27) & (s^2+s-77) & -3 \\ 5s-31 & 7s-76 & (s^2-10s+4) \end{bmatrix}$$

$$\text{Therefore, } G(s) = \frac{-44s^2 + 291s + 1814}{s^3 - s^2 - 91s + 67}.$$

Factoring the denominator, or using  $\det(s\mathbf{I} - \mathbf{A})$ , we find the poles to be 9.683, 0.7347, -9.4179.

11.

**Program:**

```
A=[8 -4 1;-3 2 0;5 7 -9]
B=[-4;-3;4]
C=[2 8 -3]
D=0
[numg,deng]=ss2tf(A,B,C,D,1);
G=tf(numg,deng)
poles=roots(deng)
```

**Computer response:**

A =

```
8      -4      1
-3      2      0
5      7     -9
```

B =

```
-4
-3
4
```

C =

$$2 \quad 8 \quad -3$$

D =

$$0$$

Transfer function:

$$-44 s^2 + 291 s + 1814$$

-----

$$s^3 - s^2 - 91 s + 67$$

poles =

$$-9.4179$$

$$9.6832$$

$$0.7347$$

12.

Writing the node equation at the capacitor,  $V_C(s) \left( \frac{1}{R_2} + \frac{1}{Ls} + Cs \right) + \frac{V_C(s) - V(s)}{R_1} = 0$ .

Hence,  $\frac{V_C(s)}{V(s)} = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{Ls} + Cs} = \frac{10s}{s^2 + 20s + 500}$ . The step response is  $\frac{10}{s^2 + 20s + 500}$ . The poles

are at

$-10 \pm j20$ . Therefore,  $v_C(t) = Ae^{-10t} \cos(20t + \phi)$ .

13.

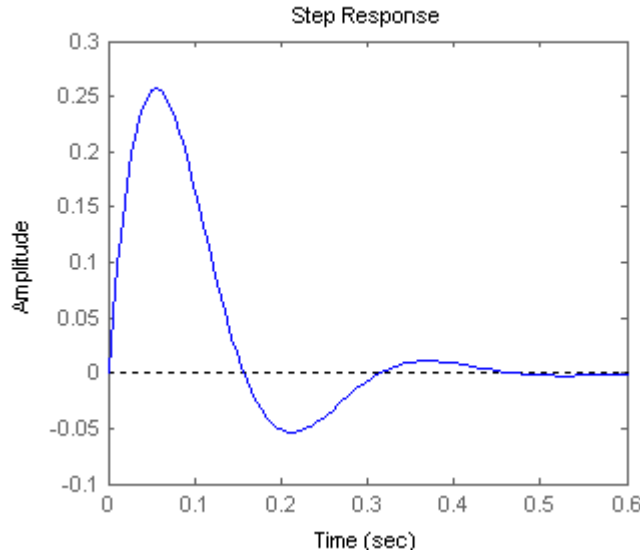
**Program:**

```
num=[10 0];
den=[1 20 500];
G=tf(num,den)
step(G)
```

**Computer response:**

Transfer function:

$$\frac{10 s}{s^2 + 20 s + 500}$$



14.

The equation of motion is:  $(Ms^2 + f_v s + K_s)X(s) = F(s)$ . Hence,  $\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + f_v s + K_s} = \frac{1}{s^2 + s + 5}$ .

The step response is now evaluated:  $X(s) = \frac{1}{s(s^2 + s + 5)} = \frac{1/5}{s} - \frac{\frac{1}{5}s + \frac{1}{5}}{(s + \frac{1}{2})^2 + \frac{19}{4}} =$

$$\frac{\frac{1}{5}(s + \frac{1}{2}) + \frac{1}{5\sqrt{19}} \frac{\sqrt{19}}{2}}{(s + \frac{1}{2})^2 + \frac{19}{4}}.$$

Taking the inverse Laplace transform,  $x(t) = \frac{1}{5} - \frac{1}{5} e^{-0.5t} \left( \cos \frac{\sqrt{19}}{2} t + \frac{1}{\sqrt{19}} \sin \frac{\sqrt{19}}{2} t \right)$

$$= \frac{1}{5} \left[ 1 - 2\sqrt{\frac{5}{19}} e^{-0.5t} \cos \left( \frac{\sqrt{19}}{2} t - 12.92^\circ \right) \right].$$

15.

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2 - \zeta^2\omega_n^2}$$

$$= \frac{1}{s} - \frac{(s + \zeta\omega_n) + \zeta\omega_n}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2} = \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta\omega_n}{\omega_n\sqrt{1 - \zeta^2}} \omega_n\sqrt{1 - \zeta^2}}{(s + \zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2}$$

$$\text{Hence, } c(t) = 1 - e^{-\zeta\omega_n t} \left( \cos \omega_n\sqrt{1 - \zeta^2} t + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_n\sqrt{1 - \zeta^2} t \right)$$



$$= 1 - e^{-\zeta \omega_n t} \sqrt{1 + \frac{\zeta^2}{1 - \zeta^2}} \cos(\omega_n \sqrt{1 - \zeta^2} t - \phi) = 1 - e^{-\zeta \omega_n t} \frac{1}{\sqrt{1 - \zeta^2}} \cos(\omega_n \sqrt{1 - \zeta^2} t - \phi),$$

$$\text{where } \phi = \tan^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}}$$

16.

$$\%OS = e^{-\zeta \pi} / \sqrt{1 - \zeta^2} \times 100. \text{ Dividing by 100 and taking the natural log of both sides,}$$

$$\ln\left(\frac{\%OS}{100}\right) = -\frac{\zeta \pi}{\sqrt{1 - \zeta^2}}. \text{ Squaring both sides and solving for } \zeta^2, \zeta^2 = \frac{\ln^2\left(\frac{\%OS}{100}\right)}{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}. \text{ Taking the}$$

$$\text{negative square root, } \zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}}.$$

17.

a.

$$C(s) = \frac{2}{s(s+2)}$$

$$C(s) = \frac{1}{s} - \frac{1}{s+2}$$

$$c(t) = 1 - e^{-2t}$$

b.

$$C(s) = \frac{5}{s(s+3)(s+6)}$$

$$C(s) = \frac{5}{18} \frac{1}{s} - \frac{5}{9} \frac{1}{s+3} + \frac{5}{18} \frac{1}{s+6}$$

$$c(t) = \frac{5}{18} - \frac{5}{9} e^{-3t} + \frac{5}{18} e^{-6t}$$

c.

$$C(s) = \frac{10(s+7)}{s(s+10)(s+20)}$$

$$C(s) = \frac{7}{20} \frac{1}{s} + \frac{3}{10} \frac{1}{s+10} - \frac{13}{20} \frac{1}{s+20}$$

$$c(t) = \frac{7}{20} + \frac{3}{10} e^{-10t} - \frac{13}{20} e^{-20t}$$

d.

$$C(s) = \frac{20}{s(s^2 + 6s + 144)}$$

$$C(s) = \frac{5}{36} \frac{1}{s} - \frac{5}{36} \frac{s+6}{s^2 + 6s + 144}$$

$$C(s) = \frac{5}{36} \frac{1}{s} - \frac{5}{36} \frac{(s+3) + \frac{3}{\sqrt{135}} \sqrt{135}}{(s+3)^2 + 135}$$

$$c(t) = \frac{5}{36} - \frac{5}{36} e^{-3t} \left( \cos[\sqrt{135}]t + \frac{3}{\sqrt{135}} \sin[\sqrt{135}]t \right)$$

e.

$$C(s) = \frac{s+2}{s(s^2+9)}$$

$$C(s) = \frac{2}{9} \frac{1}{s} + \frac{1}{9} \frac{-2s+9}{s^2+9}$$

$$C(s) = \frac{2}{9} \frac{1}{s} + \frac{1}{9} \frac{-2s+3 \cdot 3}{s^2+9}$$

$$c(t) = \frac{2}{9} - \left( \frac{2}{9} \cos 3t - \frac{1}{3} \sin 3t \right)$$

f.

$$C(s) = \frac{s+5}{s(s+10)^2}$$

$$C(s) = \frac{1}{20} \frac{1}{s} - \frac{1}{20} \frac{1}{s+10} + \frac{1}{2} \frac{1}{(s+10)^2}$$

$$c(t) = \frac{1}{20} - \frac{1}{20} e^{-10t} + \frac{1}{2} t e^{-10t}$$

18.

a. N/A

b.  $s^2+9s+18$ ,  $\omega_n^2 = 18$ ,  $2\zeta\omega_n = 9$ , Therefore  $\zeta = 1.06$ ,  $\omega_n = 4.24$ , overdamped.

c.  $s^2+30s+200$ ,  $\omega_n^2 = 200$ ,  $2\zeta\omega_n = 30$ , Therefore  $\zeta = 1.06$ ,  $\omega_n = 14.14$ , overdamped.

d.  $s^2+6s+144$ ,  $\omega_n^2 = 144$ ,  $2\zeta\omega_n = 6$ , Therefore  $\zeta = 0.25$ ,  $\omega_n = 12$ , underdamped.

e.  $s^2+9$ ,  $\omega_n^2 = 9$ ,  $2\zeta\omega_n = 0$ , Therefore  $\zeta = 0$ ,  $\omega_n = 3$ , undamped.

f.  $s^2+20s+100$ ,  $\omega_n^2 = 100$ ,  $2\zeta\omega_n = 20$ , Therefore  $\zeta = 1$ ,  $\omega_n = 10$ , critically damped.

19.

$$X(s) = \frac{100^2}{s(s^2+100s+100^2)} = \frac{1}{s} - \frac{s+100}{(s+50)^2+7500} = \frac{1}{s} - \frac{(s+50)+50}{(s+50)^2+7500} = \frac{1}{s} - \frac{(s+50) + \frac{50}{\sqrt{7500}} \sqrt{7500}}{(s+50)^2+7500}$$

$$\text{Therefore, } x(t) = 1 - e^{-50t} \left( \cos \sqrt{7500} t + \frac{50}{\sqrt{7500}} \sin \sqrt{7500} t \right)$$

$$= 1 - \frac{2}{\sqrt{3}} e^{-50t} \cos \left( 50\sqrt{3} t - \tan^{-1} \frac{1}{\sqrt{3}} \right)$$

20.

a.  $\omega_n^2 = 16$  r/s,  $2\zeta\omega_n = 3$ . Therefore  $\zeta = 0.375$ ,  $\omega_n = 4$ .  $T_s = \frac{4}{\zeta\omega_n} = 2.667$  s;  $T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} =$

$$0.8472 \text{ s; } \%OS = e^{-\zeta\pi} / \sqrt{1-\zeta^2} \times 100 = 28.06 \%; \omega_n T_r = (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1) = 1.4238;$$

therefore,  $T_r = 0.356$  s.

b.  $\omega_n^2 = 0.04$  r/s,  $2\zeta\omega_n = 0.02$ . Therefore  $\zeta = 0.05$ ,  $\omega_n = 0.2$ .  $T_s = \frac{4}{\zeta\omega_n} = 400$  s;  $T_P = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} =$

15.73 s; %OS =  $e^{-\zeta\pi} / \sqrt{1-\zeta^2} \times 100 = 85.45$  %;  $\omega_n T_r = (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1)$ ; therefore,

$T_r = 5.26$  s.

c.  $\omega_n^2 = 1.05 \times 10^7$  r/s,  $2\zeta\omega_n = 1.6 \times 10^3$ . Therefore  $\zeta = 0.247$ ,  $\omega_n = 3240$ .  $T_s = \frac{4}{\zeta\omega_n} = 0.005$  s;  $T_P =$

$\frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 0.001$  s; %OS =  $e^{-\zeta\pi} / \sqrt{1-\zeta^2} \times 100 = 44.92$  %;  $\omega_n T_r = (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta +$

1); therefore,  $T_r = 3.88 \times 10^{-4}$  s.

21.

**Program:**

```
'(a)'
clf
numa=16;
dena=[1 3 16];
Ta=tf(numa,dena)
omegana=sqrt(dena(3))
zetaa=dena(2)/(2*omegana)
Tsa=4/(zetaa*omegana)
Tpa=pi/(omegana*sqrt(1-zetaa^2))
Tra=(1.76*zetaa^3 - 0.417*zetaa^2 + 1.039*zetaa + 1)/omegana
percenta=exp(-zetaa*pi/sqrt(1-zetaa^2))*100
subplot(221)
step(Ta)
title('(a)')
'(b)'
numb=0.04;
denb=[1 0.02 0.04];
Tb=tf(numb,denb)
omeganb=sqrt(denb(3))
zetab=denb(2)/(2*omeganb)
Tsb=4/(zetab*omeganb)
Tpb=pi/(omeganb*sqrt(1-zetab^2))
Trb=(1.76*zetab^3 - 0.417*zetab^2 + 1.039*zetab + 1)/omeganb
percentb=exp(-zetab*pi/sqrt(1-zetab^2))*100
subplot(222)
step(Tb)
title('(b)')
'(c)'
numc=1.05E7;
denc=[1 1.6E3 1.05E7];
Tc=tf(numc,denc)
omeganc=sqrt(denc(3))
zetac=denc(2)/(2*omeganc)
Tsc=4/(zetac*omeganc)
Tpc=pi/(omeganc*sqrt(1-zetac^2))
Trc=(1.76*zetac^3 - 0.417*zetac^2 + 1.039*zetac + 1)/omeganc
percentc=exp(-zetac*pi/sqrt(1-zetac^2))*100
subplot(223)
step(Tc)
title('(c)')
```

**Computer response:**

ans =

(a)

Transfer function:

$$\frac{16}{s^2 + 3s + 16}$$

omegana =

4

zetaa =

0.3750

Tsa =

2.6667

Tpa =

0.8472

Tra =

0.3559

percenta =

28.0597

ans =

(b)

Transfer function:

$$\frac{0.04}{s^2 + 0.02s + 0.04}$$

omeganb =

0.2000

zetaab =

0.0500

Tsb =

400

Tpb =

15.7276

Trb =

5.2556

percentb =

85.4468

ans =

(c)

Transfer function:

1.05e007

-----  
s^2 + 1600 s + 1.05e007

omeganc =

3.2404e+003

zetac =

0.2469

Tsc =

0.0050

Tpc =

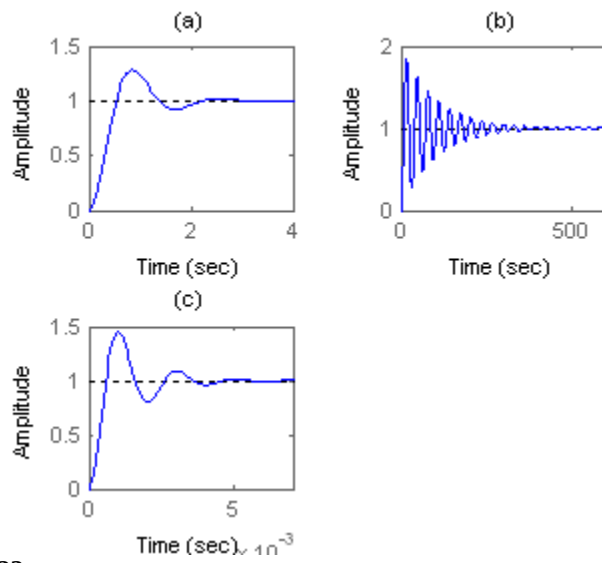
0.0010

Trc =

3.8810e-004

percentc =

44.9154



22.

**Program:**

```
T1=tf(16,[1 3 16])
T2=tf(0.04,[1 0.02 0.04])
T3=tf(1.05e7,[1 1.6e3 1.05e7])
ltiview
```

**Computer response:**

Transfer function:

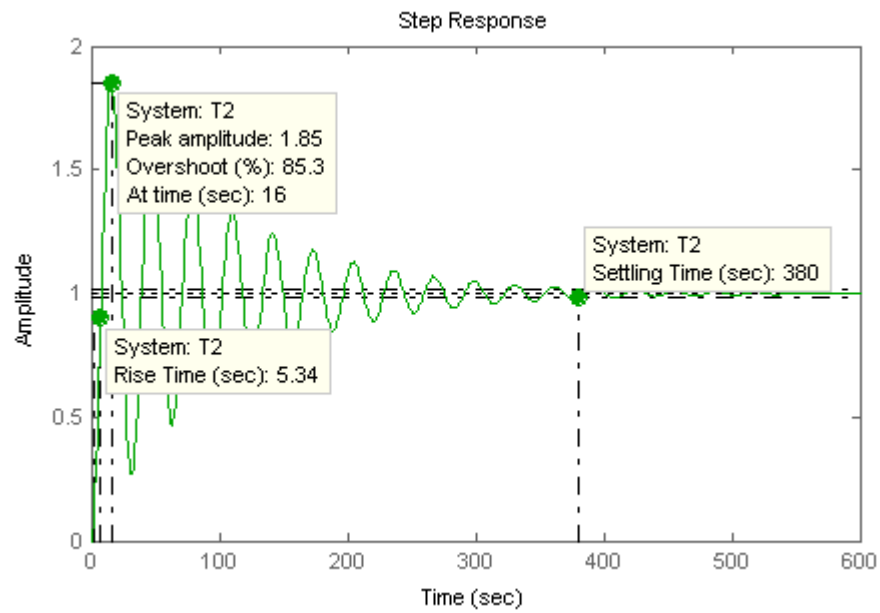
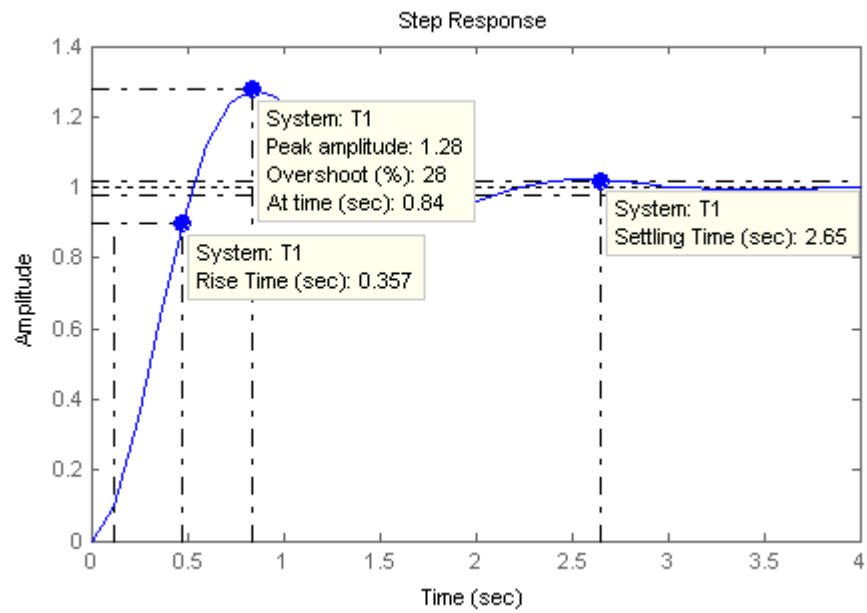
```
16
-----
s^2 + 3 s + 16
```

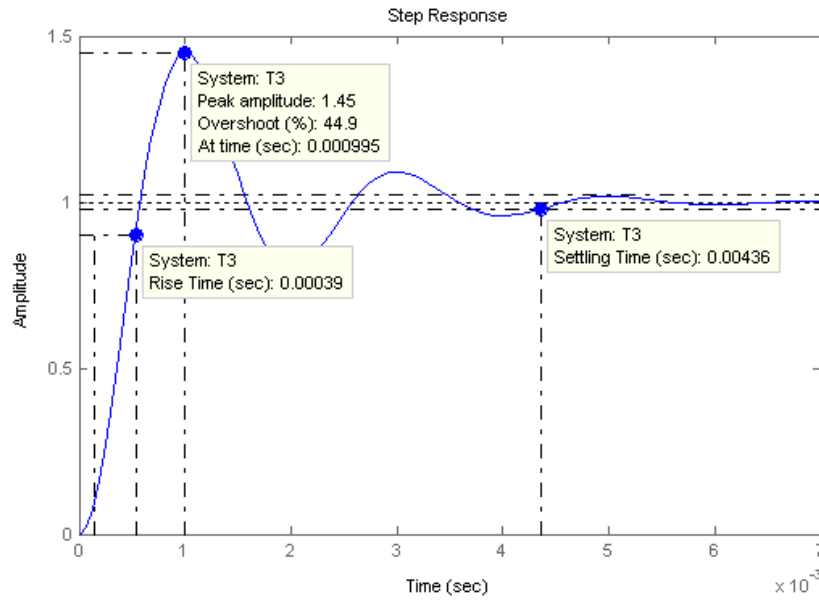
Transfer function:

```
0.04
-----
s^2 + 0.02 s + 0.04
```

Transfer function:

```
1.05e007
-----
s^2 + 1600 s + 1.05e007
```





23.

$$\text{a. } \zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}} = 0.56, \omega_n = \frac{4}{\zeta T_s} = 11.92. \text{ Therefore, poles} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$= -6.67 \pm j9.88.$$

$$\text{b. } \zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}} = 0.591, \omega_n = \frac{\pi}{T_p\sqrt{1-\zeta^2}} = 0.779.$$

$$\text{Therefore, poles} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -0.4605 \pm j0.6283.$$

$$\text{c. } \zeta\omega_n = \frac{4}{T_s} = 0.571, \omega_n\sqrt{1-\zeta^2} = \frac{\pi}{T_p} = 1.047. \text{ Therefore, poles} = -0.571 \pm j1.047.$$

24.

$$\text{Re} = \frac{4}{T_s} = 4; \quad \zeta = \frac{-\ln(12.3/100)}{\sqrt{\pi^2 + \ln^2(12.3/100)}} = 0.5549$$

$$\text{Re} = \zeta\omega_n = 0.5549\omega_n = 4; \quad \therefore \omega_n = 7.21$$

$$\text{Im} = \omega_n\sqrt{1-\zeta^2} = 6$$

$$\therefore G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{51.96}{s^2 + 8s + 51.96}$$



25.

a. Writing the equation of motion yields,  $(5s^2 + 5s + 28)X(s) = F(s)$

Solving for the transfer function,

$$\frac{X(s)}{F(s)} = \frac{1/5}{s^2 + s + \frac{28}{5}}$$

b.  $\omega_n^2 = 28/5$  r/s,  $2\zeta\omega_n = 1$ . Therefore  $\zeta = 0.211$ ,  $\omega_n = 2.37$ .  $T_s = \frac{4}{\zeta\omega_n} = 8.01$  s;  $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} =$

1.36 s; %OS =  $e^{-\zeta\pi} / \sqrt{1-\zeta^2} \times 100 = 50.7$  %;  $\omega_n T_r = (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1)$ ; therefore,  $T_r = 0.514$  s.

26.

Writing the loop equations,

$$(1.07s^2 + 1.53s)\theta_1(s) - 1.53\theta_2(s) = T(s)$$

$$-1.53s\theta_1(s) + (1.53s + 1.92)\theta_2(s) = 0$$

Solving for  $\theta_2(s)$ ,

$$\theta_2(s) = \frac{\begin{vmatrix} (1.07s^2 + 1.53s) & T(s) \\ -1.53s & 0 \end{vmatrix}}{\begin{vmatrix} (1.07s^2 + 1.53s) & -1.53s \\ -1.53s & (1.53s + 1.92) \end{vmatrix}} = \frac{0.935T(s)}{s^2 + 1.25s + 1.79}$$

Forming the transfer function,

$$\frac{\theta_2(s)}{T(s)} = \frac{0.935}{s^2 + 1.25s + 1.79}$$

Thus  $\omega_n = 1.34$ ,  $2\zeta\omega_n = 1.25$ . Thus,  $\zeta = 0.467$ . From Eq. (4.38), %OS = 19.0%. From Eq. (4.42),  $T_s = 6.4$  seconds. From Eq. (4.34),  $T_p = 2.66$  seconds.

27.

$$\text{a. } \frac{24.542}{s(s^2 + 4s + 24.542)} = \frac{1}{s} - \frac{s+4}{(s+2)^2 + 20.542} = \frac{1}{s} - \frac{(s+2) + \frac{2}{4.532}}{(s+2)^2 + 20.542}.$$

Thus  $c(t) = 1 - e^{-2t} (\cos 4.532t + 0.441 \sin 4.532t) = 1 - 1.09e^{-2t} \cos(4.532t - 23.8^\circ)$ .

b.

$$\frac{245.42}{s(s+10)(s^2 + 4s + 24.542)} = \frac{1}{s} - 0.29029 \frac{1}{s+10} - \frac{0.70971s + 5.7418}{s^2 + 4s + 24.542}$$

$$\frac{245.42}{s(s+10)(s^2 + 4s + 24.542)} = \frac{1}{s} - 0.29029 \frac{1}{s+10} - \frac{0.70971s + 5.7418}{(s+2)^2 + 20.542}$$

$$\frac{245.42}{s(s+10)(s^2+4s+24.542)} = \frac{1}{s} - 0.29029 \frac{1}{s+10} - \frac{0.70971(s+2) + \frac{4.3223}{\sqrt{20.542}} \sqrt{20.542}}{(s+2)^2 + 20.542}$$

$$\frac{245.42}{s(s+10)(s^2+4s+24.542)} = \frac{1}{s} - 0.29029 \frac{1}{s+10} - \frac{0.70971(s+2) + 0.95367 \sqrt{20.542}}{(s+2)^2 + 20.542}$$

Therefore,  $c(t) = 1 - 0.29e^{-10t} - e^{-2t}(0.71 \cos 4.532t + 0.954 \sin 4.532t)$

$$= 1 - 0.29e^{-10t} - 1.189 \cos(4.532t - 53.34^\circ).$$

c.

$$\begin{aligned} \frac{73.626}{s(s+3)(s^2+4s+24.542)} &= \frac{1}{s} - 1.1393 \frac{1}{s+3} + \frac{0.13926s - 2.8607}{s^2 + 4s + 24.542} \\ \frac{73.626}{s(s+3)(s^2+4s+24.542)} &= \frac{1}{s} - 1.1393 \frac{1}{s+3} + \frac{0.13926(s+2) - \frac{3.1393}{\sqrt{20.542}} \sqrt{20.542}}{(s+2)^2 + 20.542} \\ \frac{73.626}{s(s+3)(s^2+4s+24.542)} &= \frac{1}{s} - 1.1393 \frac{1}{s+3} + \frac{0.13926(s+2) - 0.69264 \sqrt{20.542}}{(s+2)^2 + 20.542} \end{aligned}$$

Therefore,  $c(t) = 1 - 1.14e^{-3t} + e^{-2t}(0.14 \cos 4.532t - 0.69 \sin 4.532t)$

$$= 1 - 1.14e^{-3t} + 0.704 \cos(4.532t + 78.53^\circ).$$

28.

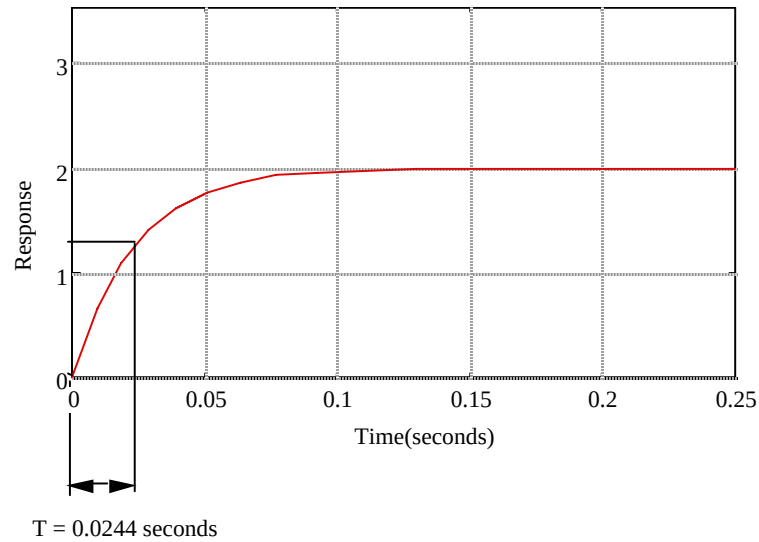
Since the third pole is more than five times the real part of the dominant pole,  $s^2 + 0.842s + 2.829$  determines the transient response. Since  $2\zeta\omega_n = 0.842$ , and  $\omega_n = \sqrt{2.829} = \omega_n = 1.682$ ,  $\zeta = 0.25$ ,

$$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100 = 44.4\%, T_s = \frac{4}{\zeta\omega_n} = 9.50 \text{ sec}, T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 1.93 \text{ sec}; \omega_n T_r =$$

$$(1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1) = 1.26, \text{ therefore, } T_r = 0.75.$$

29.

a. Measuring the time constant from the graph,  $T = 0.0244$  seconds.

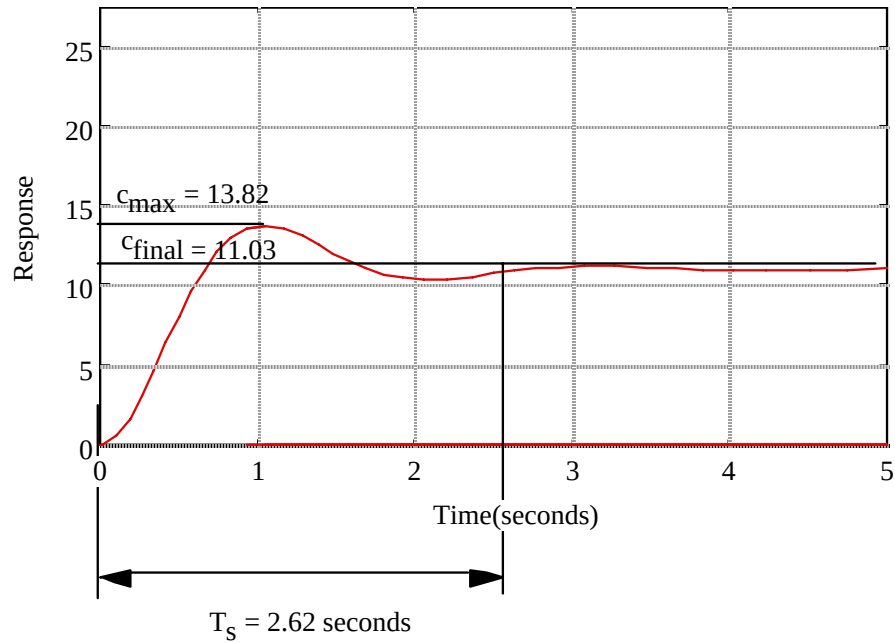


Estimating a first-order system,  $G(s) = \frac{K}{s+a}$ . But,  $a = 1/T = 40.984$ , and  $\frac{K}{a} = 2$ . Hence,  $K = 81.967$ .

Thus,

$$G(s) = \frac{81.967}{s+40.984}$$

b. Measuring the percent overshoot and settling time from the graph:  $\%OS = (13.82-11.03)/11.03 = 25.3\%$ ,



and  $T_s = 2.62$  seconds. Estimating a second-order system, we use Eq. (4.39) to find  $\zeta = 0.4$ , and Eq. (4.42) to find  $\omega_n = 3.82$ . Thus,  $G(s) = \frac{K}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ . Since  $C_{\text{final}} = 11.03$ ,  $\frac{K}{\omega_n^2} = 11.03$ . Hence,

$K = 160.95$ . Substituting all values,

$$G(s) = \frac{160.95}{s^2 + 3.056s + 14.59}$$

c. From the graph, %OS = 40%. Using Eq. (4.39),  $\zeta = 0.28$ . Also from the graph,

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = 4. \text{ Substituting } \zeta = 0.28, \text{ we find } \omega_n = 0.818.$$

Thus,

$$G(s) = \frac{K}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{0.669}{s^2 + 0.458s + 0.669}.$$

30.

a.

$$\frac{s+3}{s(s+2)(s^2+3s+10)} = \frac{3}{20} \frac{1}{s} - \frac{1}{16} \frac{1}{s+2} - \frac{1}{80} \frac{7s+31}{\left(s+\frac{3}{2}\right)^2 + \frac{31}{4}}$$

$$\frac{s+3}{s(s+2)(s^2+3s+10)} = \frac{3}{20} \frac{1}{s} - \frac{1}{16} \frac{1}{s+2} - \frac{1}{80} \frac{7\left(s+\frac{3}{2}\right) + 7.3638\sqrt{\frac{31}{4}}}{\left(s+\frac{3}{2}\right)^2 + \frac{31}{4}}$$

Since the amplitude of the sinusoids are of the same order of magnitude as the residue of the pole at -2, pole-zero cancellation cannot be assumed.

b.

$$\frac{s+2.5}{s(s+2)(s^2+4s+20)} = \frac{1}{16} \frac{1}{s} - \frac{1}{64} \frac{1}{s+2} - \frac{1}{64} \frac{3s+14}{s^2+4s+20}$$

$$\frac{s+2.5}{s(s+2)(s^2+4s+20)} = \frac{1}{16} \frac{1}{s} - \frac{1}{64} \frac{1}{s+2} - \frac{1}{64} \frac{3\left(s+2\right) + 2\sqrt{16}}{(s+2)^2 + 16}$$

Since the amplitude of the sinusoids are of the same order of magnitude as the residue of the pole at -2, pole-zero cancellation cannot be assumed.

c.

$$\frac{s+2.1}{s(s+2)(s^2+s+5)} = 0.21 \frac{1}{s} - 0.0071429 \frac{1}{s+2} - \frac{0.20286s+0.21714}{s^2+s+5}$$

$$\frac{s+2.1}{s(s+2)(s^2+s+5)} = 0.21 \frac{1}{s} - 0.0071429 \frac{1}{s+2} - \frac{0.20286\left(s+\frac{1}{2}\right) + 0.053093\sqrt{\frac{19}{4}}}{\left(s+\frac{1}{2}\right)^2 + \frac{19}{4}}$$

Since the amplitude of the sinusoids are of two orders of magnitude larger than the residue of the pole at -2, pole-zero cancellation can be assumed. Since  $2\zeta\omega_n = 1$ , and  $\omega_n = \sqrt{5} = 2.236$ ,  $\zeta = 0.224$ ,

$$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100 = 48.64\%, T_s = \frac{4}{\zeta\omega_n} = 8 \text{ sec}, T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 1.44 \text{ sec}; \omega_n T_r = 1.23,$$

therefore,  $T_r = 0.55$ .

d.

$$\frac{s+2.01}{s(s+2)(s^2+5s+20)} = 0.05025 \frac{1}{s} - 0.00035714 \frac{1}{s+2} - \frac{0.049893s+0.25018}{\left(s+\frac{5}{2}\right)^2 + \frac{55}{4}}$$

$$\frac{s+2.01}{s(s+2)(s^2+5s+20)} = 0.05025 \frac{1}{s} - 0.00035714 \frac{1}{s+2} - \frac{0.049893\left(s+\frac{5}{2}\right) + 0.03383\sqrt{\frac{55}{4}}}{\left(s+\frac{5}{2}\right)^2 + \frac{55}{4}}$$

Since the amplitude of the sinusoids are of two orders of magnitude larger than the residue of the pole at -2, pole-zero cancellation can be assumed. Since  $2\zeta\omega_n = 5$ , and  $\omega_n = \sqrt{20} = 4.472$ ,  $\zeta = 0.559$ ,

$$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100 = 12.03\%, T_s = \frac{4}{\zeta\omega_n} = 1.6 \text{ sec}, T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 0.847 \text{ sec}; \omega_n T_r =$$

1.852, therefore,  $T_r = 0.414$ .

31.

**Program:**

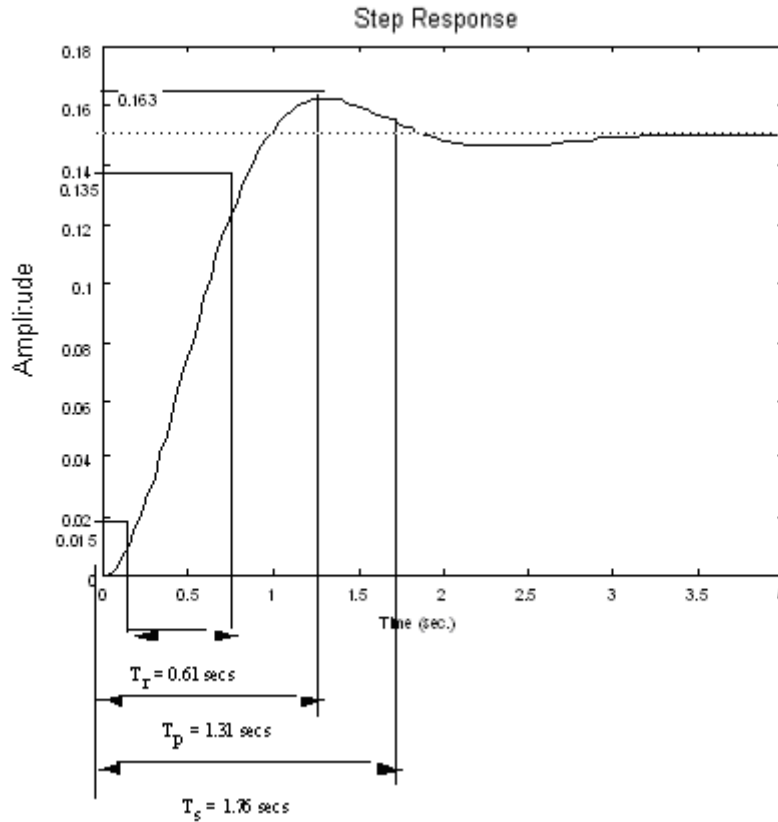
```
%Form sC(s) to get transfer function
clf
num=[1 3];
den=conv([1 3 10],[1 2]);
T=tf(num,den)
step(T)
```

**Computer response:**

Transfer function:

```

      s + 3
-----
s^3 + 5 s^2 + 16 s + 20
```



$$\%OS = \frac{(0.163 - 0.15)}{0.15} = 8.67\%$$

32.

Part c can be approximated as a second-order system. From the exponentially decaying cosine, the poles are located at  $s_{1,2} = -2 \pm j9.796$ . Thus,

$$T_s = \frac{4}{|\text{Re}|} = \frac{4}{2} = 2 \text{ s}; T_p = \frac{\pi}{|\text{Im}|} = \frac{\pi}{9.796} = 0.3207 \text{ s}$$

Also,  $\omega_n = \sqrt{2^2 + 9.796^2} = 10$  and  $\zeta\omega_n = |\text{Re}| = 2$ . Hence,  $\zeta = 0.2$ , yielding 52.66 percent overshoot.

Part d can be approximated as a second-order system. From the exponentially decaying cosine, the poles are located at  $s_{1,2} = -2 \pm j9.951$ . Thus,

$$T_s = \frac{4}{|\text{Re}|} = \frac{4}{2} = 2 \text{ s}; T_p = \frac{\pi}{|\text{Im}|} = \frac{\pi}{9.951} = 0.3157 \text{ s}$$

Also,  $\omega_n = \sqrt{2^2 + 9.951^2} = 10.15$  and  $\zeta\omega_n = |\text{Re}| = 2$ . Hence,  $\zeta = 0.197$ , yielding 53.19 percent overshoot.

33.

a.

$$(1) \quad C_{a1}(s) = \frac{1}{s^2 + 3s + 36} = \frac{\frac{1}{\sqrt{33.75}} \sqrt{33.75}}{(s + 1.5)^2 + 33.75} = \frac{0.17213 \sqrt{33.75}}{(s + 1.5)^2 + 33.75} = \frac{0.17213 \cdot 5.8095}{(s + 1.5)^2 + 33.75}$$

Taking the inverse Laplace transform

$$\begin{aligned}
 C_{a1}(t) &= 0.17213 e^{-1.5t} \sin 5.8095t \\
 (2) \quad C_{a2}(s) &= \frac{2}{s(s^2 + 3s + 36)} = \frac{1}{18} \frac{1}{s} - \frac{\frac{1}{18}s + \frac{1}{6}}{s^2 + 3s + 36} = \\
 &= \frac{1}{18} \frac{1}{s} - \frac{\frac{1}{18} \left(s + \frac{3}{2}\right) + \frac{0.083333}{\sqrt{33.75}}}{\left(s + \frac{3}{2}\right)^2 + 33.75} \\
 &= 0.055556 \frac{1}{s} - \frac{0.055556 \left(s + \frac{3}{2}\right) + 0.014344 \sqrt{33.75}}{\left(s + \frac{3}{2}\right)^2 + 33.75}
 \end{aligned}$$

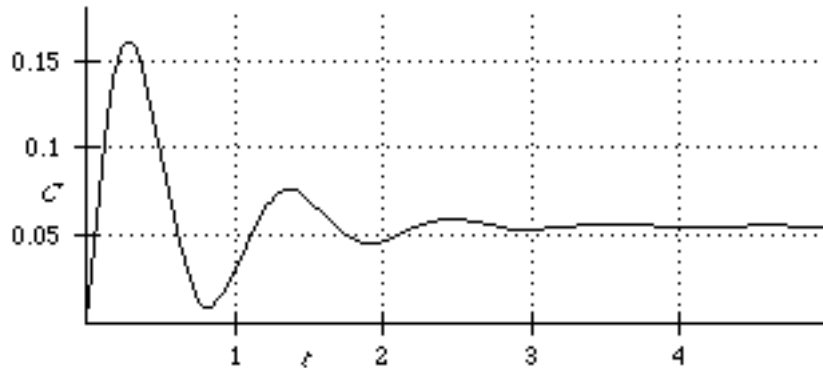
Taking the inverse Laplace transform

$$C_{a2}(t) = 0.055556 - e^{-1.5t} (0.055556 \cos 5.809t + 0.014344 \sin 5.809t)$$

The total response is found as follows:

$$C_{at}(t) = C_{a1}(t) + C_{a2}(t) = 0.055556 - e^{-1.5t} (0.055556 \cos 5.809t - 0.157786 \sin 5.809t)$$

Plotting the total response:



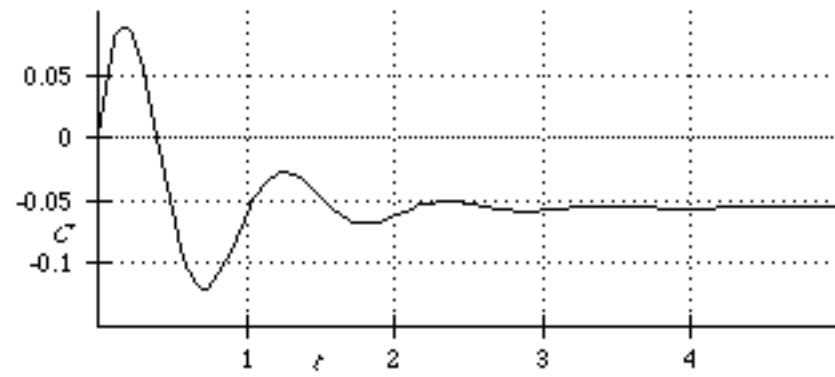
**b.**

(1) Same as (1) from part (a), or  $C_{b1}(t) = C_{a1}(t)$

(2) Same as the negative of (2) of part (a), or  $C_{b2}(t) = -C_{a2}(t)$

The total response is

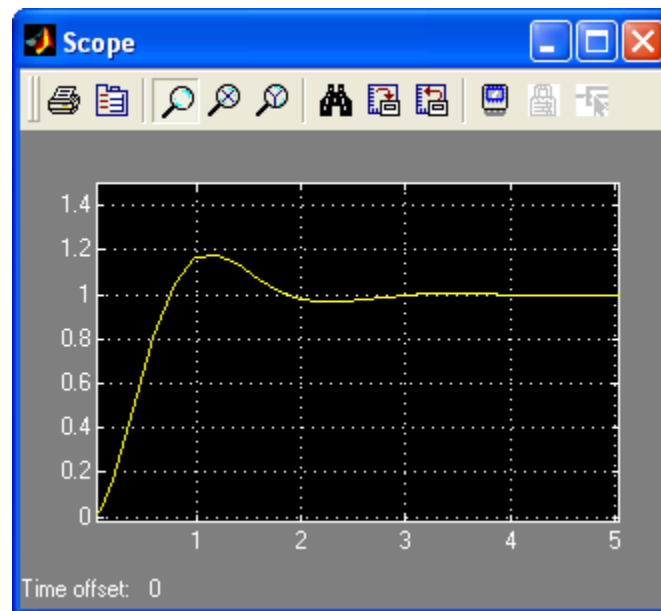
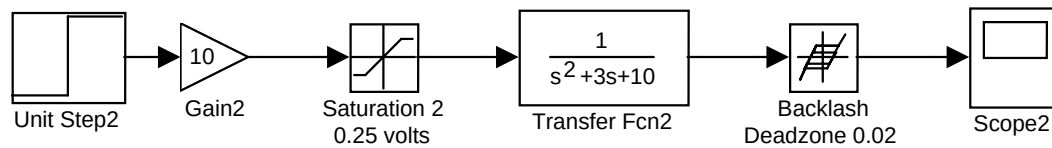
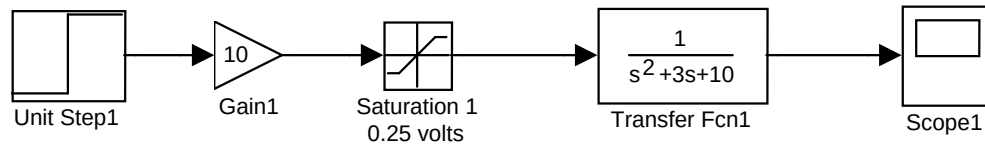
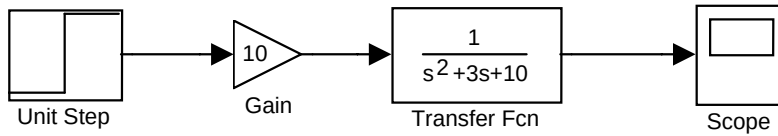
$$\begin{aligned}
 C_{bt}(t) &= C_{b1}(t) + C_{b2}(t) = C_{a1}(t) - C_{a2}(t) = -0.055556 + e^{-1.5t} (0.055556 \cos 5.809t + 0.186474 \sin 5.809t)
 \end{aligned}$$

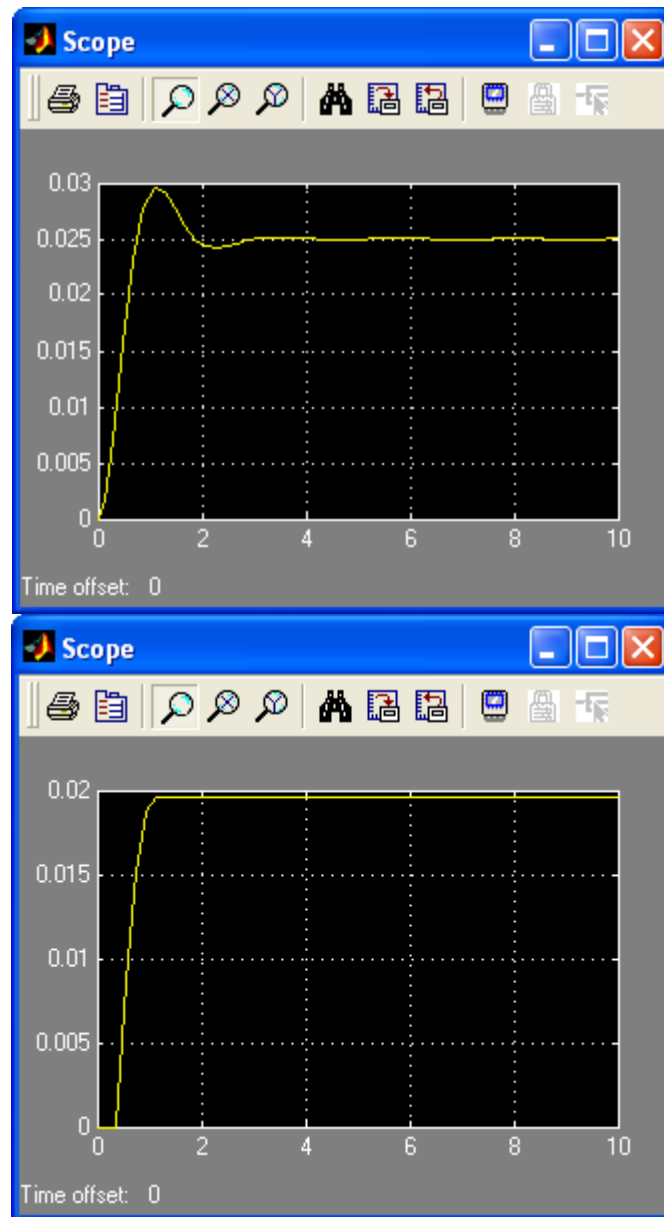


Notice the nonminimum phase behavior for  $C_{bt}(t)$ .



34.





35.

$$s\mathbf{I} - \mathbf{A} = s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -2 & -1 \\ -3 & -5 \end{bmatrix} = \begin{bmatrix} (s+2) & 1 \\ 3 & (s+5) \end{bmatrix}$$

$$|s\mathbf{I} - \mathbf{A}| = s^2 + 7s + 7$$

Factoring yields poles at  $-5.7913$  and  $-1.2087$ .

36.

a.

$$s\mathbf{I} - \mathbf{A} = s \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & 0 & 6 & 5 \\ 0 & 0 & 1 & 1 & 4 & 2 \end{array} \right] = \left[ \begin{array}{ccc|ccc} s & -2 & -3 & 0 & 0 & 0 \\ 0 & (s-6) & -5 & 0 & 0 & 0 \\ -1 & -4 & (s-2) & 0 & 0 & 0 \end{array} \right]$$

$$|s\mathbf{I} - \mathbf{A}| = s^3 - 8s^2 - 11s + 8$$

b. Factoring yields poles at 9.111, 0.5338, and -1.6448.

37.

$$\mathbf{x} = (s\mathbf{I} - \mathbf{A})^{-1}(\mathbf{x}_0 + \mathbf{B}u)$$

$$\mathbf{X} = \left( s \left[ \begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] - \left[ \begin{array}{cc} 1 & 2 \\ -3 & -1 \end{array} \right] \right)^{-1} \left( \left[ \begin{array}{c} 3 \\ 1 \end{array} \right] + \left[ \begin{array}{c} 1 \\ 1 \end{array} \right] \cdot 3 \frac{1}{s^2 + 9} \right)$$

$$\mathbf{X} = \begin{pmatrix} \frac{3s^3 + 5s^2 + 30s + 54}{[s^2 + 5][s^2 + 9]} \\ \frac{s^3 - 10s^2 + 12s - 102}{[s^2 + 5][s^2 + 9]} \end{pmatrix}$$

$$Y(s) = [1 \quad 2]\mathbf{X}$$

$$Y(s) = \left( \frac{5s^3 - 15s^2 + 54s - 150}{[s^2 + 9][s^2 + 5]} \right)$$

38.

$$\mathbf{x} = (s\mathbf{I} - \mathbf{A})^{-1}(\mathbf{x}_0 + \mathbf{B}u)$$

$$\mathbf{x} = \left( s \left[ \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] - \left[ \begin{array}{ccc} 0 & 1 & 0 \\ -2 & -4 & 1 \\ 0 & 0 & -6 \end{array} \right] \right)^{-1} \left( \left[ \begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right] + \left[ \begin{array}{c} 0 \\ 0 \\ 1 \end{array} \right] \frac{1}{s+1} \right)$$

$$\mathbf{x} = \begin{pmatrix} \frac{1}{[s+6][s+1][s+0.58579][s+3.4142]} \\ \frac{s}{[s+6][s+1][s+0.58579][s+3.4142]} \\ \frac{s^2 + 4s + 2}{[s+6][s+1][s+0.58579][s+3.4142]} \end{pmatrix}$$

$$Y(s) = [0 \quad 0 \quad 1]\mathbf{x}$$

$$Y(s) = \frac{s^2 + 4s + 2}{[s+6][s+1][s+0.58579][s+3.4142]}$$

39.

$$\mathbf{x} = (s\mathbf{I} - \mathbf{A})^{-1}(\mathbf{x}_0 + \mathbf{B}u)$$

$$\mathbf{X} = \left( s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -2 & 0 \\ -1 & -1 \end{bmatrix} \right)^{-1} \left( \begin{bmatrix} 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \frac{1}{s} \right)$$

$$\mathbf{X} = \begin{pmatrix} \frac{3s+1}{s[s+2]} \\ \frac{1-2s}{s[s+1][s+2]} \end{pmatrix}$$

$$Y(s) = [0 \quad 1]\mathbf{X}$$

$$Y(s) = \left( \frac{1-2s}{s[s+1][s+2]} \right)$$

Applying partial fraction decomposition,

$$Y(s) = \left( \frac{1}{2} \frac{1}{s} - \frac{3}{s+1} + \frac{5}{2} \frac{1}{s+2} \right)$$

$$y(t) = \frac{1}{2}u(t) - 3e^{-t} + \frac{5}{2}e^{-2t}$$

40.

$$\mathbf{x} = (s\mathbf{I} - \mathbf{A})^{-1}(\mathbf{x}_0 + \mathbf{B}u)$$

$$\mathbf{x} = \left( s \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} -3 & 1 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & -5 \end{bmatrix} \right)^{-1} \left( \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \frac{1}{s} \right)$$

$$\mathbf{x} = \begin{bmatrix} \frac{1}{s(s+3)(s+5)} \\ \frac{1}{s(s+5)} \\ \frac{1}{s(s+5)} \end{bmatrix}$$

$$\mathbf{x}(t) = \begin{bmatrix} \frac{1}{15} - \frac{1}{6}e^{-3t} + \frac{1}{10}e^{-5t} \\ \frac{1}{5} - \frac{1}{5}e^{-5t} \\ \frac{1}{5} - \frac{1}{5}e^{-5t} \end{bmatrix}$$

$$y(t) = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \mathbf{x} = \frac{2}{5} - \frac{2}{5} e^{-5t}$$

41.

**Program:**

```
A=[-3 1 0;0 -6 1;0 0 -5];
B=[0;1;1];
C=[0 1 1];
D=0;
S=ss(A,B,C,D)
step(S)
```

**Computer response:**

a =

|    | x1 | x2 | x3 |
|----|----|----|----|
| x1 | -3 | 1  | 0  |
| x2 | 0  | -6 | 1  |
| x3 | 0  | 0  | -5 |

b =

|    | u1 |
|----|----|
| x1 | 0  |
| x2 | 1  |
| x3 | 1  |

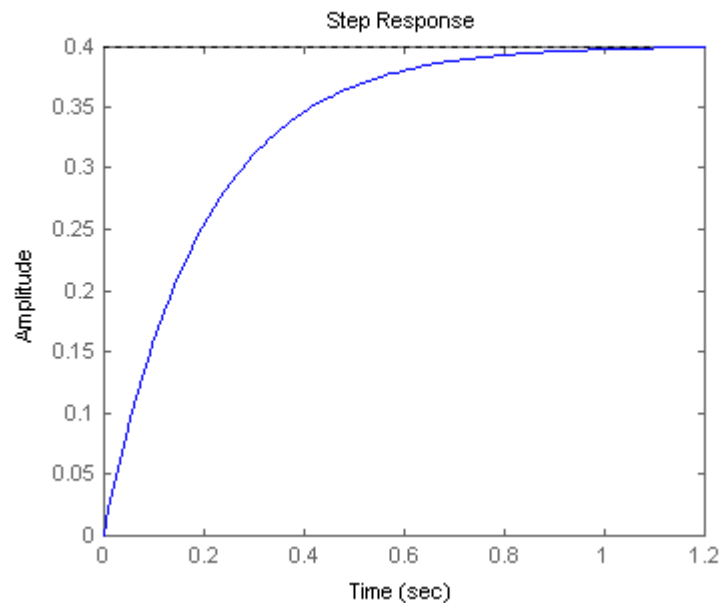
c =

|    | x1 | x2 | x3 |
|----|----|----|----|
| y1 | 0  | 1  | 1  |

d =

|    | u1 |
|----|----|
| y1 | 0  |

Continuous-time model.



42.

**Program:**

```

syms s                                %Construct symbolic object for
                                     %frequency variable 's'.
'a'                                   %Display label
A=[-3 1 0;0 -6 1;0 0 -5]             %Create matrix A.
B=[0;1;1];                           %Create vector B.
C=[0 1 1];                           %Create C vector
X0=[1;1;0]                           %Create initial condition vector,X(0).
U=1/s;                                %Create U(s).
I=[1 0 0;0 1 0;0 0 1];               %Create identity matrix.
X=((s*I-A)^-1)*(X0+B*U);              %Find Laplace transform of state vector.
x1=ilaplace(X(1))                    %Solve for X1(t).
x2=ilaplace(X(2))                    %Solve for X2(t).
x3=ilaplace(X(3))                    %Solve for X3(t).
y=C*[x1;x2;x3]                       %Solve for output, y(t).
y=simplify(y)                        %Simplify y(t).
'y(t)'                               %Display label.
pretty(y)                            %Pretty print y(t).

```

**Computer response:**

```

ans =

a

A =

    -3     1     0
     0    -6     1
     0     0    -5

X0 =

     1
     1
     0

x1 =

7/6*exp(-3*t)-1/3*exp(-6*t)+1/15+1/10*exp(-5*t)

x2 =

exp(-6*t)+1/5-1/5*exp(-5*t)

x3 =

1/5-1/5*exp(-5*t)

y =

2/5+exp(-6*t)-2/5*exp(-5*t)

y =

2/5+exp(-6*t)-2/5*exp(-5*t)

ans =

y(t)

2/5 + exp(-6 t) - 2/5 exp(-5 t)

```

43.

$$|\lambda \mathbf{I} - \mathbf{A}| = \lambda^2 + 5\lambda + 1$$

$$|\lambda \mathbf{I} - \mathbf{A}| = (\lambda + 0.20871)(\lambda + 4.7913)$$

$$\mathbf{P} = -0.20871$$

$$\begin{aligned} Q &= -4.7913 \\ \Phi &= \begin{pmatrix} A_1 e^{-0.20871t} + A_2 e^{-4.7913t} & A_5 e^{-0.20871t} + A_6 e^{-4.7913t} \\ A_3 e^{-0.20871t} + A_4 e^{-4.7913t} & A_7 e^{-0.20871t} + A_8 e^{-4.7913t} \end{pmatrix} \\ \Phi_0 &= \begin{pmatrix} A_2 + A_1 & A_6 + A_5 \\ A_4 + A_3 & A_8 + A_7 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial t} \Phi &= \begin{pmatrix} -0.20871 A_1 e^{-0.20871t} - 4.7913 A_2 e^{-4.7913t} & -0.20871 A_5 e^{-0.20871t} - 4.7913 A_6 e^{-4.7913t} \\ -0.20871 A_3 e^{-0.20871t} - 4.7913 A_4 e^{-4.7913t} & -0.20871 A_7 e^{-0.20871t} - 4.7913 A_8 e^{-4.7913t} \end{pmatrix} \\ \frac{d}{dt} \Phi_0 &= \begin{pmatrix} -4.7913 A_2 - 0.20871 A_1 & -4.7913 A_6 - 0.20871 A_5 \\ -4.7913 A_4 - 0.20871 A_3 & -4.7913 A_8 - 0.20871 A_7 \end{pmatrix} \end{aligned}$$

Therefore,

$$\begin{aligned} \begin{pmatrix} A_2 + A_1 & A_6 + A_5 \\ A_4 + A_3 & A_8 + A_7 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} -4.7913 A_2 - 0.20871 A_1 & -4.7913 A_6 - 0.20871 A_5 \\ -4.7913 A_4 - 0.20871 A_3 & -4.7913 A_8 - 0.20871 A_7 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ -1 & -5 \end{pmatrix} \end{aligned}$$

Solving for  $A_i$ 's two at a time, and substituting into the state-transition matrix

$$\Phi = \begin{pmatrix} 1.0455 e^{-0.20871t} - 0.045545 e^{-4.7913t} & 0.21822 e^{-0.20871t} - 0.21822 e^{-4.7913t} \\ -0.21822 e^{-0.20871t} + 0.21822 e^{-4.7913t} & -0.045545 e^{-0.20871t} + 1.0455 e^{-4.7913t} \end{pmatrix}$$

To find  $x(t)$ ,

$$\begin{aligned} x &= \Phi x_0 \\ x &= \begin{pmatrix} 1.0455 e^{-0.20871t} - 0.045545 e^{-4.7913t} & 0.21822 e^{-0.20871t} - 0.21822 e^{-4.7913t} \\ -0.21822 e^{-0.20871t} + 0.21822 e^{-4.7913t} & -0.045545 e^{-0.20871t} + 1.0455 e^{-4.7913t} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ x &= \begin{pmatrix} 1.0455 e^{-0.20871t} - 0.045545 e^{-4.7913t} \\ -0.21822 e^{-0.20871t} + 0.21822 e^{-4.7913t} \end{pmatrix} \end{aligned}$$

To find the output,

$$\begin{aligned} y &= (1, 2)x \\ y &= (1, 2) \begin{pmatrix} 1.0455 e^{-0.20871t} - 0.045545 e^{-4.7913t} \\ -0.21822 e^{-0.20871t} + 0.21822 e^{-4.7913t} \end{pmatrix} \\ y &= (0.60911 e^{-0.20871t} + 0.39089 e^{-4.7913t}) \end{aligned}$$

44.

$$\lambda I - A = \begin{pmatrix} \lambda & -1 \\ 1 & \lambda \end{pmatrix}$$

$$|\lambda I - A| = \lambda^2 + 1$$

$$\Phi = \begin{pmatrix} A_1 \cos[t] + A_2 \sin[t] & A_5 \cos[t] + A_6 \sin[t] \\ A_3 \cos[t] + A_4 \sin[t] & A_7 \cos[t] + A_8 \sin[t] \end{pmatrix}$$

$$\begin{aligned}\frac{d}{dt}\Phi &= \begin{pmatrix} A_2 \cos[t] - A_1 \sin[t] & A_6 \cos[t] - A_5 \sin[t] \\ A_4 \cos[t] - A_3 \sin[t] & A_8 \cos[t] - A_7 \sin[t] \end{pmatrix} \\ \Phi_0 &= \begin{pmatrix} A_1 & A_5 \\ A_3 & A_7 \end{pmatrix} \\ \frac{d}{dt}\Phi_0 &= \begin{pmatrix} A_2 & A_6 \\ A_4 & A_8 \end{pmatrix} \\ \begin{pmatrix} A_1 & A_5 \\ A_3 & A_7 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{pmatrix} A_2 & A_6 \\ A_4 & A_8 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\end{aligned}$$

Solving for the  $A_i$ 's and substituting into the state-transition matrix,

$$\Phi = \begin{pmatrix} \cos[t] & \sin[t] \\ -\sin[t] & \cos[t] \end{pmatrix}$$

To find the state vector,

$$\begin{aligned}x &= \int_0^t (\Phi[t-\tau] B u[\tau]) d\tau \\ x &= \int_0^t \left( \begin{bmatrix} \cos(t-\tau) & \sin(t-\tau) \\ -\sin(t-\tau) & \cos(t-\tau) \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) d\tau \\ x &= \int_0^t \begin{pmatrix} \sin[t-\tau] \\ \cos[t-\tau] \end{pmatrix} d\tau \\ t-\tau &= \theta \\ x &= \int_t^0 \begin{pmatrix} -\sin[\theta] \\ -\cos[\theta] \end{pmatrix} d\theta \\ x &= \begin{pmatrix} 1 - \cos[t] \\ \sin[t] \end{pmatrix} \\ y &= (3, 4)x \\ \Delta y &= (3, 4) \begin{pmatrix} 1 - \cos[t] \\ \sin[t] \end{pmatrix} \\ y &= (-3\cos[t] + 4\sin[t] + 3)\end{aligned}$$

45.

$$|\lambda \mathbf{I} - \mathbf{A}| = (\lambda + 2)(\lambda + 0.5 - 2.3979i)(\lambda + 0.5 + 2.3979i)$$

Let the state-transition matrix be

$$\Phi = \begin{bmatrix} A_1 e^{-.5t} \cos(2.3979t) + A_2 e^{-.5t} \sin(2.3979t) + A_3 e^{-2t} & A_{10} e^{-.5t} \cos(2.3979t) + A_{11} e^{-.5t} \sin(2.3979t) + A_{12} & \bullet \\ A_4 e^{-.5t} \cos(2.3979t) + A_5 e^{-.5t} \sin(2.3979t) + A_6 e^{-2t} & A_{13} e^{-.5t} \cos(2.3979t) + A_{14} e^{-.5t} \sin(2.3979t) + A_{15} & \bullet \\ A_7 e^{-.5t} \cos(2.3979t) + A_8 e^{-.5t} \sin(2.3979t) + A_9 e^{-2t} & A_{16} e^{-.5t} \cos(2.3979t) + A_{17} e^{-.5t} \sin(2.3979t) + A_{18} & \bullet \end{bmatrix}$$



Since  $\phi(0) = \mathbf{I}$ ,  $\dot{\phi}(0) = \mathbf{A}$ , and  $\ddot{\phi}(0) = \mathbf{A}^2$ , we can evaluate the coefficients,  $A_i$ 's. Thus,

$$\begin{pmatrix} A_3 + A_1 & A_{12} + A_{10} & A_{21} + A_{19} \\ A_6 + A_4 & A_{15} + A_{13} & A_{24} + A_{22} \\ A_9 + A_7 & A_{18} + A_{16} & A_{27} + A_{25} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -2A_3 + 2.3979A_2 - 0.5A_1 & -2A_{12} + 2.3979A_{11} - 0.5A_{10} & -2A_{21} + 2.3979A_{20} - 0.5A_{19} \\ -2A_6 + 2.3979A_5 - 0.5A_4 & -2A_{15} + 2.3979A_{14} - 0.5A_{13} & -2A_{24} + 2.3979A_{23} - 0.5A_{22} \\ -2A_9 + 2.3979A_8 - 0.5A_7 & -2A_{18} + 2.3979A_{17} - 0.5A_{16} & -2A_{27} + 2.3979A_{26} - 0.5A_{25} \end{pmatrix} = \begin{pmatrix} -2 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -6 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 4A_3 - 2.3979A_2 - 5.4999A_1 & 4A_{12} - 2.3979A_{11} - 5.4999A_{10} & 4A_{21} - 2.3979A_{20} - 5.4999A_{19} \\ 4A_6 - 2.3979A_5 - 5.4999A_4 & 4A_{15} - 2.3979A_{14} - 5.4999A_{13} & 4A_{24} - 2.3979A_{23} - 5.4999A_{22} \\ 4A_9 - 2.3979A_8 - 5.4999A_7 & 4A_{18} - 2.3979A_{17} - 5.4999A_{16} & 4A_{27} - 2.3979A_{26} - 5.4999A_{25} \end{pmatrix} = \begin{pmatrix} 4 & -2 & 1 \\ 0 & -6 & -1 \\ 0 & 6 & -5 \end{pmatrix}$$

Solving for the  $A_i$ 's taking three equations at a time,

$$\phi = \begin{pmatrix} e^{-2t} & 0.125e^{-0.5t}\cos[2.3979t] + 0.33884e^{-0.5t}\sin[2.3979t] - 0.125e^{-2t} & -0.125e^{-0.5t}\cos[2.3979t] + 0.078194e^{-0.5t}\sin[2.3979t] + 0.125e^{-2t} \\ 0 & e^{-0.5t}\cos[2.3979t] + 0.20851e^{-0.5t}\sin[2.3979t] & 0.41703e^{-0.5t}\sin[2.3979t] \\ 0 & -2.502e^{-0.5t}\sin[2.3979t] & e^{-0.5t}\cos[2.3979t] - 0.20852e^{-0.5t}\sin[2.3979t] \end{pmatrix}$$

Using  $\mathbf{x}(t) = \phi(t)\mathbf{x}(0) + \int_0^t \phi(t-\tau)\mathbf{B}\mathbf{u}(\tau)d\tau$ , and  $\mathbf{y} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}(t)$ ,

$$\begin{aligned} y &= \int_0^t e^{-2(t-\tau)} d\tau \\ &= \frac{1}{2} - \frac{1}{2} e^{-2t} \end{aligned}$$

46.

**Program:**

```
syms s t tau
```

```
'a'
```

```
A=[-2 1 0;0 0 1;0 -6 -1]
```

```
B=[1;0;0]
```

```
C=[1 0 0]
```

```
X0=[1;1;0]
```

```
I=[1 0 0;0 1 0;0 0 1];
```

```
'E=(s*I-A)^-1'
```

```
E=((s*I-A)^-1)
```

```
Fi11=ilaplace(E(1,1));
```

```
Fi12=ilaplace(E(1,2));
```

```
Fi13=ilaplace(E(1,3));
```

```
Fi21=ilaplace(E(2,1));
```

```
Fi22=ilaplace(E(2,2));
```

```
Fi23=ilaplace(E(2,3));
```

```
%Construct symbolic object for
```

```
%frequency variable 's', 't', and 'tau.'
```

```
%Display label.
```

```
%Create matrix A.
```

```
%Create vector B.
```

```
%Create vector C.
```

```
%Create initial condition vector,X(0).
```

```
%Create identity matrix.
```

```
%Display label.
```

```
%Find Laplace transform of state
```

```
%transition matrix, (sI-A)^-1.
```

```
%Take inverse Laplace transform
```

```
%of each element
```

```

Fi31=ilaplace(E(3,1));
Fi32=ilaplace(E(3,2)); %to find state transition matrix.
Fi33=ilaplace(E(3,3)); %of (sI-A)^-1.
'Fi(t)' %Display label.
Fi=[Fi11 Fi12 Fi13 %Form Fi(t).
    Fi21 Fi22 Fi23
    Fi31 Fi32 Fi33];
pretty(Fi) %Pretty print state transition matrix, Fi.
Fitmtau=subs(Fi,t,t-tau); %Form Fi(t-tau).
'Fi(t-tau)' %Display label.
pretty(Fitmtau) %Pretty print Fi(t-tau).
x=Fi*X0+int(Fitmtau*B*1,tau,0,t); %Solve for x(t).
x=simple(x); %Collect terms.
x=simplify(x); %Simplify x(t).
x=vpa(x,3);
'x(t)' %Display label.
pretty(x) %Pretty print x(t).
y=C*x; %Find y(t)
y=simplify(y);
y=vpa(simple(y),3);
y=collect(y);
'y(t)'
pretty(y) %Pretty print y(t).

```

**Computer response:**

ans =

a

A =

$$\begin{bmatrix} -2 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -6 & -1 \end{bmatrix}$$

B =

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

C =

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

X0 =

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

ans =

E=(s\*I-A)^-1

E =

$$\begin{bmatrix} \frac{1}{s+2}, & \frac{(s+1)}{(s+2)(s^2+s+6)}, & \frac{1}{(s+2)(s^2+s+6)} \\ 0, & \frac{(s+1)}{(s^2+s+6)}, & \frac{1}{(s^2+s+6)} \\ 0, & \frac{-6}{(s^2+s+6)}, & \frac{s}{(s^2+s+6)} \end{bmatrix}$$

ans =

Fi(t)

$$\begin{aligned} & \left[ \begin{aligned} & \exp(-2t), -\frac{1}{8} \exp(-2t) + \frac{1}{8} \%1 + \frac{13}{184} \%2, \\ & \frac{1}{8} \exp(-2t) - \frac{1}{8} \%1 + \frac{3}{184} \%2 \end{aligned} \right] \\ & \left[ \begin{aligned} & 0, \frac{1}{23} \%2 + \%1, -\frac{1}{23} \\ & (-23)^{1/2} (\exp((-1/2 + 1/2 (-23)^{1/2}) t) - \exp((-1/2 - 1/2 (-23)^{1/2}) t)) \\ & \end{aligned} \right] \\ & \left[ \begin{aligned} & 0, \frac{6}{23} \\ & (-23)^{1/2} (\exp((-1/2 + 1/2 (-23)^{1/2}) t) - \exp((-1/2 - 1/2 (-23)^{1/2}) t)) \\ & , -\frac{1}{23} \%2 + \%1 \end{aligned} \right] \end{aligned}$$

$$\%1 := \exp(-1/2 t) \cos(1/2 \sqrt{23} t)$$

$$\%2 := \exp(-1/2 t) \sqrt{23} \sin(1/2 \sqrt{23} t)$$

ans =

Fi(t-tau)

$$\begin{aligned} & \left[ \begin{aligned} & \exp(-2t + 2\tau), \\ & -\frac{1}{8} \exp(-2t + 2\tau) + \frac{1}{8} \%2 \cos(\%1) + \frac{13}{184} \%2 \sqrt{23} \sin(\%1), \\ & \frac{1}{8} \exp(-2t + 2\tau) - \frac{1}{8} \%2 \cos(\%1) + \frac{3}{184} \%2 \sqrt{23} \sin(\%1) \end{aligned} \right] \\ & \left[ \begin{aligned} & 0, \frac{1}{23} \%2 \sqrt{23} \sin(\%1) + \%2 \cos(\%1), -\frac{1}{23} (-23)^{1/2} ( \\ & \exp((-1/2 + 1/2 (-23)^{1/2}) (t - \tau)) \\ & - \exp((-1/2 - 1/2 (-23)^{1/2}) (t - \tau))) \end{aligned} \right] \\ & \left[ \begin{aligned} & 0, \frac{6}{23} (-23)^{1/2} (\exp((-1/2 + 1/2 (-23)^{1/2}) (t - \tau)) \\ & - \exp((-1/2 - 1/2 (-23)^{1/2}) (t - \tau))) , \end{aligned} \right] \end{aligned}$$

```

- 1/23 %2 231/2 sin(%1) + %2 cos(%1)]
%1 := 1/2 231/2 (t - tau)
%2 := exp(- 1/2 t + 1/2 tau)
ans =
x(t)
[.375 exp(-2. t) + .125 exp(-.500 t) cos(2.40 t)
+ .339 exp(-.500 t) sin(2.40 t) + .500]
[.209 exp(-.500 t) sin(2.40 t) + exp(-.500 t) cos(2.40 t)]
[1.25 i (exp((- .500 + 2.40 i) t) - 1. exp((- .500 - 2.40 i) t))]
ans =
y(t)
.375 exp(-2. t) + .125 exp(-.500 t) cos(2.40 t)
+ .339 exp(-.500 t) sin(2.40 t) + .500

```

47.

The state-space representation used to obtain the plot is,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -1 & -0.8 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t); \quad y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}$$

Using the Step Response software,

Step Response - Data Entry

About

$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u(t)$   
 $y = \mathbf{C}\mathbf{x}$

Order: 2

**A =**

|   |    |      |
|---|----|------|
|   | 1  | 2    |
| 1 | 0  | 1    |
| 2 | -1 | -0.8 |

**B =**

|   |   |
|---|---|
| 1 | 0 |
| 2 | 1 |

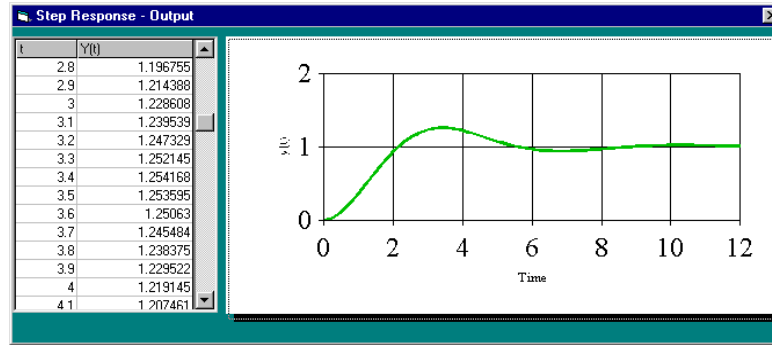
**C =**

|   |   |
|---|---|
| 1 | 2 |
| 1 | 0 |

**x(0) =**

|   |   |
|---|---|
| 1 | 0 |
| 2 | 0 |

Iteration Interval: .001    Print Interval: .1    Total Time: 12    Calculate



Calculating % overshoot, settling time, and peak time,

$$2\zeta\omega_n = 0.8, \omega_n = 1, \zeta = 0.4. \text{ Therefore, } \%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100 = 25.38\%, T_s = \frac{4}{\zeta\omega_n} = 10 \text{ sec,}$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 3.43 \text{ sec.}$$

48.

Step Response - Data Entry

About

Order: 3

$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u(t)$   
 $y = \mathbf{C}\mathbf{x}$

**A =**

|   |     |    |
|---|-----|----|
|   | 1   | 2  |
| 1 | 0   | 1  |
| 2 | -10 | -7 |
| 3 | 0   | 0  |

**B =**

|   |   |
|---|---|
| 1 | 0 |
| 2 | 0 |
| 3 | 1 |

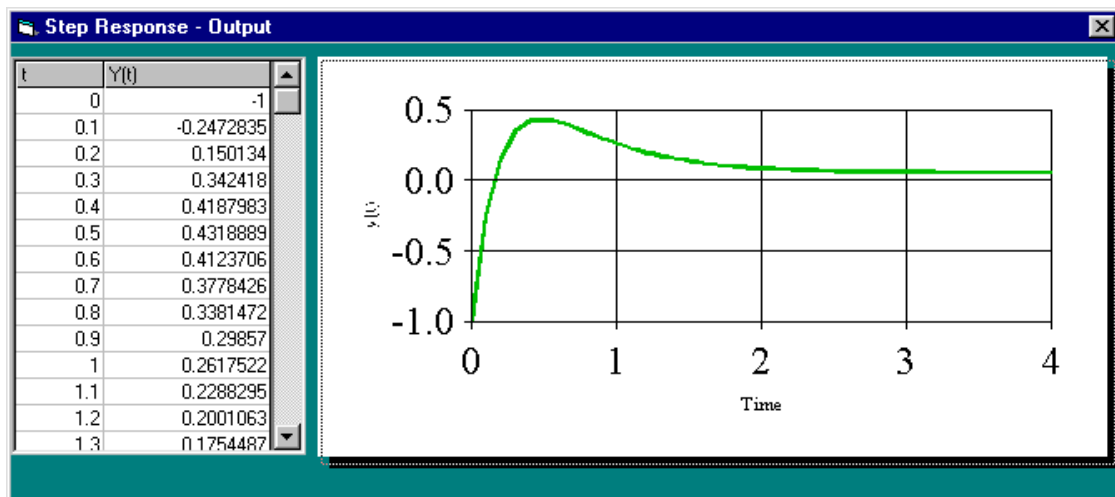
**C =**

|   |   |
|---|---|
| 1 | 2 |
| 1 | 1 |

**x(0) =**

|   |    |
|---|----|
| 1 | -1 |
| 2 | 0  |
| 3 | 0  |

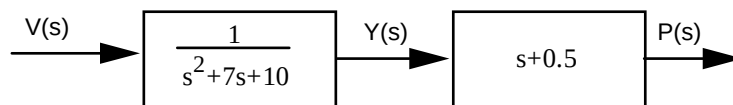
Iteration Interval: .001    Print Interval: .1    Total Time: 4    Calculate



49.

a.  $P(s) = \frac{s+0.5}{s(s+2)(s+5)} = \frac{1/20}{s} + \frac{1/4}{s+2} - \frac{3/10}{s+5}$ . Therefore,  $p(t) = \frac{1}{20} + \frac{1}{4} e^{-2t} - \frac{3}{10} e^{-5t}$ .

b. To represent the system in state space, draw the following block diagram.



For the first block,

$$\ddot{y} + 7\dot{y} + 10y = v(t)$$

Let  $x_1 = y$ , and  $x_2 = \dot{y}$ . Therefore,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -10x_1 - 7x_2 + v(t)$$

Also,

$$p(t) = 0.5y + \dot{y} = 0.5x_1 + x_2$$

Thus,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -10 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} v(t); \quad p(t) = \begin{bmatrix} 0.5 & 1 \end{bmatrix} \mathbf{x}$$

c.

**Program:**

$A = \begin{bmatrix} 0 & 1 \\ -10 & -7 \end{bmatrix};$

$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix};$

$C = \begin{bmatrix} 0.5 & 1 \end{bmatrix};$

$D = 0;$

$S = ss(A, B, C, D)$

`step(S)`

**Computer response:**

a =

|    |     |    |
|----|-----|----|
|    | x1  | x2 |
| x1 | 0   | 1  |
| x2 | -10 | -7 |

b =

|    |    |
|----|----|
|    | u1 |
| x1 | 0  |
| x2 | 1  |

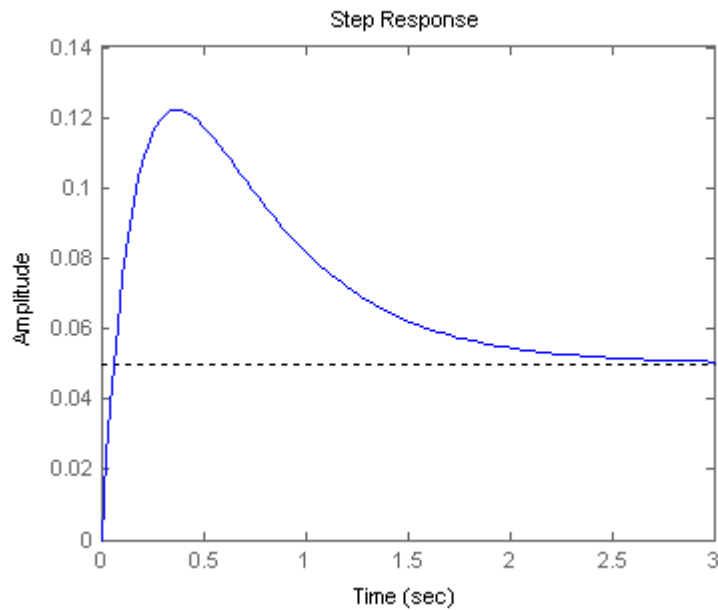
c =

|    |     |    |
|----|-----|----|
|    | x1  | x2 |
| y1 | 0.5 | 1  |

d =

|    |    |
|----|----|
|    | u1 |
| y1 | 0  |

Continuous-time model.

**50.**

a.  $\omega_n = \sqrt{10} = 3.16$ ;  $2\zeta\omega_n = 4$ . Therefore  $\zeta = 0.632$ .  $\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} * 100 = 7.69\%$ .

$T_s = \frac{4}{\zeta\omega_n} = 2$  seconds.  $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 1.28$  seconds. From Figure 4.16,  $T_r\omega_n = 1.93$ .

Thus,  $T_r = 0.611$  second. To justify second-order assumption, we see that the dominant poles are at –

$2 \pm j2.449$ . The third pole is at -10, or 5 times further. The second-order approximation is valid.

b.  $G_e(s) = \frac{K}{(s+10)(s^2+4s+10)} = \frac{K}{s^3+14s^2+50s+100}$ . Representing the system in phase-variable form:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -100 & -50 & -14 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ K \end{bmatrix}; \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

c.

**Program:**

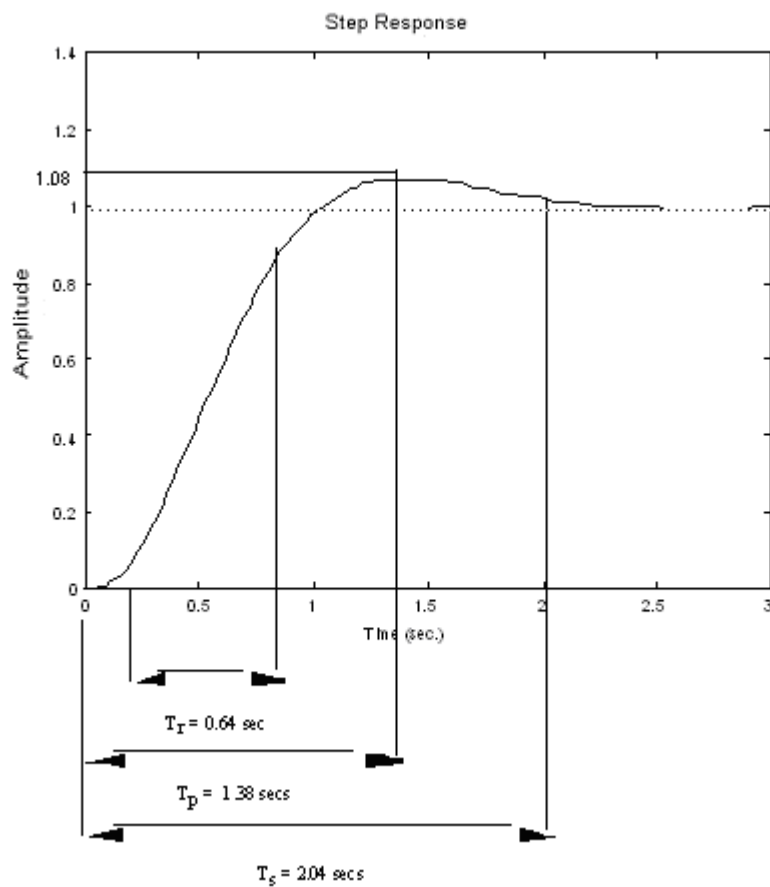
```
numg=100;
deng=conv([1 10],[1 4 10]);
G=tf(numg,deng)
step(G)
```

**Computer response:**

Transfer function:

$$\frac{100}{s^3 + 14s^2 + 50s + 100}$$

$$s^3 + 14s^2 + 50s + 100$$



$$\%OS = \frac{(1.08-1)}{1} * 100 = 8\%$$



51.

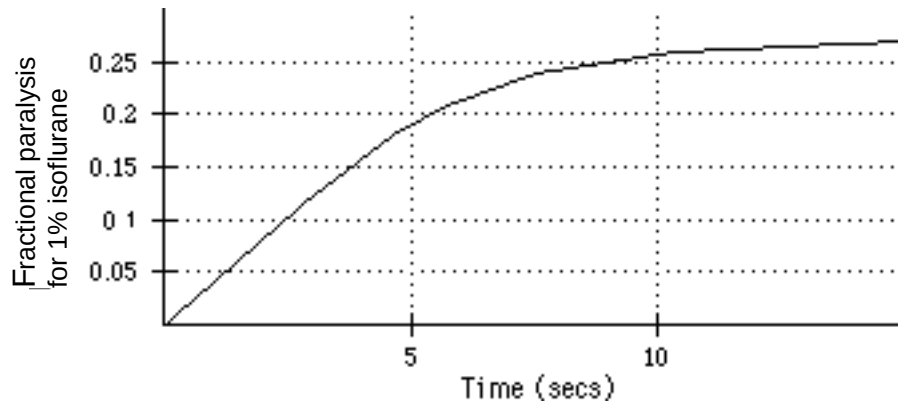
a.  $\omega_n = \sqrt{0.28} = 0.529$ ;  $2\zeta\omega_n = 1.15$ . Therefore  $\zeta = 1.087$ .

b.  $P(s) = U(s) \frac{7.63 \times 10^{-2}}{s^2 + 1.15s + 0.28}$ , where  $U(s) = \frac{2}{s}$ . Expanding by partial fractions,  $P(s) = \frac{0.545}{s} +$

natural response terms. Thus percent paralysis = 54.5%.

c.  $P(s) = \frac{7.63 \times 10^{-2}}{s(s^2 + 1.15s + 0.28)} = \frac{0.2725}{s} - \frac{0.48444}{s + 0.35} + \frac{0.21194}{s + 0.8}$ .

Hence,  $p(t) = 0.2725 - 0.48444e^{-0.35t} + 0.21194e^{-0.8t}$ . Plotting,



d.  $P(s) = \frac{K}{s} * \frac{7.63 \times 10^{-2}}{s^2 + 1.15s + 0.28} = \frac{1}{s} +$  natural response terms. Therefore,  $\frac{7.63 \times 10^{-2} K}{0.28} = 1$ . Solving

for K,  $K = 3.67\%$ .

52.

a. Writing the differential equation,

$$\frac{dc(t)}{dt} = -k_{10}c(t) + \frac{i(t)}{V_d}$$

Taking the Laplace transform and rearranging,

$$(s + k_{10})C(s) = \frac{I(s)}{V_d}$$

from which the transfer function is found to be

$$\frac{C(s)}{I(s)} = \frac{1}{V_d(s + k_{10})}$$

For a step input,  $I(s) = \frac{I_0}{s}$ . Thus the response is

$$C(s) = \frac{\frac{I_0}{V_d}}{s(s + k_{10})} = \frac{I_0}{k_{10}V_d} \left( \frac{1}{s} - \frac{1}{s + k_{10}} \right)$$

Taking the inverse Laplace transform,

$$c(t) = \frac{I_0}{k_{10}V_d}(1 - e^{-k_{10}t})$$

where the steady-state value,  $C_D$ , is

$$C_D = \frac{I_0}{k_{10}V_d}$$

Solving for  $I_R = I_0$ ,

$$I_R = C_D k_{10} V_d$$

**b.**  $T_R = \frac{2.2}{k_{10}} ; T_S = \frac{4}{k_{10}}$

**c.**  $I_R = C_D k_{10} V_d = 12 \frac{\mu\text{g}}{\text{ml}} \times 0.07 \text{ hr}^{-1} \times 0.6 \text{ liters} = 0.504 \frac{\text{mg}}{\text{h}}$

**d.** Using the equations of part b, where  $k_{10} = 0.07$ ,  $T_R = 31.43 \text{ hrs}$ , and  $T_S = 57.14 \text{ hrs}$ .

**53.** Consider the un-shifted Laplace transform of the output

$$Y(s) = \frac{2.5(1 + 0.172s)(1 + 0.008s)}{s(1 + 0.07s)^2(1 + 0.05s)^2} = \frac{280.82(s + 5.814)(s + 125)}{s(s + 14.286)^2(s + 20)^2}$$

$$= \frac{A}{s} + \frac{B}{(s + 14.286)^2} + \frac{C}{(s + 14.286)} + \frac{D}{(s + 20)^2} + \frac{E}{(s + 20)}$$

$$A = \frac{280.82(s + 5.814)(s + 125)}{(s + 14.286)^2(s + 20)^2} \Big|_{s=0} = 2.5$$

$$B = \frac{280.82(s + 5.814)(s + 125)}{s(s + 20)^2} \Big|_{s=-14.286} = 564.7$$

$$C = \frac{d}{ds} \frac{280.82(s + 5.814)(s + 125)}{s(s + 20)^2} \Big|_{s=-14.286} = \frac{d}{ds} \frac{280.82s^2 + 36735.2s + 204085.94}{s^3 + 40s^2 + 400s} \Big|_{s=-14.286}$$

$$= \frac{(s^3 + 40s^2 + 400s)(561.64s + 36735.2) - (280.82s^2 + 36735.2s + 204085.94)(3s^2 + 80s + 400)}{(s^3 + 40s^2 + 400s)^2} \Big|_{s=-14.286}$$

$$= -219.7$$

$$D = \frac{280.82(s + 5.814)(s + 125)}{s(s + 14.286)^2} \Big|_{s=-20} = 640.57$$

$$E = \frac{d}{ds} \frac{280.82(s + 5.814)(s + 125)}{s(s + 14.286)^2} \Big|_{s=-20} = \frac{d}{ds} \frac{280.82s^2 + 36735.2s + 204085.94}{s^3 + 28.572s^2 + 204.09s} \Big|_{s=-20}$$

$$= \frac{(s^3 + 28.572s^2 + 204.09s)(561.64s + 36735.2) - (280.82s^2 + 36735.2s + 204085.94)(3s^2 + 57.144s + 204.09)}{(s^3 + 28.572s^2 + 204.09s)^2} \Big|_{s=-20}$$

$$= 217.18$$

thus

$$Y(s) = \frac{2.5}{s} + \frac{564.7}{(s + 14.286)^2} - \frac{219.7}{(s + 14.286)} + \frac{640.57}{(s + 20)^2} + \frac{217.18}{(s + 20)}$$

Obtaining the inverse Laplace transform of the latter and delaying the equation in time domain we get

$$y(t) = [2.5 + 564.7(t - 0.008)e^{-14.286(t-0.008)} - 219.7e^{-14.286(t-0.008)} + 640.57(t - 0.008)e^{-20(t-0.008)} + 217.18e^{-20(t-0.008)}]u(t - 0.008)$$

54.

a. The transfer function can be written as

$$\frac{\theta}{I}(s) = \frac{2.5056(s + 3.33)e^{-0.1s}}{(s + 1)(s^2 + 0.72s + 1.44)}$$

It has poles at  $s = -0.36 \pm j1.145$  and  $s = -1$ . A zero at  $s = -3.33$

The 'far away' pole at  $-1$  is relatively close to the complex conjugate poles as  $0.36 \cdot 5 > 1$  so a dominant pole approximation can't be applied.

b) In time domain the input can be expressed as:

$$i(t) = 250 \mu A (u(t) - u(t - 0.15))$$

Obtaining Laplace transforms this can be expressed as

$$I(s) = 250\mu \frac{1 - e^{-0.15s}}{s}$$

We first obtain the response to an unshifted unit step:

$$\theta(s) = \frac{2.5056(s + 3.33)}{s(s + 1)(s^2 + 0.72s + 1.44)} = \frac{A}{s} + \frac{B}{s + 1} + \frac{Cs + D}{s^2 + 0.72s + 1.44}$$

$$A = \frac{2.5056(s + 3.33)}{(s + 1)(s^2 + 0.72s + 1.44)} \Big|_{s=0} = 5.8$$

$$B = \frac{2.5056(s + 3.33)}{s(s^2 + 0.72s + 1.44)} \Big|_{s=-1} = \frac{2.5056(2.33)}{(-1)(1.72)} = -3.4$$

We will get C and D by equating coefficients. Substituting these two values and multiplying both sides by the denominator we get.

$$2.5056(s + 3.33) = 5.8(s + 1)(s^2 + 0.72s + 1.44) - 3.4s(s^2 + 0.72s + 1.44) + (Cs + D)s(s + 1)$$

$$2.5056(s + 3.33) = 5.8(s^3 + 1.72s^2 + 2.16s + 1.44) - 3.4(s^3 + 0.72s^2 + 1.44s) + (Cs^3 + (C + D)s^2 + Ds)$$

$$2.5056(s + 3.33) = (2.4 + C)s^3 + (7.528 + C + D)s^2 + (10.632 + D)s + 8.352$$

We immediately get C=-2.4 and D=-5.128

So

$$\theta(s) = \frac{5.8}{s} - \frac{3.4}{s + 1} - \frac{2.4s + 5.128}{s^2 + 0.72s + 1.44} = \frac{5.8}{s} - \frac{3.4}{s + 1} - \frac{2.4s + 5.128}{(s + 0.36)^2 + 1.3104}$$

$$= \frac{5.8}{s} - \frac{3.4}{s+1} - \frac{2.4(s+0.15)+4.768}{(s+0.36)^2+1.3104} = \frac{5.8}{s} - \frac{3.4}{s+1} - \frac{2.4(s+0.15)}{(s+0.36)^2+1.3104} - 4.164 \frac{1.145}{(s+0.36)^2+1.3104}$$

Obtaining inverse Laplace transform we get

$$\begin{aligned}\theta(t) &= 5.8 - 3.4e^{-t} - 2.4e^{-0.36t} \cos(1.145t) - 4.164e^{-0.36t} \sin(1.145t) \\ &= 5.8 - 3.4e^{-t} - 2.4e^{-0.36t} \sin(1.145t + 30^\circ)\end{aligned}$$

So the actual (shifted) unit step response is given by

$$\theta(t) = [5.8 - 3.4e^{-(t-0.1)} - 2.4e^{-0.36(t-0.1)} \sin(1.145(t-0.1) + 30^\circ)]u(t-0.1)$$

The response to the pulse is given by:

$$\begin{aligned}\theta(t) &= [1.45m - 0.85me^{-(t-0.1)} - 0.6me^{-0.36(t-0.1)} \sin(1.145(t-0.1) + 30^\circ)]u(t-0.1) - \\ &[1.45m - 0.85me^{-(t-0.25)} - 0.6me^{-0.36(t-0.25)} \sin(1.145(t-0.25) + 30^\circ)]u(t-0.25)\end{aligned}$$

55.

At steady state the input is  $\approx 9V$  and the output is  $\approx 6V$  Thus  $G(0)=6/9=0.667$

The maximum peak is achieved at  $\approx 285\mu$  with a %OS =  $(7.5/6-1)*100 = 25\%$

This corresponds to a damping factor of

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} = \frac{1.3863}{\sqrt{\pi^2 + 1.9218}} \approx 0.4$$

$$\omega_n = \frac{\pi}{T_p \sqrt{1-\zeta^2}} = \frac{\pi}{(285\mu)(0.9165)} = 12027.2$$

So the approximated transfer function is

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{0.667 * 12027.2^2}{s^2 + 2 * 0.4 * 12027.2s + 12027.2^2} = \frac{96.5 * 10^6}{s^2 + 9622s + 14.5 * 10^7}$$

56.

The oscillation period is

$$\frac{2\pi}{T} = \omega_n \sqrt{1 - \zeta^2} \text{ and from the figure } \frac{T}{2} = 0.0675s - 0.0506s = 0.0169s$$

$$\text{Thus } T = 0.0338s \text{ from which we get } \omega_n \sqrt{1 - \zeta^2} = 185.8931$$

The peaks of the response occur when the 'cos' term of the step response is  $\pm 1$  thus from the figure we have:

$$1 + \frac{e^{-\zeta\omega_n(0.0506)}}{\sqrt{1 - \zeta^2}} = 1.1492 \text{ and } 1 - \frac{e^{-\zeta\omega_n(0.0675)}}{\sqrt{1 - \zeta^2}} = 0.9215$$

From which we get

$$\frac{e^{-\zeta\omega_n(0.0506)}}{e^{-\zeta\omega_n(0.0675)}} = \frac{0.1492}{0.0785} = 1.9006 \text{ or } e^{-\zeta\omega_n(0.0169)} = 1.9006 \text{ or } \zeta\omega_n = 38$$

$$\text{Substituting this result we get } \omega_n \sqrt{1 - \zeta^2} = \frac{38}{\zeta} \sqrt{1 - \zeta^2} = 185.8931$$

$$\text{or } \frac{1444}{\zeta^2} (1 - \zeta^2) = 34556.2284 \text{ or } \zeta^2 = 0.0436 \text{ or } \zeta = 0.21$$

$$\text{Finally } \omega_n = \frac{38}{\zeta} = 180.9$$

57.

The step input amplitude is the same for both responses so it will just be assumed to be unitary.

For the 'control' response we have:

$$c_{final} = 0.018, M_{pt} = 0.024 \text{ from which we get}$$

$$\%OS = \frac{c_{max} - c_{final}}{c_{final}} \times 100\% = \frac{0.024 - 0.018}{0.018} \times 100\% = 33.33\%$$

$$\zeta = \frac{-\ln(\%OS / 100)}{\sqrt{\pi^2 + \ln^2(\%OS / 100)}} = \frac{-\ln(0.333)}{\sqrt{\pi^2 + \ln^2(0.333)}} = 0.33$$

$$\omega_n = \frac{\pi}{T_p \sqrt{1 - \zeta^2}} = \frac{\pi}{0.1 \sqrt{1 - 0.333^2}} = 33.3$$

Leading a transfer function

$$G_c(s) = \frac{1108.9}{s^2 + 22s + 1108.9}$$

Similarly for the 'hot tail':

$$C_{final} = 0.023, M_{pt} = 0.029$$

$$\%OS = \frac{0.029 - 0.023}{0.023} \times 100\% = 26.1\%$$

$$\zeta = \frac{-\ln(0.261)}{\sqrt{\pi^2 + \ln^2(0.261)}} = 0.393$$

$$\omega_n = \frac{\pi}{T_p \sqrt{1 - \zeta^2}} = \frac{\pi}{0.1 \sqrt{1 - 0.261^2}} = 34.17$$

$$G_h(s) = \frac{1167.6}{s^2 + 26.9s + 1167.6}$$

Using MATLAB:

```
>> syms s
```

```
>> s=tf('s')
```

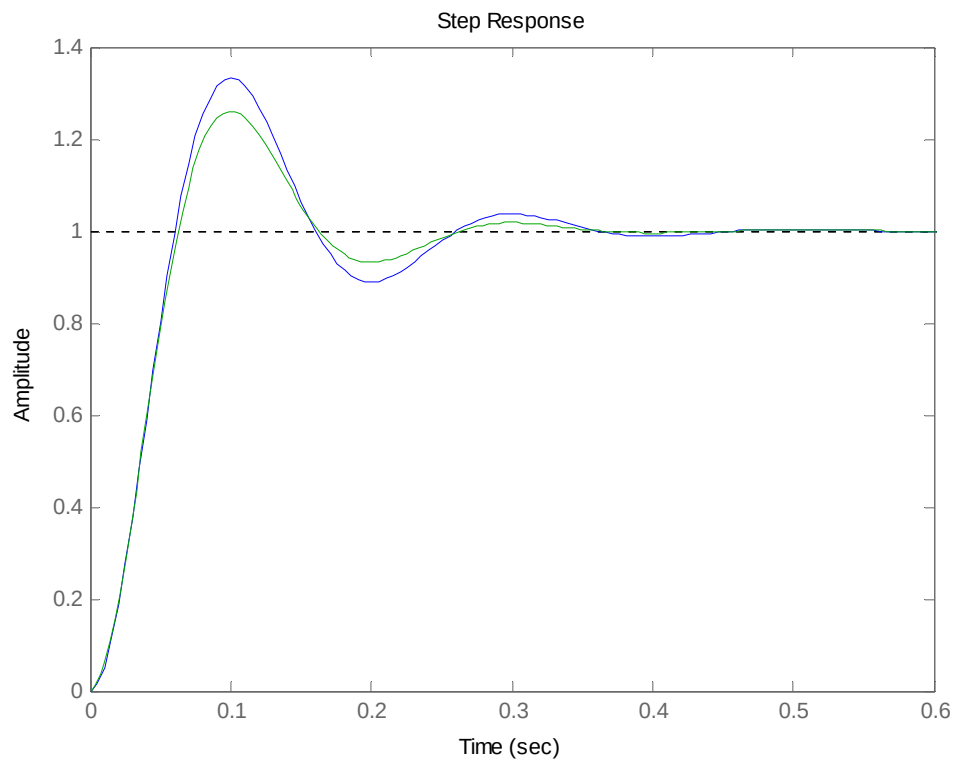
Transfer function:

```
s
```

```
>> Gc = 1108.89/(s^2+22*s+1108.89);
```

```
>> Gh = 1167.6/(s^2+26.9*s+1167.6);
```

```
>> step(Gc,Gh)
```



Both responses are equivalent if error tolerances are considered.

58.

The original transfer function has zeros at  $s = -7200 \pm j7400$

And poles at  $s = -1900 \pm j4500$ ;  $s = -120 \pm j1520$

With  $G(0) = 0.1864$

The dominant poles are those with real parts at -120, so a real pole is added at

-1200 giving the following approximation:

$$G(s) \approx 0.1864 \frac{(1200)(2324.8 \times 10^3)}{106.6 \times 10^6} \frac{(s^2 - 14400s + 106.6 \times 10^6)}{(s^2 + 240s + 2324.8 \times 10^3)(s + 1200)}$$

$$= \frac{4.8782(s^2 - 14400s + 106.6 \times 10^6)}{(s^2 + 240s + 2324.8 \times 10^3)(s + 1200)}$$

Using MATLAB:

```
>> syms s
```

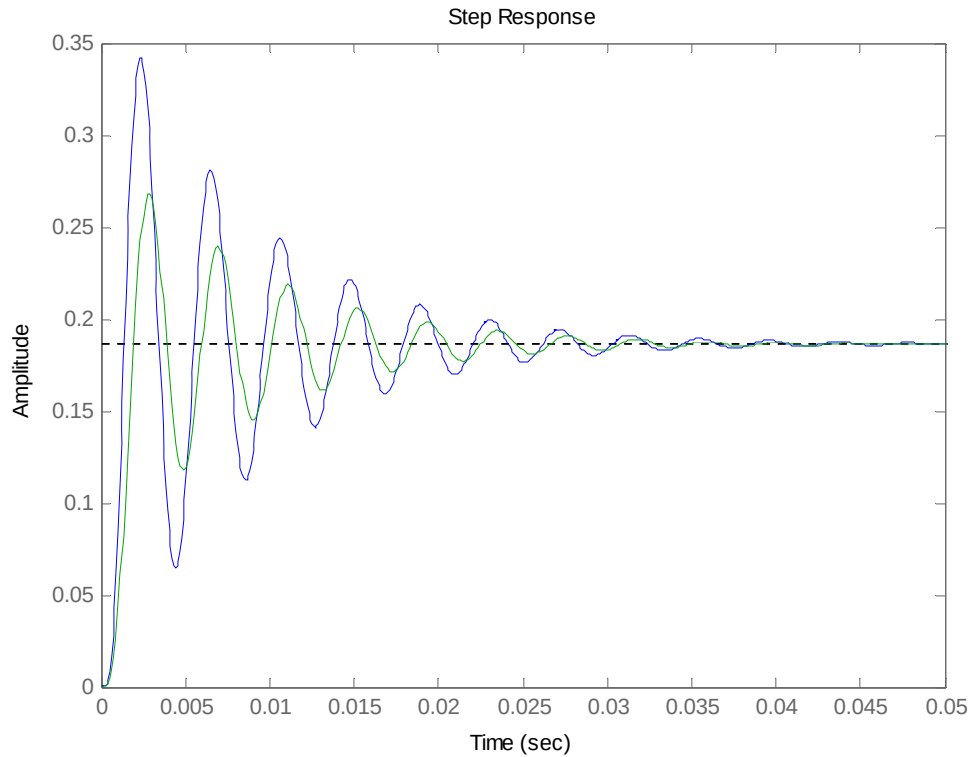
```
>> s=tf('s');
```



```

>>G=9.7e4*(s^2-14400*s+106.6e6)...
    /(s^2+3800*s+23.86e6)/(s^2+240*s+2324.8e3);
>> Gdp=4.8782*(s^2-14400*s+106.6e6)/(s^2+240*s+2324.8e3)/(s+1200);
>> step(G,Gdp)

```



Both responses differ because the original non-dominant poles are very close to the complex pair of zeros.

59.

$M(s)$  requires at least 4 'far away' poles that are added a decade beyond all original poles and zeros.

This gives

$$M(s) = \frac{(s + 0.009)^2 (s^2 + 0.018s + 0.0001)}{9.72 \times 10^{-8} (s + 0.0001)(1 + s/0.1)^4} = \frac{1028.81(s + 0.009)^2 (s^2 + 0.018s + 0.0001)}{(s + 0.0001)(s + 0.1)^4}$$

60.

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} = \frac{-\ln(0.30)}{\sqrt{\pi^2 + \ln^2(0.30)}} = 0.36$$

$$\omega_n = \frac{\pi}{T_p \sqrt{1 - \zeta^2}} = \frac{\pi}{127 \sqrt{1 - 0.30^2}} = 0.026$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{0.00067}{s^2 + 0.0187s + 0.00067}$$

61.

a. Let the impulse response of  $T(s)$  be  $h(t)$ . We have that

$$H(s) = \frac{450}{(s+5)(s+20)} = \frac{A}{s+5} + \frac{B}{s+20}$$

$$A = \frac{450}{s+20} \Big|_{s=-5} = 30; \quad B = \frac{450}{s+5} \Big|_{s=-20} = -30$$

$$H(s) = \frac{30}{s+5} - \frac{30}{s+20}. \text{ Obtaining the inverse Laplace transform we get}$$

$$h(t) = 30e^{-5t} - 30e^{-20t}$$

b. Let the step response of the system be  $g(t)$ . We have that

$$g(t) = \int_0^t h(t) dt = \int_0^t 30e^{-5t} dt - \int_0^t 30e^{-20t} dt = -\frac{30}{5}e^{-5t} \Big|_0^t - \frac{30}{-20}e^{-20t} \Big|_0^t$$

$$= -6(e^{-5t} - 1) + 1.5(e^{-20t} - 1) = 4.5 - 6e^{-5t} + 1.5e^{-20t}$$

$$\text{c. } G(s) = \frac{450}{s(s+5)(s+20)} = \frac{A}{s} + \frac{B}{s+5} + \frac{C}{s+20}$$

$$A = \frac{450}{(s+5)(s+20)} \Big|_{s=0} = 4.5; \quad B = \frac{450}{s(s+20)} \Big|_{s=-5} = -6; \quad C = \frac{450}{s(s+5)} \Big|_{s=-20} = 1.5$$

$$\text{Leading } G(s) = \frac{4.5}{s} - \frac{6}{s+5} + \frac{1.5}{s+20}. \text{ After the inverse Laplace we get}$$

$$g(t) = 4.5 - 6e^{-5t} + 1.5e^{-20t}$$

62.

a. The poles given by  $s^2 + 8.99 \times 10^{-3}s + 3.97 \times 10^{-3} = 0$  have an  $\omega_n = 0.063 \text{ rad/sec}$  and  $\zeta = 0.0714$

The poles given by  $s^2 + 4.21s + 18.23 = 0$  have an  $\omega_n = 4.27 \text{ rad/sec}$  and  $\zeta = 0.493$ . Thus the former represent the Phugoid and the latter the Short Period modes.

**b.** In the original we have  $\frac{\theta}{\delta_e}(0) = -4.85$  so the Phugoid approximation is given by:

$$\frac{\theta}{\delta_e} \approx -\frac{1.965(s + 0.0098)}{(s^2 + 8.99 \times 10^{-3}s + 3.97 \times 10^{-3})}$$

**c.**

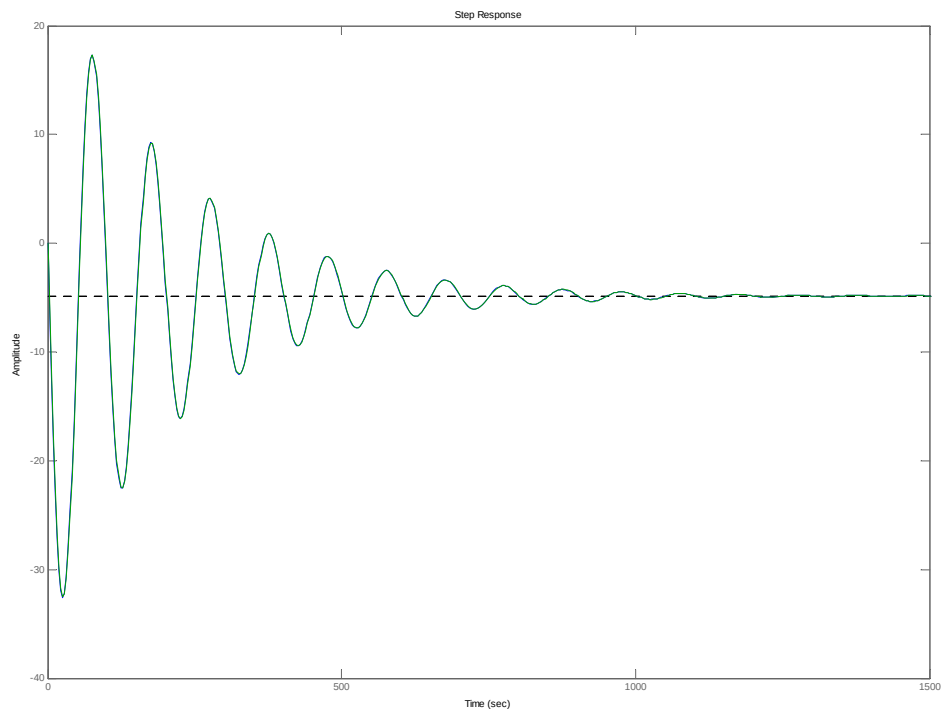
```
>> syms s
```

```
>> s=tf('s');
```

```
>> G=-26.12*(s+0.0098)*(s+1.371)/(s^2+8.99e-3*s+3.97e-3)/(s^2+4.21*s+18.23);
```

```
>> Gphug=-1.965*(s+0.0098)/(s^2+8.99e-3*s+3.97e-3);
```

```
>> step(G,Gphug)
```



Both responses are indistinguishable.

63.

a.

Program

```

numg=[33 202 10061 24332 170704];
deng=[1 8 464 2411 52899 167829 913599 1076555];
G=tf(numg,deng)
[K,p,k]=residue(numg,deng)

```

**Computer Response**

K =

```

0.0018 + 0.0020i
0.0018 - 0.0020i
-0.1155 - 0.0062i
-0.1155 + 0.0062i
0.0077 - 0.0108i
0.0077 + 0.0108i
0.2119

```

p =

```

-1.6971 +16.4799i
-1.6971 -16.4799i
-0.5992 +12.1443i
-0.5992 -12.1443i
-1.0117 + 4.2600i
-1.0117 - 4.2600i
-1.3839

```

k =

[]

b.

Therefore, an approximation to  $G(s)$  is:

$$G(s) = \frac{0.2119}{s + 1.3839}$$

c.

Program

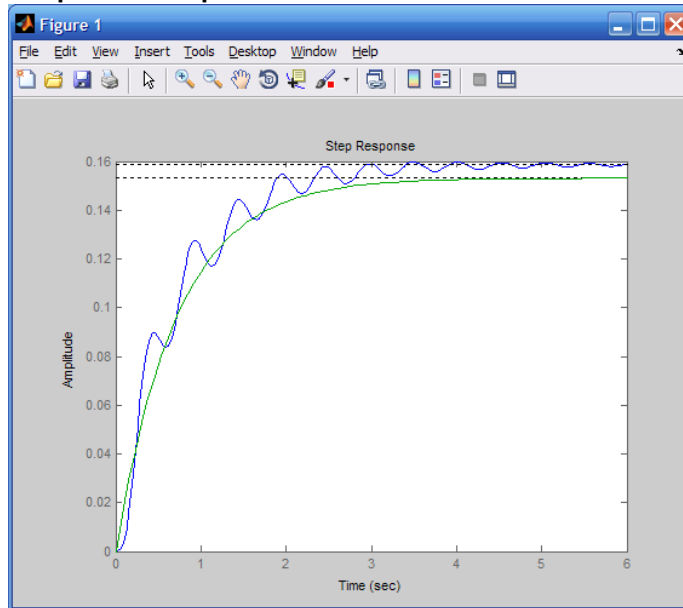
```

numg=[33 202 10061 24332 170704];
deng=[1 8 464 2411 52899 167829 913599 1076555];
G=tf(numg,deng);
numga=0.2119;
denga=[1 1.3839];

```

```
Ga=tf(numga,denga);
step(G)
hold on
step(Ga)
```

### Computer Response



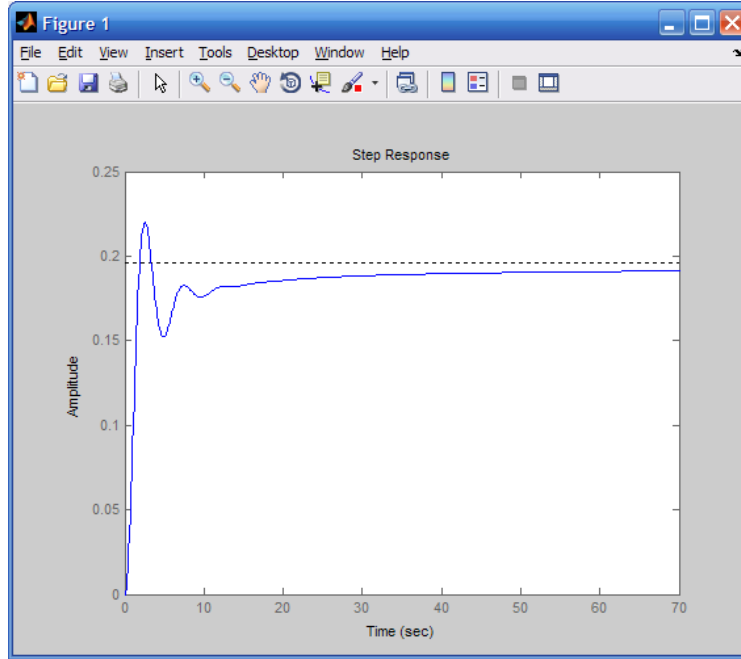
Approximation does not show oscillations and is slightly off of final value.

64.

### Computer Response

Transfer function:

$$\begin{aligned}
 & s^{15} + 1775 s^{14} + 1.104e006 s^{13} + 2.756e008 s^{12} + 2.272e010 s^{11} \\
 & + 7.933e011 s^{10} + 1.182e013 s^9 + 6.046e013 s^8 + 1.322e014 s^7 \\
 & + 1.238e014 s^6 + 3.977e013 s^5 + 5.448e012 s^4 + 3.165e011 s^3 \\
 & + 6.069e009 s^2 + 4.666e007 s + 1.259e005 \\
 & \hline
 & 31.62 s^{17} + 4.397e004 s^{16} + 1.929e007 s^{15} + 2.941e009 s^{14} \\
 & + 1.768e011 s^{13} + 4.642e012 s^{12} + 5.318e013 s^{11} + 2.784e014 s^{10} \\
 & + 7.557e014 s^9 + 1.238e015 s^8 + 1.356e015 s^7 + 8.985e014 s^6 \\
 & + 2.523e014 s^5 + 3.179e013 s^4 + 1.732e012 s^3 + 3.225e010 s^2 \\
 & + 2.425e008 s + 6.414e005
 \end{aligned}$$



65.

a. To find the step responses for these two processes,  $y_a(t)$  and  $y_p(t)$ , we consider first the un-shifted Laplace transform of their outputs for  $X_a(s) = X_p(s) = 1/s$ :

$$Y_a^*(s) = \frac{14.49}{s(1478.26s + 1)} = \frac{9.8 \times 10^{-3}}{s(s + 6.77 \times 10^{-4})} = \frac{A}{s} + \frac{B}{(s + 6.77 \times 10^{-4})} \quad (1),$$

$$\text{where } A = \left. \frac{9.8 \times 10^{-3}}{s + 6.77 \times 10^{-4}} \right|_{s=0} = 14.49 \text{ and}$$

$$B = \left. \frac{9.8 \times 10^{-3}}{s} \right|_{s=-6.77 \times 10^{-4}} = -14.49 \quad (2)$$

Substituting the values of  $A$  and  $B$  into equation (1) gives:

$$Y_a^*(s) = \frac{A}{s} + \frac{B}{(s + 6.76 \times 10^{-4})} = 14.49 \left( \frac{1}{s} - \frac{1}{(s + 6.76 \times 10^{-4})} \right) \quad (3)$$

Taking the inverse Laplace transform of  $Y_a^*(s)$  and delaying the resulting response in the time domain by 4 seconds, we get:

$$y_a(t) = 14.49[1 - e^{-6.76 \times 10^{-4}(t-4)}] u(t-4) \quad (4)$$

Noting that the denominator of  $G_p(s)$  can be factored into

$(s + 0.174 \times 10^{-3})(s + 6.814 \times 10^{-3})$ , we have:

$$Y_p^*(s) = \frac{1.716 \times 10^{-5}}{s(s + 0.174 \times 10^{-3})(s + 6.814 \times 10^{-3})} = \frac{C}{s} + \frac{D}{(s + 0.174 \times 10^{-3})} + \frac{E}{(s + 6.814 \times 10^{-3})} \quad (5),$$

$$\text{where: } C = \left. \frac{1.716 \times 10^{-5}}{(s + 0.174 \times 10^{-3})(s + 6.814 \times 10^{-3})} \right|_{s=0} = 14.48;$$

$$D = \left. \frac{1.716 \times 10^{-5}}{s(s + 6.814 \times 10^{-3})} \right|_{s=-0.174 \times 10^{-3}} = -14.85;$$

$$E = \left. \frac{1.716 \times 10^{-5}}{s(s + 0.174 \times 10^{-3})} \right|_{s=-6.814 \times 10^{-3}} = 0.37. \quad (6)$$

Substituting the values of  $C$ ,  $D$  and  $E$  into equation (5) and simplifying gives:

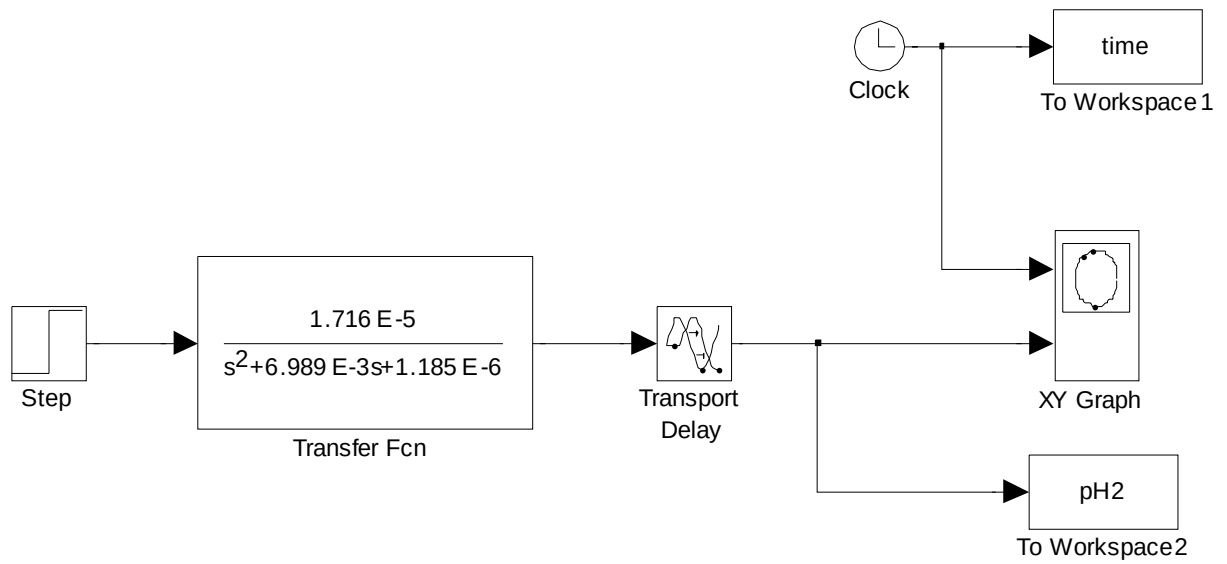
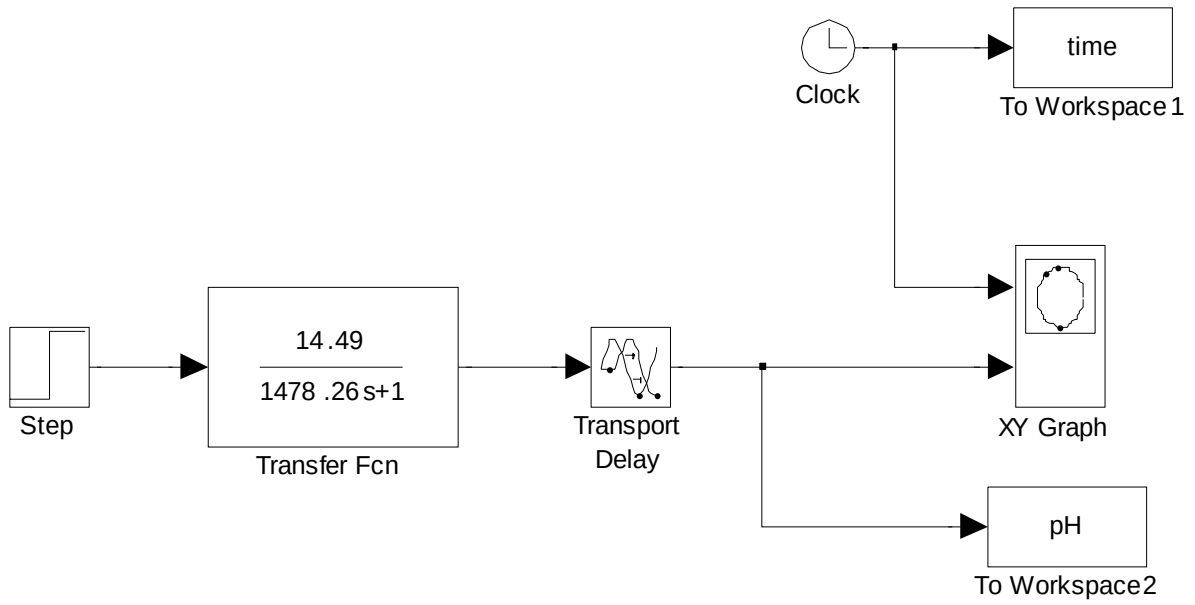
$$Y_p^*(s) = \frac{14.48}{s} - \frac{14.85}{(s + 0.174 \times 10^{-3})} + \frac{0.37}{(s + 6.814 \times 10^{-3})} = 14.48 \left( \frac{1}{s} - \frac{1.0256}{(s + 0.174 \times 10^{-3})} + \frac{0.0256}{(s + 6.814 \times 10^{-3})} \right) \quad (7)$$

Taking the inverse Laplace transform of  $Y_p^*(s)$  and delaying the resulting response in the time

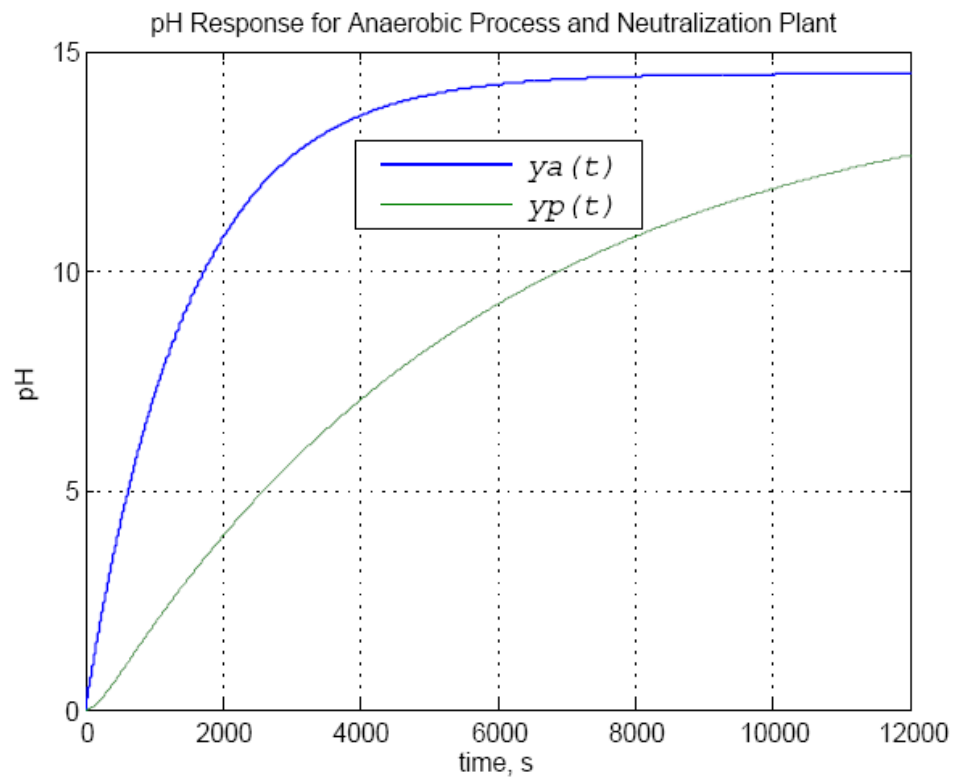
domain by 30 seconds, we get:

$$y_p(t) = 14.48[1 - 1.0256e^{-0.174 \times 10^{-3}(t-30)} + 0.0256e^{-6.814 \times 10^{-3}(t-30)}]u(t-30) \quad (8)$$

b. Using Simulink to model the two processes described above,  $y_a(t)$  and  $y_p(t)$  were output to the “workspace.” Matlab plot commands were then utilized to plot  $y_a(t)$  and  $y_p(t)$  on a single graph, which is shown below.







66.

a.

```
>> A=[-8.792e-3 0.56e-3 -1e-3 -13.79e-3; -0.347e-3 -11.7e-3 -0.347e-3 0; 0.261 -20.8e-3 -96.6e-3
0; 0 0 1 0]
```

A =

```
-0.0088  0.0006 -0.0010 -0.0138
-0.0003 -0.0117 -0.0003      0
0.2610 -0.0208 -0.0966      0
      0      0 1.0000      0
```

```
>> eig(A)
```

ans =

```
-0.1947
0.0447 + 0.1284i
0.0447 - 0.1284i
-0.0117
```

b.

Given the eigenvalues, the state-transition matrix will be of the form

$$\Phi(t) = \begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{21} & K_{22} & K_{23} & K_{24} \\ K_{31} & K_{32} & K_{33} & K_{34} \\ K_{41} & K_{42} & K_{43} & K_{44} \end{bmatrix} \quad \text{with}$$

$$K_{ij} = K_{ij1} e^{-0.1947 t} + K_{ij2} e^{-0.0117 t} + K_{ij3} e^{+0.0447 t} \sin(0.1284 t) + K_{ij4} e^{+0.0447 t} \cos(0.1284 t)$$

Thus 64 constants have to be found.

67.

a. The equations are rewritten as

$$\begin{aligned}\frac{di_L}{dt} &= -\frac{1-d}{L}u_C + \frac{1}{L}E_s \\ \frac{du_C}{dt} &= \frac{1-d}{C}i_L - \frac{1}{RC}u_C\end{aligned}$$

from which we obtain

$$\begin{aligned}\begin{bmatrix} \frac{di_L}{dt} \\ \frac{du_C}{dt} \end{bmatrix} &= \begin{bmatrix} 0 & -\frac{1-d}{L} \\ \frac{1-d}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ u_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} E_s \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ u_C \end{bmatrix}\end{aligned}$$

b. To obtain the transfer function we first calculate

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{bmatrix} s & \frac{1-d}{L} \\ -\frac{1-d}{C} & s + \frac{1}{RC} \end{bmatrix}^{-1} = \frac{\begin{bmatrix} s + \frac{1}{RC} & -\frac{1-d}{L} \\ \frac{1-d}{C} & s \end{bmatrix}}{s(s + \frac{1}{RC}) + \frac{(1-d)^2}{LC}}$$

So

$$\begin{aligned}G(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \begin{bmatrix} 0 & 1 \end{bmatrix} \frac{\begin{bmatrix} s + \frac{1}{RC} & -\frac{1-d}{L} \\ \frac{1-d}{C} & s \end{bmatrix}}{s(s + \frac{1}{RC}) + \frac{(1-d)^2}{LC}} \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} \\ &= \frac{\begin{bmatrix} \frac{1-d}{C} & s \end{bmatrix}}{s^2 + \frac{1}{RC}s + \frac{(1-d)^2}{LC}} \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} = \frac{\frac{1-d}{LC}}{s^2 + \frac{1}{RC}s + \frac{(1-d)^2}{LC}}\end{aligned}$$

68.

a. We have  $(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s + 8.34 & 2.26 \\ -1 & s \end{bmatrix}$  and

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\begin{bmatrix} s & -2.26 \\ 1 & s + 8.34 \end{bmatrix}}{s^2 + 8.34s + 2.26} = \frac{\begin{bmatrix} s & -2.26 \\ 1 & s + 8.34 \end{bmatrix}}{(s + 0.28)(s + 8.06)}$$

We first find  $\mathcal{L}^{-1} \left\{ \frac{1}{(s + 0.28)(s + 8.06)} \right\}$

$$\frac{1}{(s + 0.28)(s + 8.06)} = \frac{K_1}{s + 0.28} + \frac{K_2}{s + 8.06}$$

$$K_1 = \frac{1}{s + 8.06} \Big|_{s=-0.28} = 0.129; K_2 = \frac{1}{s + 0.28} \Big|_{s=-8.06} = -0.129 \text{ so}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s + 0.28)(s + 8.06)} \right\} = 0.129e^{-0.28t} - 0.129e^{-8.06t}$$

Follows that

$$\mathcal{L}^{-1} \left\{ \frac{-2.26}{(s + 0.28)(s + 8.06)} \right\} = -2.26 \mathcal{L}^{-1} \left\{ \frac{1}{(s + 0.28)(s + 8.06)} \right\} = -0.292e^{-0.28t} + 0.292e^{-8.06t}$$

$$\mathcal{L}^{-1} \left\{ \frac{s}{(s + 0.28)(s + 8.06)} \right\} = \frac{d}{dt} \mathcal{L}^{-1} \left\{ \frac{1}{(s + 0.28)(s + 8.06)} \right\} = -0.036e^{-0.28t} + 1.04e^{-8.06t}$$

And

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{s + 8.34}{(s + 0.28)(s + 8.06)} \right\} &= \mathcal{L}^{-1} \left\{ \frac{s}{(s + 0.28)(s + 8.06)} \right\} + 8.34 \mathcal{L}^{-1} \left\{ \frac{1}{(s + 0.28)(s + 8.06)} \right\} \\ &= -0.036e^{-0.28t} + 1.04e^{-8.06t} + 1.076e^{-0.28t} - 1.076e^{-8.06t} = 1.04e^{-0.28t} - 0.036e^{-8.06t} \end{aligned}$$

Finally the state transition matrix is given by:

$$\Phi(t) = \begin{bmatrix} -0.036e^{-0.28t} + 1.04e^{-8.06t} & -0.292e^{-0.28t} + 0.292e^{-8.06t} \\ 0.129e^{-0.28t} - 0.129e^{-8.06t} & 1.04e^{-0.28t} - 0.036e^{-8.06t} \end{bmatrix}$$

b.

$$\Phi(t)\mathbf{B} = \begin{bmatrix} -0.036e^{-0.28t} + 1.04e^{-8.06t} \\ 0.129e^{-0.28t} - 0.129e^{-8.06t} \end{bmatrix}$$

$$\mathbf{C}\Phi(t)\mathbf{B} = -0.451e^{-0.28t} + 13.04e^{-8.06t} + 0.292e^{-0.28t} - 0.292e^{-8.06t} = -0.159e^{-0.28t} + 12.748e^{-8.06t}$$

Since  $u(t) = 1$ 

$$\begin{aligned} y(t) &= \int_0^t \mathbf{C}\Phi(t-\tau)\mathbf{B}d\tau = \int_0^t [-0.159e^{-0.28(t-\tau)} + 12.748e^{-8.06(t-\tau)}]d\tau \\ &= -0.159e^{-0.28t} \int_0^t e^{0.28\tau} d\tau + 12.748e^{-8.06t} \int_0^t e^{8.06\tau} d\tau \\ &= \frac{-0.159}{0.28} e^{-0.28t} e^{0.28\tau} + \frac{12.748}{8.06} e^{-8.06t} e^{8.06\tau} \Big|_0^t \\ &= -0.568[1 - e^{-0.28t}] + 1.582[1 - e^{-8.06t}] = 1.014 + 0.568e^{-0.28t} - 1.582e^{-8.06t} \end{aligned}$$

c.

&gt;&gt; A=[-8.34 -2.26; 1 0];

&gt;&gt; B = [1; 0];

&gt;&gt; C = [12.54 2.26];

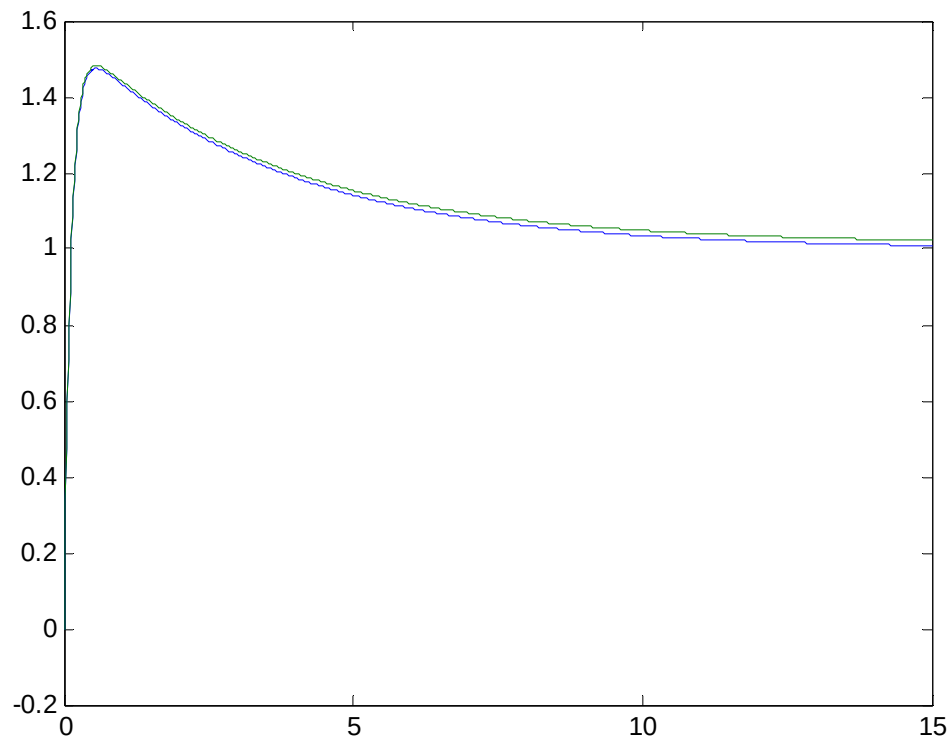
&gt;&gt; D = 0;

&gt;&gt; t = linspace(0,15,1000);

&gt;&gt; y1 = step(A,B,C,D,1,t);

&gt;&gt; y2 = 1.014+0.568\*exp(-0.28.\*t)-1.582\*exp(-8.06.\*t);

&gt;&gt; plot(t,y1,t,y2)



## SOLUTIONS TO DESIGN PROBLEMS

69.

Writing the equation of motion,  $(f_v s + 2)X(s) = F(s)$ . Thus, the transfer function is

$$\frac{X(s)}{F(s)} = \frac{1/f_v}{s + \frac{2}{f_v}}. \text{ Hence, } T_s = \frac{4}{a} = \frac{4}{\frac{2}{f_v}} = 2 f_v, \text{ or } f_v = \frac{T_s}{2}.$$

70.

The transfer function is,  $F(s) = \frac{1/M}{s^2 + \frac{1}{M}s + \frac{K}{M}}$ . Now,  $T_s = 4 = \frac{4}{|\text{Re}|} = \frac{4}{\frac{1}{2M}}$ . Thus,

$M = \frac{1}{2}$ . Substituting the value of M in the denominator of the transfer function yields,

$s^2 + 2s + 2K$ . Identify the roots  $s_{1,2} = -1 \pm j\sqrt{2K-1}$ . Using the imaginary part and substituting

into the peak time equation yields  $T_p = 1 = \frac{\pi}{|\text{Im}|} = \frac{\pi}{\sqrt{2K-1}}$ , from which  $K = 5.43$ .

71.

Writing the equation of motion,  $(Ms^2 + f_v s + 1)X(s) = F(s)$ . Thus, the transfer function is  $\frac{X(s)}{F(s)} = \frac{1/M}{s^2 + \frac{f_v}{M}s + \frac{1}{M}}$ . Since  $T_s = 10 = \frac{4}{\zeta\omega_n}$ ,  $\zeta\omega_n = 0.4$ . But,  $\frac{f_v}{M} = 2\zeta\omega_n = 0.8$ . Also,

from Eq. (4.39) 17% overshoot implies  $\zeta = 0.491$ . Hence,  $\omega_n = 0.815$ . Now,  $1/M = \omega_n^2 = 0.664$ .

Therefore,  $M = 1.51$ . Since  $\frac{f_v}{M} = 2\zeta\omega_n = 0.8$ ,  $f_v = 1.21$ .

72.

Writing the equation of motion:  $(Js^2 + s + K)\theta(s) = T(s)$ . Therefore the transfer function is

$$\frac{\theta(s)}{T(s)} = \frac{\frac{1}{J}}{s^2 + \frac{1}{J}s + \frac{K}{J}}$$

$$\zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}} = 0.358.$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{1}{2J}} = 8J = 3.$$

Therefore  $J = \frac{3}{8}$ . Also,  $T_s = 3 = \frac{4}{\zeta\omega_n} = \frac{4}{(0.358)\omega_n}$ . Hence,  $\omega_n = 3.724$ . Now,  $\frac{K}{J} = \omega_n^2 = 13.868$ . Finally,  $K = 5.2$ .

73.

Writing the equation of motion

$$[s^2 + D(5)s + \frac{1}{4}(10)^2]\theta(s) = T(s)$$

The transfer function is

$$\frac{\theta(s)}{T(s)} = \frac{1}{s^2 + 25Ds + 25}$$

Also,

$$\zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}} = 0.358$$

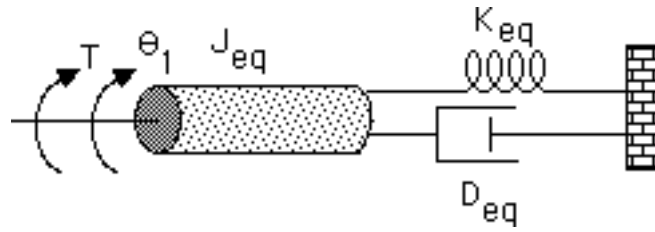
and

$$2\zeta\omega_n = 2(0.358)(5) = 25D$$

Therefore  $D = 0.14$ .

74.

The equivalent circuit is:



where  $J_{eq} = 1 + (\frac{N_1}{N_2})^2$ ;  $D_{eq} = (\frac{N_1}{N_2})^2$ ;  $K_{eq} = (\frac{N_1}{N_2})^2$ . Thus,

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{J_{eq}s^2 + D_{eq}s + K_{eq}}. \text{ Letting } \frac{N_1}{N_2} = n \text{ and substituting the above values into the transfer}$$

function,

$$\frac{\theta_1(s)}{T(s)} = \frac{1}{s^2 + \frac{n^2}{1+n^2}s + \frac{n^2}{1+n^2}}. \text{ Therefore, } \zeta\omega_n = \frac{n^2}{2(1+n^2)}. \text{ Finally, } T_s = \frac{4}{\zeta\omega_n} = \frac{8(1+n^2)}{n^2} = 16. \text{ Thus}$$

$$n = 1.$$

75.

Let the rotation of the shaft with gear  $N_2$  be  $\theta_L(s)$ . Assuming that all rotating load has been reflected

to the  $N_2$  shaft,  $(J_{eqL}s^2 + D_{eqL}s + K)\theta_L(s) + F(s)r = T_{eq}(s)$ , where  $F(s)$  is the force from the translational system,  $r = 2$  is the radius of the rotational member,  $J_{eqL}$  is the equivalent inertia at the  $N_2$  shaft, and  $D_{eqL}$  is the equivalent damping at the  $N_2$  shaft. Since  $J_{eqL} = 1(2)^2 + 1 = 5$  and  $D_{eqL} = 1(2)^2 = 4$ , the equation of motion becomes,  $(5s^2 + 4s + K)\theta_L(s) + 2F(s) = T_{eq}(s)$ . For the translational system  $(Ms^2 + s)X(s) = F(s)$ . Substituting  $F(s)$  into the rotational equation of motion,  $(5s^2 + 4s + K)\theta_L(s) + (Ms^2 + s)2X(s) = T_{eq}(s)$ .

$$\text{But, } \theta_L(s) = \frac{X(s)}{r} = \frac{X(s)}{2} \text{ and } T_{eq}(s) = 2T(s). \text{ Substituting these quantities in the equation}$$

above yields  $((5 + 4M)s^2 + 8s + K)\frac{X(s)}{4} = T(s)$ . Thus, the transfer function is

$$\frac{X(s)}{T(s)} = \frac{4/(5 + 4M)}{s^2 + \frac{8}{(5 + 4M)}s + \frac{K}{(5 + 4M)}}. \text{ Now, } T_s = 15 = \frac{4}{\text{Re}} = \frac{4}{\frac{8}{2(5 + 4M)}} = (5 + 4M).$$

Hence,  $M = 5/2$ . For 10% overshoot,  $\zeta = 0.5912$  from Eq. (4.39). Hence,



$$2\zeta\omega_n = \frac{8}{(5+4M)} = 0.5333. \text{ Solving for } \omega_n \text{ yields } \omega_n = 0.4510. \text{ But,}$$

$$\omega_n = \sqrt{\frac{K}{(5+4M)}} = \sqrt{\frac{K}{15}} = 0.4510. \text{ Thus, } K = 3.051.$$

76.

The transfer function for the capacitor voltage is  $\frac{V_C(s)}{V(s)} = \frac{\frac{1}{Cs}}{R+Ls+\frac{1}{Cs}} = \frac{10^6}{s^2+Rs+10^6}.$

For 20% overshoot,  $\zeta = \frac{-\ln(\frac{\%OS}{100})}{\sqrt{\pi^2 + \ln^2(\frac{\%OS}{100})}} = 0.456.$  Therefore,  $2\zeta\omega_n = R = 2(0.456)(10^3) =$

912Ω.

77.

Solving for the capacitor voltage using voltage division,  $V_C(s) = V_i(s) \frac{1/(CS)}{R+LS+\frac{1}{CS}}.$  Thus, the

transfer function is  $\frac{V_C(s)}{V_i(s)} = \frac{1/(LC)}{s^2 + \frac{R}{L}s + \frac{1}{LC}}.$  Since  $T_s = \frac{4}{|\text{Re}|} = 10^{-3}, |\text{Re}| = \frac{R}{2L} = 4000.$  Thus

$R = 8 \text{ K}\Omega.$  Also, since 20% overshoot implies a damping ratio of 0.46 and

$2\zeta\omega_n = 8000, \omega_n = 8695.65 = \frac{1}{\sqrt{LC}}.$  Hence,  $C = 0.013 \mu\text{F}.$

78.

Using voltage division the transfer function is,

$$\frac{V_C(s)}{V_i(s)} = \frac{\frac{1}{Cs}}{R+Ls+\frac{1}{Cs}} = \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

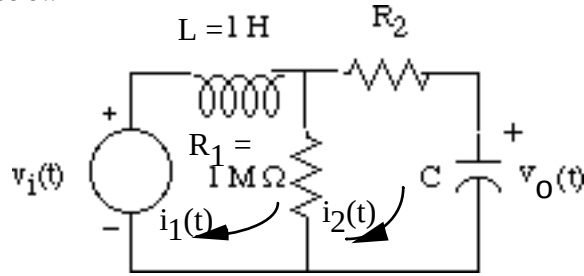
Also,  $T_s = 7 \times 10^{-3} = \frac{4}{\text{Re}} = \frac{4}{\frac{R}{2L}} = \frac{8L}{R}.$  Thus,  $\frac{R}{L} = 1143.$  Using Eq. (4.39) with 15% overshoot,  $\zeta$

$= 0.5169.$  But,  $2\zeta\omega_n = R/L.$  Thus,  $\omega_n = 1105.63 = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{L(10^{-5})}}.$  Therefore,  $L = 81.8 \text{ mH}$

and  $R = 98.5 \Omega.$

79.

For the circuit shown below



write the loop equations as

$$\begin{aligned}(R_1 + Ls)I_1(s) - R_1 I_2(s) &= V_i(s) \\ -R_1 I_1(s) + \left(R_1 + R_2 + \frac{1}{Cs}\right)I_2(s) &= 0\end{aligned}$$

Solving for  $I_2(s)$ 

$$I_2(s) = \frac{\begin{vmatrix} R_1 + Ls & V_i(s) \\ -R_1 & 0 \end{vmatrix}}{\begin{vmatrix} R_1 + Ls & -R_1 \\ -R_1 & R_1 + R_2 + \frac{1}{Cs} \end{vmatrix}}$$

But,  $V_o(s) = \frac{1}{Cs} I_2(s)$ . Thus,

$$\frac{V_o(s)}{V_i(s)} = \frac{R_1}{(R_2 + R_1)Cs^2 + (CR_2R_1 + L)s + R_1}$$

Substituting component values,

$$\frac{V_o(s)}{V_i(s)} = 1000000 \frac{1}{s^2 + \frac{(1000000CR_2 + 1)}{(R_2 + 1000000)C}s + 1000000 \frac{1}{(R_2 + 1000000)C}}$$

For 8% overshoot,  $\zeta = 0.6266$ . For  $T_s = 0.001$ ,  $\zeta\omega_n = \frac{4}{0.001} = 4000$ . Hence,  $\omega_n = 6383.66$ . Thus,

$$1000000 \frac{1}{(R_2 + 1000000)C} = 6383.66^2$$

or,

$$C = 0.0245 \frac{1}{R_2 + 1000000} \quad (1)$$

Also,

$$\frac{1000000 C R_2 + 1}{(R_2 + 1000000) C} = 8000 \quad (2)$$

Solving (1) and (2) simultaneously,  $R_2 = 8023 \, \Omega$ , and  $C = 2.4305 \times 10^{-2} \, \mu\text{F}$ .

80.

$$\begin{aligned} s\mathbf{I} - \mathbf{A} &= \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} (3.45 - 14000K_c) & -0.255 \times 10^{-9} \\ 0.499 \times 10^{11} & -3.68 \end{bmatrix} \\ &= \begin{bmatrix} s - (3.45 - 14000K_c) & 0.255 \times 10^{-9} \\ -0.499 \times 10^{11} & s + 3.68 \end{bmatrix} \\ |s\mathbf{I} - \mathbf{A}| &= s^2 + (0.23 + 0.14 \times 10^5 K_c)s + (51520K_c + 0.0285) \\ (2\zeta\omega_n)^2 &= [2 \times 0.9]^2 \times (51520K_c + 0.0285) = (0.23 + 0.14 \times 10^5 K_c)^2 \end{aligned}$$

or

$$K_c^2 - 8.187 \times 10^{-4} K_c - 2.0122 \times 10^{-10} = 0$$

Solving for  $K_c$ ,

$$K_c = 8.189 \times 10^{-4}$$

81.

a. The transfer function from Chapter 2 is,

$$\frac{Y_h(s) - Y_{cat}(s)}{F_{up}(s)} = \frac{0.7883(s + 53.85)}{(s^2 + 15.47s + 9283)(s^2 + 8.119s + 376.3)}$$

The dominant poles come from  $s^2 + 8.119s + 376.3$ . Using this polynomial,

$2\zeta\omega_n = 8.119$ , and  $\omega_n^2 = 376.3$ . Thus,  $\omega_n = 19.4$  and  $\zeta = 0.209$ . Using Eq. (4.38), %OS =

51.05%. Also,  $T_s = \frac{4}{\zeta\omega_n} = 0.985 \, \text{s}$ , and  $T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = 0.166 \, \text{s}$ . To find rise time, use

Figure 4.16. Thus,  $\omega_n T_r = 1.2136$  or  $T_r = 0.0626 \, \text{s}$ .

b. The other poles have a real part of  $15.47/2 = 7.735$ . Dominant poles have a real part of  $8.119/2 = 4.06$ . Thus,  $7.735/4.06 = 1.91$ . This is not at least 5 times.

c.

**Program:**

```
syms s
numg=0.7883*(s+53.85);
deng=(s^2+15.47*s+9283)*(s^2+8.119*s+376.3);
'G(s) transfer function'
G=vpa(numg/deng,3);
pretty(G)
numg=sym2poly(numg);
deng=sym2poly(deng);
G=tf(numg,deng)
step(G)
```

**Computer response:**

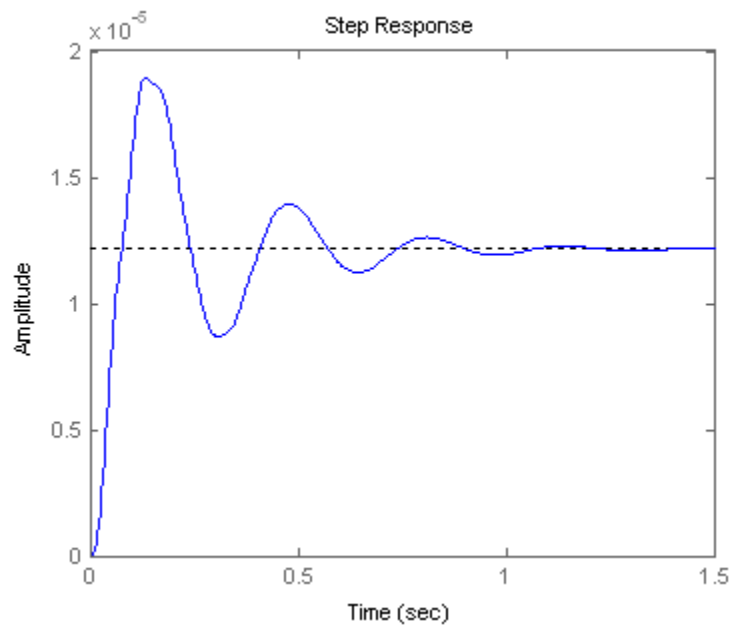
ans =

G(s) transfer function

$$\frac{.788 s + 42.4}{(s^2 + 15.5 s + 9280.) (s^2 + 8.12 s + 376.)}$$

Transfer function:

$$\frac{0.7883 s + 42.45}{s^4 + 23.59 s^3 + 9785 s^2 + 8.119e004 s + 3.493e006}$$



The time response shows 58 percent overshoot,  $T_s = 0.86$  s,  $T_p = 0.13$  s,  $T_r = 0.05$  s.

82.

a. In Problem 3.30 we had

$$\begin{bmatrix} \dot{T} \\ \dot{T}^* \\ \dot{v} \end{bmatrix} = \begin{bmatrix} -(d + \beta v_0) & 0 & -\beta T_0 \\ \beta v_0 & -\mu & \beta T_0 \\ 0 & k & -c \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix} + \begin{bmatrix} \beta T_0 v_0 & 0 \\ -\beta T_0 v_0 & 0 \\ 0 & -k T_0^* \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix}$$

When  $u_2 = 0$  the equations are equivalent to

$$\begin{bmatrix} \dot{T} \\ \dot{T}^* \\ \dot{v} \end{bmatrix} = \begin{bmatrix} -(d + \beta v_0) & 0 & -\beta T_0 \\ \beta v_0 & -\mu & \beta T_0 \\ 0 & k & -c \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix} + \begin{bmatrix} \beta T_0 v_0 \\ -\beta T_0 v_0 \\ 0 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix}$$

Substituting parameter values one gets

$$\begin{bmatrix} \dot{T} \\ \dot{T}^* \\ \dot{v} \end{bmatrix} = \begin{bmatrix} -0.04167 & 0 & -0.0058 \\ 0.0217 & -0.24 & 0.0058 \\ 0 & 100 & -2.4 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix} + \begin{bmatrix} 5.2 \\ -5.2 \\ 0 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix}$$

**b.**

$$(s\mathbf{I} - \mathbf{A})^{-1} = \begin{bmatrix} s + 0.04167 & 0 & 0.0058 \\ -0.0217 & s + 0.24 & -0.0058 \\ 0 & -100 & s + 2.4 \end{bmatrix}^{-1} = \frac{\text{Adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}$$

$$\begin{aligned} \det(s\mathbf{I} - \mathbf{A}) &= (s + 0.04167) \begin{vmatrix} s + 0.24 & -0.0058 \\ -100 & s + 2.4 \end{vmatrix} + 0.0058 \begin{vmatrix} -0.0217 & s + 0.24 \\ 0 & -100 \end{vmatrix} \\ &= (s + 0.04167)[(s + 0.24)(s + 2.4) - 0.58] + (0.0058)(2.17) \\ &= s^3 + 2.6817s^2 + 0.11s + 0.0126 \\ &= (s + 2.6419)(s^2 + 0.0398s + 0.0048) \end{aligned}$$

To obtain the adjoint matrix we calculate the cofactors:

$$\begin{aligned} C_{11} &= \begin{vmatrix} s + 0.24 & -0.0058 \\ -100 & s + 2.4 \end{vmatrix} = s(s + 2.64) \\ C_{12} &= \begin{vmatrix} -0.0217 & -0.0058 \\ 0 & s + 2.4 \end{vmatrix} = -0.0217(s + 2.4) \\ C_{13} &= \begin{vmatrix} -0.0217 & s + 0.24 \\ 0 & -100 \end{vmatrix} = 2.17 \\ C_{21} &= \begin{vmatrix} 0 & 0.0058 \\ -100 & s + 2.4 \end{vmatrix} = 0.58 \end{aligned}$$

$$C_{22} = \begin{vmatrix} s + 0.04167 & 0.0058 \\ 0 & s + 2.4 \end{vmatrix} = (s + 0.04167)(s + 2.4)$$

$$C_{23} = \begin{vmatrix} s + 0.04167 & 0 \\ 0 & -100 \end{vmatrix} = -100(s + 0.04167)$$

$$C_{31} = \begin{vmatrix} 0 & 0.0058 \\ s + 0.24 & -0.0058 \end{vmatrix} = -0.0058(s + 0.24)$$

$$C_{32} = \begin{vmatrix} s + 0.04167 & 0.0058 \\ -0.0217 & s + 2.4 \end{vmatrix} = s^2 + 2.4117s + 0.1001$$

$$C_{33} = \begin{vmatrix} s + 0.04167 & 0 \\ -0.0217 & s + 0.24 \end{vmatrix} = s^2 + 0.2817s + 0.01$$

Then we have

$$\text{Adj}(s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s(s + 2.64) & -0.58 & -0.0058(s + 0.24) \\ 0.0217(s + 2.4) & (s + 0.04167)(s + 2.4) & -(s^2 + 2.4417s + 0.1101) \\ 2.17 & 100(s + 0.04167) & s^2 + 0.2817s + 0.01 \end{bmatrix}$$

Finally

$$\begin{aligned} \frac{Y}{U_1}(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} = \frac{\begin{bmatrix} 2.17 & 100(s + 0.04167) & s^2 + 0.2817s + 0.01 \end{bmatrix}}{(s + 2.6419)(s^2 + 0.0398s + 0.0048)} \begin{bmatrix} 5.2 \\ -5.2 \\ 0 \end{bmatrix} \\ &= \frac{-520s - 10.3844}{(s + 2.6419)(s^2 + 0.0398s + 0.0048)} = -520 \frac{s + 0.02}{(s + 2.6419)(s^2 + 0.0398s + 0.0048)} \end{aligned}$$

c. 100% effectiveness means that  $u_1 = 1$  or  $U_1(s) = \frac{1}{s}$ , so by the final value theorem

$$y(\infty) = \lim_{s \rightarrow 0} sY(s) = -\lim_{s \rightarrow 0} s \frac{520(s + 0.02)}{(s + 2.6419)(s^2 + 0.0398s + 0.0048)} \frac{1}{s} = -820.1168$$

(virus copies per mL of plasma)

The closest poles to the imaginary axis are  $-0.0199 \pm j0.0661$  so the approximate settling time

$$\text{will be } T_s \approx \frac{4}{0.0199} = 210 \text{ days.}$$

83.

a.

Substituting  $\Delta F(s) = \frac{2650}{s}$  into the transfer function and solving for  $\Delta V(s)$  gives:

$$\Delta V(s) = \frac{\Delta F(s)}{1908 \cdot s} = \frac{2650}{s(1908 \cdot s + 10)} = \frac{A}{s} + \frac{B}{(1908 \cdot s + 10)}$$

$$\text{Here: } A = \left. \frac{2650}{(1908 \cdot s + 10)} \right|_{s=0} = 265 \text{ and } B = \left. \frac{2650}{s} \right|_{s=-1/190.8} = -505,620$$

Substituting we have:

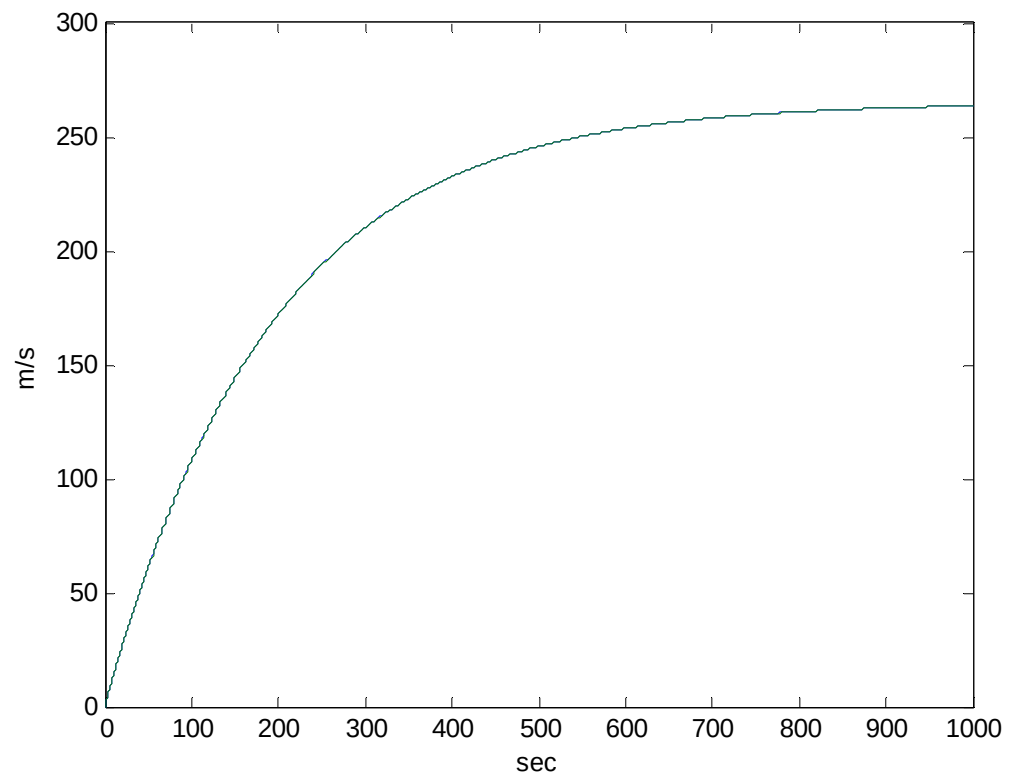
$$\Delta V(s) = \frac{265}{s} - \frac{505620}{(1908 \cdot s + 10)} = 265 \left( \frac{1}{s} - \frac{1}{(s + 5.24 \times 10^{-3})} \right)$$

Taking the inverse Laplace transform, we have:

$$\Delta v(t) = 265(1 - e^{-5.24 \times 10^{-3} t}) \cdot u(t), \text{ in m/s}$$

**b.**

```
>> s=tf('s');
>> G=1/(1908*s+10);
>> t=0:0.1:1000;
>> y1=2650*step(G,t);
>> y2=265*(1-exp(-5.24e-3.*t));
>> plot(t,y1,t,y2)
>> xlabel('sec')
>> ylabel('m/s')
```



Both plots are identical.